

Econ 210 The History & Economics of Money

Numbered Transcript

ECON 210 • Dr. Yaron Brook

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Lecture 1: The First Currencies

At a glance — Brook on the first currencies and the nature of money.

- 1.1 All right, we are here to talk about money, this stuff. I think I've got some in my pocket, this paper stuff, or at least this is how we represent it in our modern times. We're going to be asking questions like, what is money? How come this paper stuff gets to be money? What is the real value of this? And what about something like this? I mean, it's not exactly money, but it kind of connects it, kind of related. And everything in between—there are all kinds of these stuff that we call money.
- 1.2 So a big part of what we're going to be talking about is what money is. What money is, and where does it come from? Who invented it? Where did it come from? Is the creation, for example, is money a phenomena of markets, of voluntary interaction, of trade, or is it a phenomena of governments, of kings? Different theories about where money comes from historically. We'll talk about those.
- 1.3 And how did it evolve? I mean, this paper stuff, this is new. It wasn't always paper. Indeed, it used to be all kinds of other stuff that constituted money. So we'll look at the history of all that. And then, of course, I know if you follow politics a little bit, then there was a bill passed. We're now in, what, in July of 2025? And President Trump just signed a bill called the Genius Act, which is going to regulate stablecoins.
- 1.4 Stablecoins, what are those? And how are they related to money? Are they money? Are they a new kind of money? And of course, behind in the background of all of this is this new contestant to be money. Bitcoin, other cryptocurrencies, and there's a lot of competition there. So, to what extent is crypto money? Could it become money? Is that the future of money? What is the future of money? Because money has evolved. There's no reason to believe that we're at a point where it's never going to evolve again. So, how is money going to evolve into the future?
- 1.5 So, we're really going to try to cover all of that. We're going to do the history. We're going to try to understand what it is. We're going to look at the present, how money functions today, how the central banks function today, how our monetary policy is functioning today. And then we're going to try to project it into the future and talk about crypto and stablecoins and fintech and what happens next, or at least some ideas about what could happen next.
- 1.6 With everything with regard to money, both in terms of trying to understand its past, trying to understand its present, and in terms of understanding its future, every part of that is contested. There is no one theory or one explanation of what happened in the past. There are different views and different theories. We'll address some of them, not all of them, but some of them, even today, to try to understand what happens. What does the central bank today actually do in terms of money? And how does it impact us on a day-to-day basis?

- 1.7 There are many theories. There's no one consensus around what happens. So we're going to discuss that and try to untangle that and try to see what is actually going on. And certainly, when it comes to the future, who knows? And when it comes to crypto, there's a lot of disagreement about whether it's money, whether it's not money, whether it could develop into money or not. And we're going to talk about some of those conflicts.
- 1.8 And finally, underlying this, there is a sense, when it comes to money, there's a little bit of discomfort always when it comes to money. There's a contention, even with regard to the moral status, if you will, of money. I mean, there are a variety of different traditions in terms of philosophies, thoughts, religion, in terms of their view of money. Is money the root of all evil? Is the love of money the root of all evil? Is money good? Is money bad? Is money something we should celebrate? Is it a great innovation, or is it something that has caused us to descend into hell?
- 1.9 So there are different views, even from that perspective, from a philosophical, moral perspective, on, if you will, the value or this value of money and the effect it has on human beings. So we're going to touch on all of that in the eight classes that we have before us. We're going to start today by looking at the history, by trying to figure out how money came into being. And by looking at that, we're going to try to start thinking about, OK, what is this thing that came into being? What is it facilitating? What value does it provide? What is it?
- 1.10 So, through some examples of different types of money, because there have been a lot of different types of money, and we'll go through some of those examples, we're going to try to, if you will, induce, try to figure out what is common and what is it that makes money money, and what is it that makes good money good, as compared to not-so-good money, or even bad money.
- 1.11 So we're going to look, we're going to start here with history, the history of our species going back thousands of years, or hunter-gatherers, or maybe small communities of farmers that are just starting out in farming. And if we need stuff, in a tribe, in a hunter-gatherer tribe, the division of labor is probably fairly clear. There's certain people who are good at hunting. There's certain people who are good at bows and arrows, building bows and arrows, and then others that are skinning and cooking. There's a division of labor within a little tribe in terms of what each one does.
- 1.12 And there's probably some trading that's going on, some need for trading that's going on. But much of that trading happens in an informal basis. There's no formal marketplace yet in pretty early civilization. Within the little tribe, there's probably what you would call, or what some people call, social credit. I do something for you, you do something for me, and we trade off of that to some extent. There's not a lot of division of labor here. There's not a lot of specialization. There's not a lot of production. We're not producing much.

- 1.13** And when you produce very little, there's just very little trading going on. There's very little economic activity going on. But at some point, you might encounter another tribe. And that tribe might have something, I don't know, shiny and pretty that you would like. And now, you need to find some way to get that, to trade for it, to exchange for it. And of course, initially, you might have some spare spears or some spare bows and arrows, and you might try to entice the other tribe with that, and a barter system develops.
- 1.14** And indeed, as we have agriculture communities, and agriculture communities are starting to grow, you now have a little bit more specialization. You have more people who are strangers to one another, and just doing favors to one another is not going to work. There's a trust element, and now you're starting to actually need to exchange goods for goods. Somebody maybe is starting to actually be responsible for making clothes, and they can make clothes in larger quantities.
- 1.15** Specialization, remember, specialization is something that makes it possible for you to get better at whatever it is that you're doing. You specialize in it, you gain knowledge of it, you repeat it over and over again, you become better at it, and you can produce more. You become more efficient. So, with production and specialization, we now start trading a lot more with one another. And that trading now becomes more complicated.
- 1.16** I might have some eggs, and you might have some cows, and somebody else over there might have some clothes. And I assume that you want eggs for the clothes, but maybe you don't. Maybe you don't need the eggs right now. You'd like the cow. And how do we figure out how to make all this barter work and exchange work? It becomes very complicated. If you have five unique goods, that means you have 10 pairs, 10 different prices. Eggs in terms of clothes, clothes in terms of cows, cows in terms of eggs. Now, if you have five, you have 10 pairs. But if you have 100, you have 4,950 pairs. I mean, that's complicated.
- 1.17** And if you think about all the things that you have today, I don't know if you want to suggest how many things you actually, how much stuff you have, different things, unique things that you have today. Just what I'm wearing, clothes, I mean, I've got shoes, socks, trousers, jacket, shirt. Each one of those is unique. And then belt, wallet, phone, keys. It's endless. You cannot function in an economy if you're constantly trying to figure out how to barter, how to exchange good for good; there's just too many units.
- 1.18** Now, today, maybe we could run a big database on the computer to try to figure all this out, but it would be incredibly inefficient, incredibly slow, and incredibly complex. And the values would be constantly changing, given each one of our needs, desires, relative values, relative hierarchy of values, as we move along.
- 1.19** In a primitive society, how do you keep track of all of this? In Mesopotamia, what is that, 1000 BC or something like that, a little less than 1000 BC? They kept these clay

tablets where they write who owes what to whom. And it's interesting to look at that because there's actually a lot of sophisticated economic activity going on in Mesopotamia, 3000, 4000 years ago, where they're actually lending each other grain and expecting back not just the grain, but interest on the grain, like a little profit on top of what you expect back.

- 1.20 And there's a lot of exchange. There's no money yet, although they might have had some silver coins. There's no formal money, but there are already kind of sophisticated, almost debt markets that are denominated in things like grain and the things you borrow and lend. So, you get these significant economies are starting to arise, Mesopotamia may be being the first one, but throughout the Middle East, in particular. And you get larger and larger communities. You get more and more interaction between tribes, and more and more trade.
- 1.21 And the barter system breaks down. It's just not efficient. The barter system requires—there's no good unit of account. The ratio between chickens and eggs, not chickens and eggs, but eggs and cows, or eggs and clothes, or eggs and whatever, is constantly changing. And what are we measuring? And one time we're measuring it in eggs, one time we're measuring it in cows, another time we're measuring it in whatever the other products are that we have. It's inefficient.
- 1.22 It's very difficult to do in advance with strangers because we don't know what strangers will want, and we don't know what strangers, how we're going to interact with them. We can't give them credit because we don't trust them. But so how do we interact on the spot? How do we barter on the spot with them? And the huge opportunities for conflict, misunderstandings, miscalculations, it's just not an efficient system.
- 1.23 So, this idea that we have credit, we have loans to one another, we're bartering with one another. But there's no unit of account, there's no standard by which to measure the relative value of all these goods and products of them. And it makes sense for something to arise that will become that unit of account. What is that going to be? Well, it's been lots of different things throughout history, particularly in very ancient history. This is something that has evolved.
- 1.24 But what does it make sense for us to use a something that can transact the eggs and the chickens and the cows and the clothes and the grain and all these other things? Well, something, something that pretty much everybody wants. So something that everybody wants. If I can get some of this stuff, right, then I know that I can use it with the guy who makes the clothes because he wants this stuff. And I can use it with the guy who sells the eggs because he wants this stuff.
- 1.25 And part of the reason he might want this stuff is, he can use that stuff to barter with somebody else to get what he wants. So, instead of always the two parties meeting who want each other's stuff, now we can create an intermediary good, an intermediary product, that bridges that gap. And I don't want your chicken, I don't want your eggs,

but if you give me this other stuff, then I can use this other stuff to go and buy the thing that I really want.

- 1.26** Some examples. In many places, including in North America, in Native American tribes, in Africa, in parts of Asia, anywhere close to the coast, close to the sea, shells became one of these intermediary goods that could serve to facilitate exchange. Shells that you pick up by the sea, but then polish them nicely, make them pretty, and then maybe put them together into a necklace or in any other kind of configuration. They're pretty. People like wearing them. They have a value in and of themselves because people like to wear them.
- 1.27** And it turns out that there's a universal aesthetic, if you will. People generally like shells. They like certain types of jewelry. And if I can use those now to get the things that I want in trade, then I can go back to the sea, make more of these, trade for goods. And one of the reasons people might start wanting these shells in return for the goods that I receive is that they then can use those shells in trade with other people.
- 1.28** And what you find in the anthropological, archeological evidence is, you see these shells moving inland. They move away from the sea and they get traded inland. And even inland, they become a means by which people are trading. So they become this intermediary good that people want, that's saleable, that you can get rid of if you need to. But it has also other characteristics that are really useful for trade. Easy to transport. I can wear them and move around with them. They're unique, so somebody inland can't just pick up shells on the street. There's some scarcity.
- 1.29** They're maybe not scarce on the beach, but even there, labor needs to be put into them because they're not the raw shells. They need to be polished. They need to be made nice, and that makes them a little scarce. And think about a relatively primitive society where this all has to happen by hand. It's time-consuming, it's effort, so something is being produced here. And now this produced good, because it's liked by many people, and therefore it's tradable, becomes a way in which we can exchange goods. It's easy to transport. It doesn't rot. Shells can last for a very long time.
- 1.30** Now, they vary in quality and maybe prettiness and size. And that might mean some necklaces are more valuable than other necklaces, and you could trade on that. So it's not all one denomination. That is both a plus in terms of you can have variability, but also a negative in terms of there's no standard, really standard unit of account. Not all shells are the same, which is a problem. There's no way to mass-produce. And you can accumulate a bunch of these. And you can express, you can show the world how wealthy you are, and indeed, in these tribes, how many necklaces or how many armbands and stuff you have. You wear your money on yourself.
- 1.31** So, this facilitation of trade is what makes something, in this case, money—something that facilitates the barter, makes barter much, much easier. Often what happens in a particular community is the most saleable good; the good that you know you can

always sell it to somebody will become this medium of exchange, this facilitator of trade.

- 1.32** But what happens once this kind of money, if we call it that, meets technology? And this literally happened. It happened here in America. When Europeans came and they figured out that these shells were money, then they built a lot of factories where they could get the shells from the sea, polish them quickly. Out they went and bought stuff from the local communities. And what happens when you flood the market with new product? We call it inflation. We'll talk a lot more about inflation in a future class.
- 1.33** But what you get is, the value of the thing diminishes. It loses its scarcity. Everybody has this stuff. So there's nothing special about it, and it loses that ability to trade. We'll see this over and over again. One of the things that eliminates bad money is technology's ability to recreate it and to reproduce it. So, shells disappear once Europeans show up. That's true in Africa; that's true in America; that's true wherever it happens. Whenever you get industry showing up, whenever you get tools, machinery, technology that can reproduce, these kind of monies disappear very, very quickly.
- 1.34** For a while, salt, salt was used as a way to exchange. Salt is easily divisible. You can have a lot of salt, a little bit of salt. It's easy in that sense. It's durable. You can verify that it's salt. How do you do that? You taste it. Pretty straightforward. It's fungible; all pieces of salt are salt. I mean, salt to salt to salt. And it has utility. People like salt. It's essential for cooking. Not very valuable. There's a lot of it. Not very scarce.
- 1.35** Really hard to transport. Because each little grain of salt is worth very little, if you want to go buy, I don't know, a cow, you're going to have to get a truckload of salt to go buy it. It's just not easy to transport. But it is interesting that the word salary, the word salary comes from Latin for salt-denominated income. So in Rome, the soldiers were sometimes paid in salt. So again, money has taken many, many different forms.
- 1.36** In colonial Virginia, or colonial South, really, beyond Virginia, tobacco for a while served as money. It was actually made legal tender by the local governments. There was not enough metals. In those days, there was already coinage, but coming, the colonies were small. There was very little economic activity, relatively speaking, a lot of tobacco growing. And tobacco was used as the way in which we trade everything else. How many something of tobacco is that worth? And that's how we traded.
- 1.37** What's the problem with that? If it's being used as money, the tobacco is going to be worth more than just the value of the tobacco for smoking because now it has an additional utility. So it gets a premium. And once a product has a premium, what's the incentive now for everybody to do? Everybody goes out and does what? They start growing tobacco. So what you get, again, is no scarcity, a major increase in supply.
- 1.38** Now, the government doesn't like this because the government has declared this legal tender. Legal tender meaning that you have to accept tobacco in repayment of loans. Legal tender generally means that the government says, if we have some credit

arrangement between us, and I offer you to pay in that legal tender, you have to accept it. And the government accepts it, payment of taxes and so on.

- 1.39 But if that's the case, the government wants it to be legal tender, but everybody's planting it, creating more and more of it, again creating this situation where the value of it is going down, is diminished by the quantity. The government then tries to place restrictions on who can grow it. So there's a prohibition on growing too much tobacco. So what does that create, an incentive to do what? What does every prohibition create? An incentive to have a black market, to do it on the side, go figure which is authentic tobacco and which is the black market tobacco. So it's cheating.
- 1.40 There's also a problem that tobacco is not fungible in the sense that tobacco has various qualities. I guess some tobacco is relatively so-so, some tobacco is really good, so they're not always equal. And that's, again, you need to certify which is which. So it was useful for a while, but it couldn't survive for very long and didn't, indeed, survive as money.
- 1.41 One of the most interesting monies that evolved was in a South Pacific island called Yap. And on this island, they had these carved stones. These were limestones, carved stones, circular shapes with holes in them. And they came in a variety of different sizes. They had very small ones. Then they had ones that were like 10 feet wide and weighed thousands of pounds. I mean, these were big, big stones.
- 1.42 What was interesting about these stones is, they weren't native to the island. They had to go to another island to get them. So they literally would row their canoes and carve out these stones and bring them back. And then on the island, they would make them these round and they'd put the hole in it. And ownership of these stones would shift from person to person, based on the trade. But it wasn't like you carried the stone in your pocket. These are stones out there, 10-feet-wide stones.
- 1.43 But there was, in a sense, a ledger being kept, a bookkeeping done, of who owned which stone. And if I wanted to buy your land or if I wanted to buy a cow from you, if I wanted to do this, then a portion of that stone shifted to me and a portion shifted—I mean, how they did this is pretty amazing. But basically, there was these stones for a unit of account. These stones represented the way in which we transact; they represented the medium of exchange, even though they physically weren't being exchanged. But just in the minds or in the ledger being kept by the people, the ownership of the stone was constantly being exchanged.
- 1.44 They had to go 250 miles in those days, canoes, to mine these things. So scarcity, definitely scarcity, right? These are hard to find, hard to get. So the stones that you had, I guess, became incredibly valuable. And your ability to say, "I own that stone," was incredibly prestigious and incredibly important. And that made it possible to use them, that ability to say, "This is my stone," for trade. But it's interesting how, what attracted people to different types of these mediums.

- 1.45 Now, of course, again, this came up against, just like the shells, comes up against technology. In the 1800s, an Irishman named David O'Keefe came across the island, figured all this out, and said, "Hey, I could become the richest man on the island because I have the technology to go over there, go to the other island, 250 miles away, relatively easily. My boats are much more capable than their canoes. I can carve up a bunch of them and bring them here. And then suddenly, I'm the richest guy on the island."
- 1.46 He started doing that, and the natives look at that and say, "This isn't good." So then the natives start adding another feature to the ledger. Now, it's not just the size of the stone and the shape. He could, of course, also carve them really quickly, much easier with the kind of tools that he had. So now they added another feature to the ledger, the age of the stone. So all the stones, it wasn't just the size, but also the age determined value. So the new stones were less valuable than the old stones.
- 1.47 And of course, this created a situation where he had an incentive to bring on more and more stones and complexity, and of course, the whole system breaks down. The whole system breaks down. I think when the Japanese occupied the island during World War II, they started using these stones as anchors, and the whole thing fell apart. But you can see that bad money, certain types of money, put it that way, because they serve their purposes, I guess, for there, but come up against technology, they fall apart. They lose whatever quality they had.
- 1.48 And of course, in a sense, they are bad money because they're static. But one of the things that they don't facilitate is real complexity in terms of economic activity. The more the money is like this, where you have to hold the ledger and it's complicated and you have to keep track of all these things in your head or in some kind of communal library, the more difficult it is to then do more and more and more transaction.
- 1.49 And you can't transact with the outside world. You can't transact with the outside world. You can only transact within your little community. So you're limited. You're limited in ability to specialize. You're limited in how much division of labor you have, and therefore you're limited in how much you produce. Now, other things also served. There were periods in which feathers were money, and beads were money.
- 1.50 Grain, even, served as money. Of course, one of the problems with grain is, you've got a season where there's lots of it and other seasons where there's very little of it. So that fluctuates over time quite dramatically. Again, not effective. None of these are effective. Again, the incentives are there to create more, which devalues the value of all of them.
- 1.51 What ultimately evolves as the money pretty much in every, let's say, civilized part of the world, you really get the same thing evolving as money in the end over time. And that is metals, some form of metals. Metals are relatively scarce. They require effort to get. They require mining and purification. And they can be molded into various shapes.

They have utility. It's not like the rocks that are just there. The metals can be used for jewelry. It's like the shells a little bit.

- 1.52** A big appeal, I mean, the main appeal of gold in the end is what? It's shiny and pretty. I mean, it's got all these other good characteristics that we'll talk about in terms of serving as money. But at the end of the day, why do people want it? Before it's money, it's because it's pretty. They can turn it into a jewelry. But jewelry is a value. It's not an insignificant part of human life. The aesthetic part, the aesthetic side in jewelry, serves part of that function.
- 1.53** So, over time, money evolves or becomes, or metals become money. The two metals that become money most often and dominate, really, ultimately, are gold and silver, although you do get copper and bronze for very low denominations of money, like pennies. You're going to get very low denominations of money in those types.
- 1.54** One of the great advantages of metals is what appears to be a universality. So you get metal monies in China, you get them in India, you get them in Europe, you get them in the Middle East, and you get them along trading routes, so that they become a common medium of exchange, a common unit of account, across an entire globe. You start getting trade and globalization, 2000, 3000 years ago along the Silk Road, along various trade routes.
- 1.55** Originally, it's raw metal, and it's all done by weight. And then, slowly you develop a system of coinage which provides uniformity, provides a certain guarantee of fineness, a purity. Every coin represents a certain amount, a certain weight. And now, up to a point, we can start trusting it, again facilitating trade much easier. Traders don't have to take this raw stuff and assay it, figure out how pure it is.
- 1.56** You remember in the Western movies, people come with little bags of gold and they weigh them, and you actually have to look and see how fine it is and what the quality is. I mean, that's a whole skill set. And indeed, in the division of labor, one of the skill sets that developed was assaying metals, being able to tell whether a metal was pure and was good, and whether the weight was true and all of that.
- 1.57** And the same would apply to coins, as we'll see. But coins made that a lot easier, made all of that a lot easier. And people had an incentive to go out and mine the metal, and then bring it to somebody to coin for them. So you separated out the coining from the mining. And the incentive for the mining is, you're digging up money, a medium of exchange, something you can use.
- 1.58** So what are the advantages of having these things as money, particularly good money? And then we'll get into more details about coinage. So money arises to deal with frictions, to deal with inefficiencies, to deal with the challenge of barter, and of debt repayment, and of debt contracts, and dealing with debt in an effective, efficient way. It's far more efficient than any kind of other system of bartering or bookkeeping, or keeping a mental ledger. And of course, we'll get to bookkeeping later.

- 1.59 But bookkeeping in the formal sense as we do today, where you're accounting assets and liabilities and all of that, that's a modern development. That is new in the historical context. That's the last thousand years. There was no real understanding of what bookkeeping was like before then. Just as an experiment, think about what it's like to do any kind of math, any kind of bookkeeping, using Roman numerals. We'll get to that. But that was the world until about, I don't know, 1400, 1450. In the West, it was Roman numerals. It's relatively new, 600 years that we actually got the system of numbers that we do today.
- 1.60 Now, because it makes barter more efficient, it actually now actually incentivizes more specialization. So, if you take a relatively primitive society, well, we can't really overly specialize, because what if I specialize in something and then it's really hard for me to trade with everybody, because it's really hard for me to figure out how to barter with everybody else? Maybe there'll be a period where they don't want the stuff that I've done, or some people will, but some people won't. And how do I find the people who do? And it's complicated.
- 1.61 But once you create a medium of exchange, once you create a money that everybody's willing to accept, I can afford, in a sense, to start specializing, noting that some people will be willing to buy my stuff, and I can take whatever they give me and use that to buy other stuff. So money comes about as a society is getting more complex in terms of division of labor. And then what money does is, accelerates the process of division of labor and specialization.
- 1.62 And division of labor and specialization are essential for production. They're essential for growing an economy, for achieving any kind of economic growth, as Adam Smith will later formally explain to all of us in *The Wealth of Nations*. Money also suddenly allows me to trade with strangers. And if I can trade with strangers because I don't have to worry about what it is I'm bartering with them, what they want, what is the quality of what they have to offer to me, we've got now this medium of exchange that facilitates this, so we all accept.
- 1.63 Now my market has grown. Again, more incentive to specialize. I don't just rely on my little tribe. Now I can go into the marketplace, and you see the evolution beginning of marketplaces, where strangers start trading with one another. So I'm expanding my customer base, and therefore I can expand production. If I can expand production, there's potentially more customers. And money, of course, is going to serve now as a unit of account. It's much, again, easier now to measure and evaluate different products relative to one another.
- 1.64 And because it serves as a medium exchange in a unit of account, I can now start thinking about time differently. Thinking about the future is tricky. And in a primitive society, there's not a lot of thinking about the future. It's, first of all, you're grinding day to day to survive, but you don't know what's going to happen in the future. And there's no way to store value for the future, because most of the stuff we can store, like

tobacco, we didn't mention this, but tobacco will rot after a while. Most organic things, most of the stuff we have and produce in an agricultural, hunter-gatherer society, rots after a while.

- 1.65** But now, if I have these metal coins, I can start saving. I can start using money as a store of value for the future. I can now start planning to buy something next year or to save up so I can buy something next year. It changes the way we start thinking about time. We can now think in terms of the future. We can now think in terms of what I will do in the future and how I will do it and how I will trade, not just today, how I'm trading today, but how I'm going to trade over time. So, store value.
- 1.66** So, what I've described is kind of the way many economists believe that money arose, an evolution out of human barter and trade that ultimately brought us coins and metals and then coins. And it's a market. It's the idea that this came out of the market, came out of voluntary transactions, and because of our individual preferences as human beings, the universality of the prettiness of shiny metals.
- 1.67** There are alternative explanations. There are those who claim that money was, particularly metal money, money in the form of metals, money in the form of silver coins and gold coins, was actually imposed by the state. That the first such monies were established by kings who, I guess, could see that this would be a good means by which human beings could interact.
- 1.68** And of course, from a historical perspective, from actually going back and figuring out what actually happened, it's very difficult, because there are no records. It's not like people wrote anything. Language is relatively new, and there are not a lot of records of this. But what you could find is, you can find very, very old coins. And it's interesting. So some of the oldest coins we can find. And of course, I'll just mention this.
- 1.69** If you look at, before we get to the coins, if we look at other types of money, like shells, clearly those came out of people just polishing shells and creating them. There's no central authority to establish shells as money. And if you look at others, there's tobacco, it looks like the state governments had a lot to do with that, in terms of establishing money. So it could go either way. But it certainly seems like, on occasion, money has evolved from the marketplace.
- 1.70** And there's every reason to believe that that is true in terms of metal money, metal coins. The first coins that we find are from Lydia, and this is about seventh century BC. It's about 600 BC. I think that's how it works. Lydia is an area overlapping with modern-day Turkey. And what you find there are first coins that actually have figures stamped on them, so faces stamped on them. And as we said, coins are a huge advantage. They give you uniform weight and fineness.
- 1.71** But it appears, given who's on the coins, the faces on the coins, that these first coins were probably private mints, that the faces are the private entrepreneur who is running the private mint and put his face on the coin. And indeed, if you look

throughout history, you'll find private minting of money in medieval Europe. You'll find it in goldrush California. There were private mints. It wasn't just all government mints. So there have been private mints throughout, but it looks like the oldest coins that we can find were actually minted by private entrepreneurs, by private businesses.

- 1.72 But then, of course, there is a huge incentive for the authorities to take those private mints over. We'll get to that in a minute. About the same time, a little later, you also find, you start finding coins. Also, the assumption is today that they were private, although, again, there's conflict; not everybody agrees on this. In India, so you start finding coins in India at about the same time, also in the B.C.
- 1.73 There, what they did is, they'd have this big sheet of silver, and they'd cut round coins out of that. And you could, as a private person, you could go, you could do this yourself. And then what in India would happen is, you would take it to an official, I guess, government agency, and they would certify it, that it was indeed pure and everything. As far as we know, in Lydia, there were just competitive currencies, and it was what would establish the purity of your coin.
- 1.74 You put your face on it. So it wasn't anonymous. Everybody knew who created the coin. So what the guarantee for the purity of the coin was your reputation. If you started putting out lousy coins, people would stop using it. It was the fact that you signed it by putting your face on it. It's why faces of minters appear on coins, to identify who the minter is. They're not anonymous coins, so it's clear who the producer is.
- 1.75 Now, over time, different rulers in different places start monopolizing the minting of coins. They start taking this over, and really, I think, for two reasons. It's a source of revenue. So how does a mint gain revenue? Well, one is, you bring your silver, your gold, into the minter, and he produces the coins for you, but he takes a fee. So he's going to take a percentage of what you brought him as a fee in order to produce the metal. So that's a source of revenue.
- 1.76 And the fee is going to be in excess of what it costs to actually mint. There's a cost associated with the stamping and the minting and all of that. And there's a nice profit there. And since money is something that's always needed and is continuously mined, there is a steady revenue coming in. This is why private businesses went into it, but then government takes it over.
- 1.77 The second is, and this is, of course, a much greater source of potential revenue, is, the government has the ability to debase coins, is to make them worth less. We'll talk about that in a few minutes. But the debasement is a massive source of revenue for governments, and governments have been very, very good at this. We'll see this throughout all these sessions that we're going through. We'll see government debasing money, primarily to go to war.
- 1.78 War is a primary—they need a lot of revenue. It's very expensive to go to war. And one way in which you do that is by debasing the currency, making the currency worth less

that people who use it, and you having more of it. So we'll see how that works. But there's a second reason. So one is revenue. They wanted revenue.

- 1.79** The second is, the kings were starting to worry about these minters and the power they might accumulate. That minters had their face all over the place. They were well known. They were making money, and they were now a power hub, as a threat to whoever the authority was in charge. So why tolerate that?
- 1.80** Or you could view it the other way around, too. Imagine these minters seeing their face on all these coins saying, "Hey, maybe I could become king and get rid of my competitors. And monopolize the minting." And indeed, it's not clear which direction it went. It could be that the minters became the kings or that the kings became the minters. But one way or another, they became both.
- 1.81** Here's a quote from one essay, it says that the "Greek tyrants"—this is the Greek city-state tyrants—"were the first men in their various cities to realize the political possibilities of the new conditions created by the introduction of the new coinage. And to a large extent, they owed their positions as tyrants to a financial and commercial supremacy, which they had already established before they had attained supreme political power in several states." So these are probably the minters who then become the tyrants, not the other way around.
- 1.82** Or another quote: "Having arisen to power, the tyrant assumed the monopoly of a coining. This step was probably part of a policy aimed at enhancing his own power and commercial success and the hindrance of his rivals. He kicked away the ladder by which he had risen, lest others might attempt to use it to take power away from him."
- 1.83** Of course, minting coins was also, for the ruler, very prestigious. Again, you got your face on every one of those coins. You got to establish who was boss, particularly if you could say these are the only coins that can be used in the realm. They have to have my portrait on them. That was really powerful in terms of marketing and in terms of establishing who the main guy is, who the main person is running the show.
- 1.84** All right, so let's think about debasement, because this was obviously a main source of revenue for these guys and a main reason to do it. So, one simple method you can think of that was called clipping. Imagine you're a merchant and coins are coming through, and every time you get a gold coin or a silver coin, you just shave a little bit off, just a little bit, tiny bits that nobody will notice. And you have a little thing where you just shave a little bit every time. And over time, the quantity grows. You can create another coin just off of the stuff you shaved. You can get the mint and have it minted.
- 1.85** Again, coin minters come up with different ways in order to try to stop that. The portrait is one. You can start seeing erosion if it's shaved off too much. They have, the edges become rimmed. I think, to this day, if you take a quarter, it has a—because if you shave that, then it would be observable. So minting tries to keep up with that. At

the most basic level, it's just clipping away. It's actually shaving these coins away. And this continues for a very long time.

- 1.86** And one of the great advantages, ultimately, of forms of paper money is, you can't do that with paper money. There are claims against, in a gold standard, there are claims against actual gold. It can't be clipped. It can't be shaved away. But really, the more profitable way of debasing a currency is basically reissuing it with less of the metal in it. So, let's say, I'm the king. I take in my taxes. I take in 100 coins of gold as taxes, and it's pure gold. This is high quality. These are really superior coins.
- 1.87** But I'm a little stressed right now. I've got a lot of my budgets. I'm in debt. I need to be able to pay back my debt. So I plan to go to war, and I need to pay my army, and my soldiers are getting antsy. So what I do is, I take these 100 coins, I melt them, I add some other material to it. And now I can print 110 coins. And each coin is less pure. But people out there won't know. What's the chance that they know? Plus, I've made this legal tender. They have to use it, whether they like it or not.
- 1.88** I'm going to pay my soldiers with this diluted stuff. They'll go on and spend it. And yeah, over time, people will figure this out. And there are now 110 coins in the kingdom, instead of 100 coins in the kingdom; that'll raise prices. But my soldiers won't notice. They're the first ones to get the coin as they go out, and it'll take a while until prices adjust. And then maybe sometime in the future, the soldiers will come back to me and say, "Wait a minute, the price has gone up," and they'll demand a raise. But by then, maybe things will get better.
- 1.89** Maybe I'll loot somebody, some neighboring kingdom and get some more gold. Or maybe I'll take those 110 coins back, and I'll melt them down, and now I'll make 120 coins. And people won't know for a while, and it'll take time. And this could go on for years, decades. Now, this will work, particularly if I can enforce legal tender laws, if I can force my people to use my coins. Once we try trading with outsiders, it becomes more difficult. They might actually want to assay the coin. They might actually want to judge the coin, and they're going to discount.
- 1.90** If it's an inferior coin with less gold in it or less silver in it, they're going to discount its value. Then I'm going to give you as much in return. And they're going to more likely keep good coins out of my kingdom. They'll keep the good coins, and my kingdom will be flooded with the inferior coins. So it's a challenge, and there are issues here, but it's incredibly tempting. I need money right now.
- 1.91** And if I just—think about Congress right now, right? We need, I don't know, 4 trillion extra dollars. We could raise everybody's taxes and get the 4 trillion dollars. That's easy. But nobody likes that. You'll get a mutiny, right? You'll get revolution. People don't like their taxes raised, so we won't raise the taxes. We'll cut their taxes. Now, we need 7 trillion instead of 4. But the people will be happy. And in the meantime, we'll borrow the money, and thus, as we'll see, ultimately tax the people, but through

inflation rather, through the fact that the value of their money goes down, rather than through—

- 1.92** This is as ancient as can be. This goes back to the beginning of money. Governments, rulers, have always had this incentive, and they've always had the means and the tool to do it. For example, in the medieval periods, for many rulers, they got half their revenue from the mint, half from taxes, half from the mint. So this is a major source of funding for the kind of activities that government engaged in in those days. And remember, in those days, most of the activities they engaged in was war.
- 1.93** All right. Another way in which, for example, they controlled this is banning the use inside the kingdom of outside money, of coins from the outside. Also making it by law necessary that if you mined silver or gold, you had to bring it to the government mint. You couldn't take it anywhere else. You couldn't leave the territory. You couldn't go outside of the country to do it. You had to print and do it there. And that's how they created a closed system, if you will, where all the money was there, and they got to control the quality of that money, and therefore its value, and therefore how much of it was theirs.
- 1.94** OK, so let's summarize quickly where we are. We've seen a little bit of how money has evolved, how the state took control of it, but really, it evolved organically as part of trade, as part of barter, as part of specialization and division of labor. And what is common to all the kinds of money, from shell to coins to tobacco, they all served one fundamental function. They evolved in order to solve one fundamental function, and that is to be a medium of exchange.
- 1.95** Basically, money is the common medium of exchange. The thing we use to exchange stuff, that is money. It could be tobacco, it could be shells, it could be gold coins, it could be this paper. It is money. This is what we use as our medium of exchange. It doesn't mean it's good money, but it is money. That is the primary function. That is the purpose of money. That's what money is, the facilitator of trade.
- 1.96** And of course, because it's the medium of exchange, it serves as the unit of account. Every product we have in the United States today is listed in terms of dollars. That's our money. So that's the unit of account. It could be listed in terms of grams of gold. It could be listed in terms of number of shells or leaves of tobacco. Those are the unit of account. And again, that adds massively to the efficiency of the market, when we have one unit of account that everything's listed in.
- 1.97** As we'll see, most of human history, we've had more than one because the challenge with, for example, gold, is that it's really valuable. So how do you price something that's really, really cheap in terms of gold? You need something that is denominated in a lower denomination to be able to buy the cheap stuff. And that's the purpose silver served, and in some cases, copper served, as the lower denominations. But even they were all exchangeable ultimately into gold. So it was all related.

- 1.98** So it serves as a unit of account, making everything much more tradeable and much more efficient. And it serves as a store value. That is, it facilitates trading over time. It facilitates saving and facilitates planning and long-term. And this is, a crucial function here is how much it expands the human mind in terms of what is possible because it brings in the dimension of time. So a good money has to serve as a store of value.
- 1.99** So money is a medium of exchange, and as a consequence, a unit of account and a store of value. That is what money is. Next time, we'll talk about some of the characteristics of a good money, and we'll start digging a little deeper into the more modern history of how money evolved in modern times, leading up to the gold standard.
- 1.100** You mentioned quite a lot in your speech about how technology comes face to face with what is the current unit of money at the time, how, of course, shells were eventually replaced as money because of factories being able to mass-produce them. There is a concept of this concept of asteroid mining, where you would go out into space, and you would take all this precious metals that's in these asteroids, and bring them back to Earth. You think that could cause a similar change in terms of gold and silver being used as currency?
- 1.101** That's a good question for next class. But I put it this way. The answer ultimately is going to be, I think, no. And the reason is that there's a cost to mining an asteroid. First of all, the quantity they're going to bring back is going to be finite and limited, just like it is today. When mining today, there's a cost to mining today. And the quantity of gold that exists today, today we have in the marketplace pretty much all the gold we've ever mined on planet Earth.
- 1.102** I mean, there's some at the bottom of the ocean, some at the bottom of the ocean, some maybe thrown away that's used in electronics, but it's not a lot. Pretty much all the gold that's ever been mined is, you're either wearing as jewelry or it's in coins and used to save for a monetary thing. The amount we add to it is relatively small relative to the stock. Now, if at some point that changes, if suddenly, I don't know, gold has rained upon us from asteroids or whatever, it becomes super cheap and there are plenty of asteroids with gold, then yeah, just like anything else, it would lose its value; it would be inflated.
- 1.103** But the likelihood that the amount that you would be able to get, given the cost of getting it would be so as to change the fundamentals, I'm skeptical of. This is interesting. Maybe I'll mention it later in one of the classes. I even read a story now that there's, you can use—now, if this is wrong in terms of physics, then I apologize, but I read the story in credible places. There's a paper out that was published, I think, last week.
- 1.104** Basically, there's a nuclear fusion company out there that's saying that, as part of the nuclear fusion process, they can turn mercury into gold, or at least they have the

theory that suggests that that's possible. They haven't actually done it. But you can turn mercury into gold, and there's no reason not to. If you can manipulate the subatomic particles, you should be able to create any element that you want. And in a sense, alchemy was always right. Newton was not a nut to try alchemy as much as he did.

- 1.105** So, as part of this, they can, neutrons or whatever, they can reshape it, turn mercury into gold. Again, so I thought, OK, there goes the gold standard. You can now create it out of thin air, but you can't really, because the reality is, it's super expensive. So you're not going to do a lot of it. And the quantities are going to be relatively small. And so, again, it'll grow the stock of gold, but it'll probably grow it at the same kind of rate as mining does.
- 1.106** And it just proves the idea that resources are infinite. There's no such thing as finite resources. If we can't find it in the ground, we'll make it out of something different. We'll figure out how to do it. Is it theoretically possible? Yes. Is it practically feasible that something like gold, we'll get so much more of it new that it would be irrelevant and lose all its value? Probably not.
- 1.107** When you mentioned money being a store of value for the future, when was the first instance of financial planning and that as a profession, where people actually sought help for planning?
- 1.108** In terms of starting to seek help? Wow. I think in terms of really seeking help and thinking about investments, I think that's fairly late, so that you're probably looking at 16th, 17th century, maybe 18th century. Think about also the evolution of insurance. This is the time we start thinking about insurance, and there's some insurance in ancient Rome. So ancient Rome was pretty sophisticated, but I don't think there was money management and investment then.
- 1.109** But I do think you start getting that maybe with the Enlightenment, the growth of, for example, stock exchanges are created in Amsterdam. In the 18th century, you get, 17th, 18th century, you get a stock exchange, and people are starting to think about investments. Of course, again, with ships going to the other side of the world to trade and come back with goods, think about that, right? You're investing in the ship.
- 1.110** You have to buy all these goods that they're going to load up on the ship. They're going to travel around the world. They're going to trade for it. They're going to come back, sell it, and hopefully you've made a profit, if the ship even makes it back. High percentage of ships don't come back. That's where insurance comes in. But that's also—I mean, the whole idea of starting to plan that way around the international trade, so international trade is a huge impetus for that.
- 1.111** And international trade really takes off once they find a route to Asia, going around the Horn of Africa, and the Portuguese and the Spanish start trading with Asia. And then, of course, the Dutch and the English fall on top of that. That's when you really get these more sophisticated now financial arrangements. I'm thinking you get the first stock

companies around the same time, created around to facilitate this because you need equity, not just debt, for these risky ventures.

1.112 So, Age of Discovery, I'd say, 16th century on, is where you really start getting to see that. In terms of actually sitting down with people and managing their money, it's probably a 20th-century deal. I mean, maybe late 19th-century, because people had no money. A few people had money. But in terms of people having money, people, the idea of saving is a completely new idea for ordinary people. You save, maybe you had some gold, so you could buy something the next year.

1.113 But saving and growing your saving? Banks only started providing interest payments on your deposits in the 17th century. And how many people had deposits with banks? Like, the 1%. So it's only in the 19th century does wealth become something that a majority of human beings in the West at least had. And they could invest and they could put it in the bank and they could owe interest and they could do that. Wealth was something that was very, very constrained to a very small group of people before then.

Lecture 2: What Makes Good Money?

At a glance — What makes good money.

- 2.1** So, money. Talked about last time, money is our medium of exchange. It is the common medium of exchange. It's the thing people use when they go out and they buy and sell stuff. When they exchange, it increases the efficiency of trade. It basically replaces barter as the way in which we interact with one another. It also serves as a unit of account. So we have a standardized unit denominating everything. And now we are also storing value. We're now planning into the future. We can now predict, or try to predict, buying stuff in the future with money that we've saved in the past.
- 2.2** So let's talk about what makes a good money. What makes for good money? We covered a bunch of different kinds of money last time. We talked about beads. We talked about those funny rocks in Yap Island. We talked about tobacco and beads and a lot of different types of money. But what are the things that make money a good money? What are the things that make money useful, productive? What allows it to become a dominant medium of exchange and a store of value?
- 2.3** And the thing about a store value is, it's a store value because we can predict that, in the future, it will also be a medium of exchange. If the medium of exchange keeps shifting, changing all the time, there's no store value. So first, it has to be something people want. As we said, it has to be saleable. It has to be something that people desire, that are going to be willing to exchange the stuff that they produce, the stuff that they have, for it. Otherwise, it can't serve as money, if it's something that some people want, some people, they really don't care about, and it's not.
- 2.4** And of course, the larger the group of people who want it, the larger the group we can trade with. If it's just a small group that wants it, don't trade with that small group. Less useful. So it has to be acceptable. It has to be acceptable. And usually, that acceptability comes from a number of different sources. But traditional money, that acceptability usually comes, it has some utility. There's something people can use, even if they don't use it as money. There's some reason, something they can do with it.
- 2.5** To a large extent, that turned out to be, in history, to be jewelry. The utility of much of the money that we've seen evolve was jewelry. Because if it's something that I have to use, if it's something like food, I have to eat it, then it doesn't serve as a good medium of exchange because it's going to be consumed very quickly. But if it's something that's optional, I can have the jewelry, it's nice, it's pretty, but I can also trade it, and it's no big deal in a sense, then that, of course, is now, is more valuable.
- 2.6** So it can be something that I must consume, I must have, but it is something everybody wants, and it's acceptable throughout. But of course, not everything that's acceptable, not everything that people want is going to be good money. Salt is something everybody wants. It has utility, but it's not good money because it's not scarce. There's

so much of it. It's too easy to produce. It's too easy to get. So, scarcity represents the fact that it takes work to get it, and it's hard to increase the supply of it, to increase the quantity of it, all in one time.

- 2.7 And we'll see how metals, particularly gold and silver, how scarce they are and why this scarcity is a big reason why they evolved to become dominant monies. It has to be portable. We have to be able to move it around, although we saw those big rocks in Yap were not particularly portable, and that was a problem. They could only trade within Yap. There's no way to take it to Asia and go trade foreign goods in foreign locations. So it has to be something we can actually move around with, and take.
- 2.8 And we'll see that, indeed, to some extent, that's a weakness of metal. It's heavy. If you want to buy something of significance, it's a big cut of even gold coins or silver coins. It's not easy to transport, but it's still portable, particularly for day to day. If we're using silver, but once we want to buy something of more substance, it's a challenge to make it portable. And we'll see the market solution to the difficulty of portability.
- 2.9 Also, when you travel with your money, what's the other risk that you face? Yeah, if somebody steals it from you. So one of the challenges of traveling with a big load of gold is that somebody comes and steals it from me. So we'll see, again, how the market deals with that, how the market solves it.
- 2.10 It has to be divisible. So it has to be in different units, so that I can use it to buy something cheap, and I can use it to buy something expensive. And the unit has to be such that I can buy this something cheap. One of the challenges with gold coins was always, well, how do I get fractions? How do I get smaller and smaller and smaller gold coins, because gold is so valuable that to buy something cheap is difficult. It's one of the reasons that in most of history we had silver. Silver was not as valuable as gold, so it could serve as another denomination.
- 2.11 And in some places like ancient Rome, they actually had a third category, bronze, which was even cheaper than silver, for the day-to-day stuff that you bought that was really, really cheap. It has to be durable. Stable. It can be like fruit stuff or furs; sometimes, in some places, furs served as money, but they don't last that long. They deteriorate in quality over time. Same thing with tobacco, same thing with anything organic. It's not going to last.
- 2.12 And then they have to be, they have to have some fungibility. They have to be uniform. Each one of these shells need to be approximately the same. They're not going to be exactly the same, one of the challenges with shell money. But they're all coins, everyone is exactly the same. If it represents a pound sterling, then it's a pound. And they're all exactly the same. Every one of them is minted the same. Every one is produced the same.
- 2.13 It's what, as we described, makes it possible for rulers to undermine the quality because we assume it's the same, and yet maybe it's not. Maybe there's only 90% of the gold in

there, even though we assume it's 100%. And there has to be a way to verify. So if I really want to know what's really here, there has to be a way to verify that. And so, assay it in some way, to figure out, is the stamped quality real?

- 2.14** So, it has to be something people want. It has to be scarce in the sense that it's costly to produce and increases the supply. Has to be portable; has to be divisible, durable, uniform, and verifiable. Money, ultimately, is going to be the most saleable good, the thing that people want and is universally accepted.
- 2.15** Menger, Carl Menger, who is the founder of the Austrian school of economics, an economist in the late 19th century, said that money transports value across space and across time. I think that's a beautiful way to put it, right? Money is a way to transport value across space— I can take it with me, all kinds of travel, trade with foreign countries; and across time, I can use it in the future. I don't have to use it today.
- 2.16** Now, what arises is that, ultimately, gold and silver become the dominant monies because they are good by all these qualities. They fulfill all these requirements more so than any of the other monies that evolved. These monies actually do fulfill most of those, at least satisfactorily, at least up until a particular level of specialization in economic growth. And again, we'll see the market solution when we get too big an economy to be traveling with gold coins in our pockets everywhere we go. How do we deal with that?
- 2.17** They score high in pretty much every one of these. And maybe most importantly is, they score high on limited supply, on scarcity. If you think about all the gold that has ever been produced is still around and has always been around, and gold is constantly added to it through mining. But mining gold is hard. And over time, gets harder because it's deeper in the ground, it's in more remote places. The big discoveries, as far as we know, have already been made.
- 2.18** Now, our technology keeps getting better. So we can get and we can refine gold cheaper over time, and we can find it in places that we couldn't find it in the past, because maybe it's deeper in the ground, or maybe it's mixed in with all kinds of other stuff, and we can't refine it as well. So now we continuously, the technology can access gold we couldn't access 100 years ago.
- 2.19** But what actually all of that produces is, basically, there's a steady flow of gold into the supply, into the existing quantity, that's pretty constant; but add about a cent and a half of new gold every year to the gold supply. And that was true 200 years ago, and that's true today, even with all the changes in technology. Now, the periods in which that goes up, like during the gold rush in California or the big gold finds in Australia, and it goes up, and then it comes back down. But on average, over long periods of time, it's about 1.5% a year added to the gold supply every time.
- 2.20** And you could imagine a situation where some of that 1% is coming in from asteroids, or some of that 1% is being converted through fusion from Mercury. But it's unlikely to

be a huge number, vastly in excess of that 1.5%. And of course, gold doesn't rot. It doesn't rust. It doesn't corrode. It's chemically inert. It's one of the things that really allows it to be a store of value because it lasts over time. You can imagine other metals, they rust, they corrode, they're worthless after a while. Gold doesn't do that. It's one of the things that makes it so valuable as jewelry.

- 2.21** And so there's this constant ratio. They call it the stock-to-flow ratio. How much flow is new added, relative to the stock? And if the stock is 100, you add 1.5, then the stock-to-flow ratio is 67. It turns out to be the highest of any material that we mine today. So the market evolved to gold, and we now understand why. And it makes complete sense, because it has this consistency in supply. It's not going to suddenly vary. You're not going to have this huge volatility in how much gold is available and added to the total supply.
- 2.22** And of course, the utility of gold primarily is jewelry, although today it's also used for some electronics, but the electronics is a very small amount. And if there is any gold lost, if you will, then it's usually the electronics that are thrown out. Silver is very similar. Silver actually has the second highest in terms of the flow-to-stock ratio. So it's less scarce than gold, but still scarcer than pretty much anything else. And so it's valuable, but not as valuable as gold.
- 2.23** And a lot of economies adopted silver as their primary money rather than gold because it was a little bit more accessible and because it was easy to denominate in small amounts. And often, as we mentioned, you had bimetallic standards, where you had both gold and silver at the same time, trading for different segments of the market, for different areas of value.
- 2.24** And basically, over time, in Europe, in Asia, in the Middle East, you saw that the metallic, these metallic standards, these metallic monies, silver and gold, basically competed out of existence all the other inferior monies. And the places where inferior money survived are places that didn't develop economically and didn't have extended trade relationships with the rest of the world.
- 2.25** So, certain parts of Africa, the Americas, until they were discovered, those are the places in which you saw sustained use of things like beads and feathers and furs and things like that. But pretty much any economy that was exposed to international trade, if you will, evolved into a silver or gold standard. Let's take a step back for a second, and then we're going to get back into the history. What's the significance of all this? What's the significance of money? What does money actually represent? We talked a little bit about this.
- 2.26** It's this ability to increase specialization and division of labor, crucial economic role. Again, it's not an accident that places that develop good money also developed economically. It's kind of a chicken and egg, but it's an ongoing supporting one another. The good money supports more specialization and more division of labor, which

necessitates more money and better money. And as we keep spiraling, the more prosperous economies are the economies that are going to be the economies with money.

- 2.27** And we'll see that, when an economy collapses, when you actually get an economy in trouble, money disappears, because you don't need it. There's nothing to buy. And money flows to those economies that are doing well, and to those economies that do well, where there's a lot of trading going on. So money is necessary for a well-functioning division of labor, specialized economy, to function. It's essential if we want to expand the scope of trade.
- 2.28** And of course, I've got a course on trade, which is going to feed off of this. And of course, the course, anything, money and trade, they're twins, they're parts of the same thing. You can't have trade without money. Money is necessary for trade. And of course, the value of trade is enormous. It's expanding the possibilities of division of labor, specialization, win-win relationships, and just a number of minds trying to solve problems of production.
- 2.29** Today on the globe, we have 8 billion people thinking about how to make life better. That's much better than having half a billion people thinking about how to make life better. Just seven and a half more minds thinking about it. That adds value. That's the huge benefit of trade, is, I can access those seven and a half billion additional people through the mechanism of trade, which money facilitates, and other technologies. We'll talk about that when we talk about trade.
- 2.30** Money also serves as a kind of a ledger. It's keeping track of who produced and who didn't, of how much you produced. It's like those big rocks in Yap Island. Who owns what is reflective of what? How do you get to own the thing there? Well, you have to produce and trade. The more you produce, the more you will get. The more value added your production is, the more you will get. So money is a measure of production and trade.
- 2.31** The quality of that production, in a sense of how much people value what you've produced, is going to be reflected in how much money you have, in the wealth you accumulate. So it's, in a sense, kind of implicitly doing all these calculations. And it's the summation of what I've produced, what I've traded, and the net is the money I have right now. So it serves the purpose of keeping track of production and trade. It stores the value that you have created. It's a summation of the economic values that you have created.
- 2.32** If you think about money, it represents—what is it? It represents your productive effort, the time you put into it, the effort, the thinking, what you have done, what you have created, what you have produced. And in that sense, it's funny to think about people who resent money or think badly about money. That means they resent production. They resent trade. They resent human betterment, human improvement.

- 2.33** So what does it mean to say money is the root of all evil or money is evil? It basically means this representation of production, of ability, of creation, of trade, is somehow negative; it's somehow bad. And what does it even mean to say that the love of money is the root of all evil? What is love? Love is to know something and to love its nature. But if you know what money is, you understand what money is, as a reflection of production, a reflection of what you've done in your life, economically at least, then what is not to love there?
- 2.34** It's a reflection of something positive, not a reflection of something negative. It's a manifestation of your productive ability. And where does your productive ability come from? From your ability to think, your ability to engage with the world, to produce, create, build. People who make a lot of money—notice this phrase, "make money"—make money is something uniquely American. Because traditionally in history, how did you get money? You often just took it.
- 2.35** And the economies didn't grow. It was very much a zero-sum world for a very, very long time. And over the last 200 years, particularly in this country, but in other countries as well, we've suddenly grown. We suddenly made money, in the sense that we've made more stuff. We created, we built, we produced. So we make money. We don't just take money. We don't just redistribute money. We actually make it, we add it to it. And how do you make money? By adding production, by adding stuff, by producing goods, services, by offering value.
- 2.36** So money, again, is just a reflection of the value you're offering the world. So money is a recognition of production, and it's a recognition of the trader principle. The trader principle is this principle that says that trade is mutually beneficial. That we trade where we know we're going to benefit. And why would somebody else trade with us? Because they're going to benefit. So we trade when there's mutual advantage. It's win-win.
- 2.37** So again, money represents the capacity to engage efficiently and productively in win-win relationships, in win-win transactions, with people we know, but also with complete strangers. 99% of the transactions we do every day are with complete strangers. And of course, in the world in which we live today, we do it almost anonymously and over thousands of miles, and it doesn't matter. You can do it all electronically. So money is a facilitator of all of that. It's a facilitator of modern civilization. You can't have modern civilization without money. So the whole idea of being suspicious of it, we should be suspicious of.
- 2.38** As I said earlier, money is, in a sense, a sign of lots of trade and civilization. And therefore, you'd expect it to be well developed in places where there's a lot of trading going on and there's civilization. There's people doing sophisticated things, creating stuff, building things, making stuff. And maybe the most dominant ancient civilization was Rome. And if you think about Rome, Rome was massive. I mean, it constituted most of western Europe, even into central and eastern Europe, northern Europe, all the

way to England, all the way through the Middle East, up until the Persian Empire, and throughout northern Africa. All of that was Rome.

2.39 And that meant that you had to both be able to control all of that with armies, which needed feeding and needed sustenance, so you needed to be able to fund all that. But it also meant that trade was happening. You could argue a lot of things against empires, and there's a lot of negatives you can say about empires, but one thing they facilitate is trade. You have one common law over the entire geographic area, and you have the opportunity to evolve one money over the entire geographic area, and facilitate very efficient, very low-barrier, no-tariffs trade between different parts of the empire.

2.40 And indeed, trade thrives in Rome. And one of the great achievements of the Romans, for example, the Roman roads that they built, they built roads all over the place. Why are they building roads? Well, primarily, probably, to get their armies to where they need them to be. But who else uses the roads? Merchants. And you can see thriving trade in the Roman era. What we see today just archeologically is, you go to old Roman towns or Roman settlements, and you find all kinds of goods that are coming in from various parts of the empire. You find very sophisticated pottery, and pottery made not just in one location, but pottery from different locations.

2.41 And you can see this all over. Another thing you find, when you find an archeological dig of Romans is, you find lots of coins. Lots of coins. To facilitate all this trade, they needed money. And the money was coined. It was coined by the emperor. It had the emperor's picture on it. On one side, it had Romulus and what's the other founder of Rome? Remus. Romulus and Remus on the other side. You knew this was a Roman empire, guaranteed by the emperor himself. And these coins circulated all over this empire and facilitated all this trade.

2.42 And indeed, standard of living, quality of life in the Roman Empire is dramatically higher than anything that comes before it, and for a while, anything in Europe, at least, that comes after it. So you have a massive trade network. And the reality is that if you examine in the archeological ruins there, what you find is that the trade network actually goes beyond what's just the Roman empire. That is, the Romans are actually trading with India. They're certainly trading with the Persians. They're trading with, who are the enemies, but they're still trading. They're trading with India. They're trading with China.

2.43 So Chinese know the Romans exist. The Romans know the Chinese exist. This is before transportation. This would have taken months to get there. Very dangerous. But there was already trade going on across these different regions of the world and all mediated by silver and gold, which were currencies, money, in India and to some extent in China, and of course, in the Roman empire itself. You also see them trading with areas that are not part of the Roman empire in Europe, like Scandinavia and parts of north and eastern Europe.

- 2.44 And as a consequence, the Romans develop a pretty sophisticated financial system that makes it possible. One of the big challenges of civilization, one of the characteristics of civilizations, start there, one of the characteristic of civilizations is urban life. It's cities. Civilizations don't arise in rural areas. They usually—what happens, as you're becoming more civilized, becoming wealthier, having more culture, have more buildings of architectural significance, as you start getting a concentration of people coming in from the rural areas into a central location, the huge advantages there, to a large extent, more trading opportunities, more people to trade with immediately, more efficiencies of trade, more opportunities to learn, to gain knowledge.
- 2.45 So cities are a feature of civilization, every one of them. But part of the problem with the city is, how do you feed it? Before you have cities, it's easy. Every village grows its food. And you don't have to transport it, you don't have to move it, you don't have to trade for it. It's just there. And indeed, for 99% of human history, we have just grown the food that we eat, we consume, in the vicinity of where we live.
- 2.46 And suddenly you move thousands, and ultimately Rome peaks at over a million people in 200, 300 AD, there are already a million people in Rome. How do you feed a million people in Rome? In China, you have major cities; in India, you have major cities, in Mesopotamia and Persia. So there's a big challenge for how to facilitate that. Again, what makes that possible and would be impossible without it is the ability to trade on large scale, the ability to transport, but first you need to be able to trade, and the ability of creating markets. People can bring their goods into the city from the countryside, and now you interact in a marketplace.
- 2.47 And that is all facilitated by money. But also, what you start getting in this period are different bills of exchange, different forms of credit, loans. I can transact in one part of the world and settle in another part of the world. We'll get into more details on how that works in a minute. But you start getting a real deep knowledge of, if you will, finance, of how money works, in order to facilitate just feeding people. And then you've got to administer and you've got to raise taxes and you've got to bring in taxes and you've got to mint these coins.
- 2.48 The Romans were very good at debasing their coins. So Roman coins were debased extensively. And you can see, coins that were printed early on in the empire and coins that were later in the empire have different quantities of gold or silver in them significantly. Rome functioned on a three, tri-metal, gold, silver, bronze, based on how scarce they were. And in spite of the complexity, debasement happened all the time. Rome was constantly struggling.
- 2.49 Indeed, you could argue, one of the causes of Rome's decline was the constant struggle to have to keep its soldiers happy. Soldiers all over, spread all over the place, and the fact is, in spite of all the trading, in spite of all that, productive growth, as we understand it today, economic growth, as we understand it today, didn't exist. The

world was pretty—you didn't have great advances in terms of wealth creation. Very small advances in technology. The Romans were pretty good with technology.

- 2.50** But even they, in comparison to what happens in the 19th century, 20th century, it's nothing. So, the economy is fairly stagnant, and yet you've got ever-growing needs to feed an army stretched all over the place, with constant harassment from different tribes trying to break in. Feeding that army, creating the money, finding the money to do that, that becomes the biggest problem Rome has, and indeed, that is part of its demise. And this debasement, constant debasement, paying soldiers with salt, is a real challenge and a real problem that the Romans continuously face.
- 2.51** By the way, China, as early as 221 BC, we could already find standardized bronze coins in China. So China is developing a monetary system at about the same time the West is developing a monetary system. So it's interesting that these regions that probably had very little communication in 200 BC—some, but very little—were developing very similar standards across space.
- 2.52** Now, when the Roman empire goes out of existence, and Rome does fall, in spite of what some historians would have us believe. Rome falls. And you can tell the best measure of the fact that Rome falls is the fact that Rome, at its peak, has over a million people. And Rome, at about 600 AD, has how many people? Thirty, or even less. It goes from a million to 30,000. That is the fall of Rome. That's not the same empire, doesn't really exist anymore. It splinters, it breaks up, it loses almost all of its possessions.
- 2.53** Yes, the eastern Roman empire, Byzantine, remains, but pretty much everything to the west is fragmented and gone. Rome itself, again, goes to 30,000 people. And you can tell, it's stunning when you look at the archeological digs, you find almost no pottery. Pottery is disappearing. The technology to make the pottery is disappearing, particularly the more advanced, more sophisticated pottery that they were making.
- 2.54** You find that, for example, under Rome, they had tiled roofs. Roofs stopped being tiled. They become thatched. So the material completely changes, reverts back to pre-technology. So technology disappears. They have mosaic floors, beautiful mosaic floors. Disappear. No more mosaic floors, just slabs. Europeans lose the technology the Romans have to build. To this day, we're still struggling to understand how the Romans built some of the structures that they built, given the quality.
- 2.55** They produced this amazing quality cement, concrete, not cement, concrete. And that was lost. So as the Romans could build high, they could create these amazing domes. Those technologies were lost. You couldn't build a multi-storey building in Europe for a thousand years. And the dome technology was, it took the Duomo in Florence before they could figure out how to make a proper dome. And that was, what? 1400, something around there? So that's about a thousand years later. So Rome did fall.
- 2.56** And one of the things you discover in the archeology of digs is no coins. No coins. Parts of Europe, not all of Europe, but parts of Europe, and not always, the ups and downs as

these little kingdoms fight it out and city states do well and poorly and so on. But for a long time, there are particular areas and regions in Europe where we go back to barter, where coin money disappears. And trade collapses. And again, it's very different if you're living in a little village, than if you're living in a city, and if there are no cities, you need money less. If you're not trading with the rest of the world, you need money less.

- 2.57** Now, not only does Rome go from a million people to 30,000 people, when is the next time Europe has a city of a million people? 19th-century London. It takes 1,500 years before you get another city of a million people. Again, I think a symbol of civilization is your cities. So, to a large extent, money disappears, trade collapses. They're still using old Roman coins, mostly. They're still circulating, but there just isn't a lot of silver and gold.
- 2.58** Who's using the silver and gold? Who's really trading during this period? Well, there are other more successful empires, particularly the Islamic empire, which goes basically from India all the way to Spain. It's got a thriving trade. It is a trading culture. And they are using coins. They are using money. Particularly during what are called the Dark Ages, from 400, I don't know, to 1100 or something like that, they are far surpassing anything that's happening in Europe in terms of culture, in terms of trade, in terms of wealth, in terms of architecture, in terms of civilization, in terms of cities.
- 2.59** They are the envy of the Europeans. And the money is flowing to where production happens, to where trade happens. The money is in the Islamic world during this period. I mean, in parts of Europe, pepper, squirrel skins, we're reverting back to beads and shells. And so, progress, particularly in any particular geographic area, is not a one-way arrow. It doesn't have to only go in one way.
- 2.60** One of the incentives around the Crusades was not just a recap to the holy places and the spirit of Christianity and so forth. It was also to go loot, go get your hands on some money. I mean, one way in which you expanded your own wealth was to steal other people's wealth. This is, again, a zero-sum mentality, but zero-sum in a world that is, relatively speaking, zero-sum. So, a big motivation around the Crusades was to seize money from the Muslims.
- 2.61** Now, the return wasn't very high because even if you were successful, the Crusades were expensive. But there was a net positive return to it. Actually, one of the first financial networks were established by some of the orders of knights that established themselves in the Holy Land and had bases in Europe, and they could provide credit across the network.
- 2.62** And it's one of the first expansive credit arrangements within Europe that existed during the Crusades, and because of the Crusades. Because now they had people that had well-established trust relationships between them, over vast geographic areas, who could exchange letters. "Hey, I owe this. Pay this guy. I owe him X amount of money. Pay him over there." So you got a letter like that and you got paid.

- 2.63** Now, during this period, because of the fragmentation again, you got some private mints being established. So to the extent that in places where things were a little bit better, you got people privately minting because you couldn't rely on the authorities anymore, and they didn't have the power to impose their will on you. So again, you got markets solving problems that existed.
- 2.64** And by the end of the Middle Ages, you basically had three types of coins. You had royal coins, so some like Charlemagne or some of the kings of the periods did have coins that they got their face stamped on. You had church coins. So the church got into minting of money. The church did pretty much anything, when it came to making money, to try to make money; they participated.
- 2.65** So even though, for example, as we'll get to, charging interest on a loan was a mortal sin, the monks were doing plenty of it, as well as the church authorities; they were making loans and charging interest on the side. And one of the reasons, you could argue, if you were a little bit cynical, one of the reasons to make usury a mortal sin is to get rid of your competitors, right? It's to disincentivize your competition, so you could dominate the banking space, which the church, to some extent, in some places, unquestionably did.
- 2.66** During this period, because, again, the fragmentation in Europe, you had lots of different money circulating. Now, they were all gold- or silver-based, or maybe bronze or copper, but there were a lot of them. Each little kingdom had their own. There were some private mints, there was church money. You could have 20, 30 different pieces of coin, and each one of them might have different levels of purity. So it became incredibly complicated to figure out what's the money coming out of France worth, versus the money coming out of Florence, versus Venice, versus the Netherlands, versus England. What were they worth, relative to one another?
- 2.67** And indeed, a big part of medieval banking was figuring that out, money changers, figuring out changing money, how to see the equivalent between money, what were they worth, assaying the purity of the coins, and then evaluating exchange rates between them. So a big part of the energy around banking was to try to figure all that out.
- 2.68** A lot of the monetary units that arise during this period are based on weight. So think about, the pound is a pound, right? From the weight. Livre is a Latin weight unit. Peso, which is still used today, right? Latin for weight. So a lot of the words that we use have references to a particular weight. So we know immediately what it's supposed to be. It's supposed to be X amount of ounces of gold or whatever it is.
- 2.69** And as we enter the Renaissance and beyond, as Europe starts to more consolidate, the monarchs take back their monopoly power. They kick out the private minters, they take control of them, and they come again to dominate the production of money over time. So money is fragmented. There's not a lot of it in Europe. Parts of Europe have

abandoned money completely because there's just no use for it, given how primitive life is.

2.70 I mean, this is a very, very difficult time in European history. I think, justifiably, parts of it are called the Dark Ages. And there's a medieval age that leads up to the Renaissance. And the Renaissance is a period where there's definitely a lot more wealth; and the demand for money is higher. And part of that is satisfied by the fact that this is also a period, I don't think accidentally, of exploration. This is a period of sailing out there and discovering new worlds.

2.71 And of course, Spain is at the forefront of this. And Spain, of course, comes across the Americas. And what do they find in America? One of the things that they find in America are vast, large silver deposits. And they start mining these deposits, some gold and a lot of silver. And they're pretty brutal. So going back to native peoples, the Spanish exploited those people and basically enslaved them to work those mines, and they produced huge quantities of silver and gold that basically then flooded Europe.

2.72 So, if you start looking at the 16th century, suddenly there's a flood of silver and gold coming in from the Americas. 20% of it goes to the Crown in Spain, and the rest of it starts getting spent. But what do we know again happens when you get a big flood of new money coming in? Not matched by increased production, because the Spaniards are not exactly known as great producers of things. Indeed, the money came, it was easy, it was consumed. They spent it.

2.73 And what did that consumption generate? Rising prices. Prices throughout Europe go up, because suddenly there's a flood of new coin coming in. The coin is all minted in Spain. So, Spain has the royal mint. They become, basically, the money for Europe. I mean, even when America is established in the 18th century, it is the Spanish silver dollar, which is the initial official money in the United States. That is a money that everybody recognizes to relatively of high quality: the Spanish piece of eight.

2.74 It's called the Thaler. Thaler becomes what? Dollar. Thaler, dollar, similar. So Spanish piece of eight becomes, really, the first global currency. So it's not just it's gold. Gold is global money. But now it's a particular gold. It's a particular coin, a particular coin minted in a particular place, with a particular stamp of approval. But because it's of high quality, the Spanish don't have to undermine the minting because they got so much of it. They can produce pure gold, pure silver coins. And so they dominate, basically, the production of money during this period.

2.75 But there's something called the resource curse in economics. Resource curse is the idea that countries that have vast amounts of resources ultimately tend to be poor. That is, they don't live up to the resource, because they squander the resource in consumption, instead of using the resource to invest. So if you look at most—I mean, put aside the last 10, 15 years, I think it's a little different in the Middle East, but

historically, the Middle East had huge amounts of oil. They sold all that oil, they took the money, and they spent the money on frivolous stuff.

- 2.76** It's only more recently they are trying, because they're trying to think ahead, they're trying to invest all that oil money in productive activity, in ways that will generate them wealth beyond the oil. Well, Spain didn't think ahead; they spent all the money. And while they had a golden age where they were very rich, we don't think of Spain as one of the rich countries of Europe. Which are the rich countries in Europe? The rich countries of Europe are the countries that actually, ultimately, produced—northern Europe, England, Netherlands, places like that.
- 2.77** Those are the countries that actually created and produced, so that money actually flowed from Spain to England. The wealth was actually created in England, not in Spain, even though Spain mined it. They got first-level consumption benefits, and then it disappeared. What was the main thing Spain used the money for, other than consumption of luxury, something more expensive than consumption of luxury? War.
- 2.78** The Spanish were involved in wars all over Europe. Throughout this period, particularly with the Netherlands, there was a long, long war in the Netherlands, but they were also involved in a 30-year war in Germany and 100-year war elsewhere. Wars are what dominated, and again, that's what these empires use this. And ultimately, what happened in Spain is stagnation, and Spain really disappears as a world power. Certainly, as you enter into the 19th century, Spain is a dramatically weakened power.
- 2.79** If you think about Spain, when they basically controlled all of the Americas. And they have all that gold and all that silver, and yet nothing at the end. So it's the same with Portugal. Portugal, one of the first countries to explore the world, to trade with Asia, to set up colonies all over the trading routes; and yet, ultimately, they stagnate. They don't invest the proceeds in production, and they stagnate. So Portugal is not exactly an image of a rich country. It's what you do with the money.
- 2.80** So, we talked about the fact that, at the end of the day, it's difficult to transport gold around or silver around to do all this trading. And there has to be, or the market, when things are that difficult, people figure out ways around that. At the end of the day, that's what banking is. Banking is a way to facilitate trade, without having to lug around large quantities of metal coins.
- 2.81** Some of the first banking relationships are banking relationships along the Silk Road. Silk Road that starts in China and ends in the Middle East, or when Europe is prosperous, ends in Europe, northern Africa. Along the Silk Road, it develops a whole system of banking that allows people to trade, to move goods, and not have to worry about lugging money, huge quantities of money, around.
- 2.82** Should have what is equivalent today to a bank, but it could be just a family relationship, a family business, with a base somewhere close to China and maybe something in Uzbekistan, and then another one in Baghdad, and then another one in,

you know, somewhere close to Istanbul. And traders can basically say, give the money in one place, get a letter that says, "I got paid, I got this amount of money," then take that letter to the next station, trade the letter for the goods, and then the settlement between the two stations happens in the background.

- 2.83** So maybe once a year, there's a big caravan of moving the gold from the place it's owed to the—now, remember, it's all offset. So sometimes the trade goes in one direction, maybe east to west. Other times, it goes west to east, and you're canceling out the debts. So I might be traveling and give you my money, and then come to him and get the goods, and we exchange, and now you owe him the money. But then another traveler might come around and do the exact opposite thing. Then he owes you the money, and then you start carrying that on your books, and you start canceling out who owes who what.
- 2.84** And so in the end, when you have to settle once a month, once a year, probably once every pretty infrequent period, because it's vast distances on camelback or on horseback, you transport the money all at once, and it's usually heavily guarded, and you make sure that you're not going to be robbed on the way. But the stories of traders being robbed of their money along the Silk Road, or if you wanted to take—there was also a route that went along the ocean. You took boats from different places in Asia to the Middle East, and then across to Europe.
- 2.85** Pirates, lots of risks and ways. But these pieces of paper, if they steal the piece of paper, it's useless for them, because the piece of paper is one-on-one. It's me. I gave you the money. I get to use this piece of paper for the goods. If somebody steals the piece of paper, it's not redeemable by them. It's only redeemable by me. And that reduces the incentive to rob. So you get these networks. It basically becomes a ledger-based payment system that gets settled in gold, but the ledger is what carries it for weeks, months, years, sometimes.
- 2.86** Money changers. The person I give them money to is now an expert, becomes an expert in assessing the value of the money that I gave them, because every coin could be different. So they get to really assess. And the letter of credit that I get is on the, if you will, the base unit, the real quantity of gold that was given, not what was stamped on the coin that might have been debased by three different emperors.
- 2.87** So, all of this is done in gold or silver, but you get a silver-gold unit of account that is based on weight, based on the quality of the silver and gold that actually is being provided. So you get again specialization. People specialize, as we said, in assessing the value of coins. So I can bring coins from all kinds of different places, and somebody will figure out what it's worth, give me a letter of credit, I can go somewhere else, trade with it, and come back.
- 2.88** And of course, because the Silk Road was a very, very active trading route for well over a thousand years, you can imagine that these things developed over time. And you can

imagine the relationships between different vendors and different bankers or the equivalent of bankers in those days, developed over time—trust relationships, relationships of who they knew and who they didn't know, who they trusted and who they didn't trust. And some of them were business relationships; some of them were family relationships; some of them were just trust relationships that built over many, many years and sometimes generations.

- 2.89** Ultimately, this basic system came to Europe. It came to Europe primarily in the early part of the Renaissance, in the Italian banking cities. Probably the premier of those was Venice, mainly because Venice was exposed to the rest of the world in ways that the other cities in Europe were not, because Venice was a trading city. It was on the water, and its fleet sailed all over the Mediterranean, and it traded with the Muslim world. It had a close relationship with the Ottoman empire, Venice did.
- 2.90** It often played, tried to figure out where it stood, because sometimes they were with the Ottomans and sometimes they were with the Christian world. But so, slowly, these ideas coming from the Silk Road, coming from the Islamic world, started filtering into Europe and helped establish the early modern Italian banking systems.
- 2.91** There's also a lot of change here that had to happen, if you will, intellectually, to facilitate this. We mentioned last time doing math in Roman numerals, but that's the reality in Europe. Until about 1400, people were doing math in Roman numerals. You can't do advanced math in Roman numerals. Just try it sometime. Try to do division or multiplication, really anything, in Roman numerals.
- 2.92** So, one of the first innovations was to bring the Hindu-Arabic number system into the West. Fibonacci, who's known by the Fibonacci series of numbers, but Fibonacci wrote a book in 1202, the Book of Calculation, where he is the first one to really introduce the West to the numeric system of the Arab and really Indian world.
- 2.93** The concept, one of the things that the Roman numerals don't have is—which number doesn't exist in Roman numerals? Zero, which is a really, really, really important number. It doesn't exist. And zero is a concept, I think, that comes from India. So it was introduced, and Fibonacci brings all of that. And he's the first kind of bookkeeping, ideas around bookkeeping, coming again from the Muslim world, where they're trying to figure out how to keep these ledgers.
- 2.94** It's not obvious. To us, it's obvious, because we think in those terms. But for them, they had to invent it, they had to create it. So, they were struggling with it. He's introducing that, the idea of currency conversion and how to do the math around currency conversion, the idea of interest calculation, how to do interest. Fibonacci is introducing all of this into the West. I mean, I'm sure stuff was done before, but he's formalizing now, this, into a book on mathematics.
- 2.95** A couple hundred years later, a guy named Luca Pacioli writes a Summa de Arithmetica, where he, the first time he gives a detailed account of double-entry bookkeeping,

assets, liabilities, and how to make that work. Now, this was already being used in Italy, and again, from the Islamic world. But he actually is the first one to actually do the class on accounting. This is how you do it. This is what it means. And this is how it helps your business. So, this is slowly introduced into European society.

- 2.96** And of course, this now leads to more sophisticated banking. Banks now can keep track of stuff much more easily. There's a logic behind of it, and they can all standardize over the same system. So they can match up books, and they can close the books, and they can match liabilities and assets, and they can figure out who owes whom what and how to settle these issues.
- 2.97** So now, starting with the Renaissance, when a merchant goes to the marketplace, they don't go with bulging pockets filled with gold. Now they go with letters of credit. They've given the gold to the banker. The banker has given them a certificate or a number of certificates, saying this money is there. He goes to the trader, he buys the stuff. The trader then goes to the bank and gets the money.
- 2.98** And now you can do that in any particular city. But maybe I deposit my money, I live in Florence, I give my money to the Medicis, the Florentine bankers. They give me a piece of paper. That piece of paper is then taken to a bank in Genoa, because the merchants in Genoa are not going to accept a letter of credit from Florence. But the Genoa banker has a relationship with the Medicis, so he's willing to give me a letter of credit from a Genoese bank, off of the letter of credit from the Medici. I go to the marketplace in Genoa, trade, and then the two bankers settle. The Medicis and the Genoan banks settle the differences.
- 2.99** And of course, just like with the old Silk Road method, other traders are going from Genoa to Florence, so they have to figure out nets, nets, who owes what to whom over time. And of course, all of this is now recorded clearly in ledgers, in bookkeeping, in books that they can then match up and make sure that everything makes sense. So, bills of exchange start being used more significantly during this period to facilitate trade.
- 2.100** Now, of course, the next step is to actually get us to what is going to be paper money redeemable into gold at a bank, and we'll get to that. We'll start on that next class. Thank you.
- 2.101** You mentioned last lecture, government debasing money to go to war, and then mentioned debasing money a couple of times again in this lecture. What exactly is debasing money?
- 2.102** Debasing money means making it less valuable. You've got a gold coin, 100% gold. They melt it down. Now, the new coin, they print a new coin with only 90% gold. They keep the 10. That's their revenue. And they sell it to you, they give it back to you, as if it's 100% gold. So they're lying to you. And you then go into the marketplace and start

trading as if it's 100% gold, until somebody assays it, figures out it's only 90% gold, and now devalues the value of those coins.

- 2.103** But the government kept the 10%. So now they have 10% more to spend, at least until prices all adjust to the fact that the money is being debased. So debasing is lowering the quality of the money by lowering the quantity of the metal, and providing a money that is not representative of what it's stamped to be. So this is stamped to be an ounce of gold. No, it's only 90% of an ounce of gold.
- 2.104** So you're cheating. And governments do that throughout the period. And as we'll see, to this day, they're constantly debasing the money. It's just the debasement today, with fiat currency, it's a different methodology than debasement back then.
- 2.105** Does debasing hurt the growth of a country?
- 2.106** Does debasing hurt the growth of the country? Yes, primarily because it makes it very difficult to plan for the future. It hurts your ability to save, and it hurts your ability to know what a unit is going to be worth in the future, and what you actually have. Now, we have to remember that, at least in this period of time, there's very little growth. This is one of the great mysteries. It's not really a mystery, but for most people, I think it's a mystery, one of the great mysteries of human history.
- 2.107** Basically, until about 250 years ago, there was very little economic growth or any economic growth. I mean, we today complain about 2% a year. We should complain about 2% a year because it sucks. It should be a lot higher than that. But there was basically 0% growth economically, forever. So there was increased production, but it was usually matched by increased population. And in terms of net wealth to individuals, flat, basically, for 100,000 years, mostly.
- 2.108** I mean, again, there were better periods. Rome was better. Certainly, coming out of the Dark Ages or the Middle Ages, Renaissance and Enlightenment was better. But until the Enlightenment, on average, things are flat. But debasement definitely makes it more difficult to plan for the future. You don't know what coins are going to be worth. You think about it in a modern banking setting, if you have a paper certificate for X amount of gold, but you go and get from the bank, and it's not X amount of gold. It's something depreciated because the amount of gold in a coin is no longer there.
- 2.109** Now, I think, there you get a lot less debasement. Why? Because now it's private, and the bank has a reputational capital. If they're going to give you something of lesser quality, it's problematic. And indeed, as we'll see, much of the debasement goes away, particularly in the more commercial-oriented areas, England. By the 1600s, they stopped debasing. There's no more debasement going on.
- 2.110** And partially, it's because they understand that it creates inefficiency in the trade, and inefficiency in planning, and inefficiencies long-term, because you're not confident enough. And think about the fact that you have to check every coin. Every coin has to

be checked. Was it debased? Wasn't it debased? As compared to being able to trust every coin. So, all the transactions become more clunky, less efficient, less productive. Everything slows down, as compared to what it could be. So yes, it has, at multiple levels, it disrupts economic activity.

Lecture 3: Origins of Banking

At a glance — The origins of banking.

- 3.1 So, last class, we talked about the beginnings of banking in Europe and how banking started to facilitate more trade, made it easier to trade, particularly by making it more convenient, safer, reducing the threat of theft. Banking houses started to develop throughout Europe, particularly in Italy. And it's interesting why Italy, the primary reason Italy is because of its closeness to the Mediterranean, because of its trade relationships with the Ottoman empire and the Muslim world, and through that, all the way through the Silk Road into Asia.
- 3.2 So even though you get the Spanish setting up colonies all over the place and bringing in the gold, it is actually the Italians and, ultimately, the northern Europeans who actually benefit from that and figure out how to benefit from all this expanded trade and the new tools that are coming in and the knowledge that they're gaining from these other civilizations.
- 3.3 So trading expands throughout Europe and indeed, the Italian banking houses landing up, instead of relying on a different bank in Genoa, they set up a branch of their own bank in Genoa, and then they set up the branch of their own bank in Germany, one of the German states. There wasn't a Germany back then, or in other parts of northern Europe. And now you have a Medici bank and other Italian banks that are multinational almost, although there are no real nations at this point, but multi-geographical banks with branches at different places. So all this exchange can happen even more efficiently than before.
- 3.4 At the same time, there's a growing demand for credit, for lending. As the economy of Europe wakes up from the Dark Ages and enters the Middle Ages and into the Renaissance and further on, there's an ever-growing demand for money, to grow the economy, to set out for new ventures. There's a lot of trade going on now, as we described, with Asia. You know what the thing they were bringing in from Asia, what the big deal was from Asia? Silk was part of it, but even bigger than silk, much bigger than silk—spices.
- 3.5 I don't know if you know this, but European food generally sucks, until you add some spice to it. And spices from Indonesia and Malaysia and India and all these places, suddenly Europeans discovered flavor in food and became pretty good at it, ultimately. But it was spices. But that trade required credit. It required lending. And of course, money lending is difficult. It's difficult because there's a real prohibition on money lending.
- 3.6 According to Christianity, money lending is a mortal sin. And Christians are not allowed to borrow and lend money at interest, particularly not lend money at interest. Indeed, the Catholic Church only eliminates the restriction on lending money at interest in the

late 19th century. So there is a real shortage in the ability to lend. So bankers figure out ways around this. So they take fees, instead of charging interest. They take an equity position in your business, so an ownership stake instead of charging interest.

- 3.7** To this day, for example, Islamic banking, which is practiced all over the Middle East, and to some extent now they have branches in Europe, they don't charge interest. They're under the same biblical restriction of not being allowed to charge interest. In the Muslim world, there is no interest payment. Now, they still pay interest. They still get interest. They just don't call it interest. They call it something else, dividends, fees, but it is a little convoluted. They have to go out of their way to create structures that work with this prohibition.
- 3.8** And the Catholic world, particularly the Catholic world, was really challenged by this. And it wasn't a joke. I mean, Dante, famous Florentine poet, in his *Inferno*, describes, he describes the different levels of hell and the different occupants of the different levels of hell. Well, the money lender is way down there in the seventh rung, and he's got a bag of money on his neck, and the bag is so heavy that it's dragging him towards the fire. And this is the perception of the money lender; the money lender is evil, is bad. This is somehow a mortal sin.
- 3.9** And the reason comes from, I mean, the reason first is because that's what it says in the Bible. The Bible basically says that you should not, you should not charge interest on money lent, except, we'll get to the exception in a minute, but you should not charge on money lent. But it's also a consequence of a big economic misunderstanding, which even Aristotle, the Greek philosopher, suffered from.
- 3.10** And that is, the idea is, if I give you money, if I lend you money, and I expect more money in return, but the money I gave you is not going to spout out new money. Money is not going to grow out of the existing money. Money is barren. It doesn't grow. So how can you give me more money in return? On what basis do I charge more money in return? All I should expect is the same money back.
- 3.11** Now, if I give you a grain, I can expect more grain in return, because the grain I give you will generate more grain than what I gave you. So then you can give me a piece of the profit you got. But money, I give you \$100, it doesn't turn into \$110. It doesn't have babies. Grain does. So it's OK to charge interest on grain, interest on things that were not barren, on seeds, on things like that. But it was not OK to charge interest on money.
- 3.12** And indeed, it took a long time for economists or thinkers to understand what it is that money does, what it is that me providing you, why I should get compensated for the money I lend to somebody else. The fact that if I lend you money, I'm forgoing current consumption. So I'm giving something up, and I want to be compensated for the fact that I'm only going to consume the money in the future. So an ice cream today or an ice

cream in the future, what's better? Ice cream today, obviously. So if I'm going to delay my consumption of ice cream a year, I want a little bit extra ice cream.

- 3.13** So that's one reason I should be compensated. Second reason is risk. You might not be able to give me back the money. And if I lend to a number of different people, on average, I want to make money, so I have to take into account that some people won't be able to pay back. Some people will be able to pay back, and average it all out, so I get my money back. So there's risk. I need to be compensated for that.
- 3.14** Then there's the loss of the value of money, the inflation. I might not be able to buy as much in the future, so I need to take into account my expectation of inflation into the interest rates. Interest on a loan is typically a composition of what's called the real rate of interest, which is the price you pay for delayed consumption. Or another way of thinking of it is opportunity cost. I could use this money for something else, and I'm using it for this, so I need to be compensated. I need to make it better than this other alternative.
- 3.15** Risk and inflation. That is what an interest rate is composed of when it's privately engaged in. But the realization of that is relatively late, I'd say probably 18th century, around the time of Adam Smith, a French economist by the name of Turgot, who does a lot of work and figures out the components of this. And there were other thinkers. Scholastics contributed a little bit to this. But they're economists who finally figure out, OK, oh, this is why we charge interest. This is what interest, this is how money is indeed.
- 3.16** It's not barren; it actually does create money through investment, through building. It facilitates the growth, even if it doesn't grow itself. It makes growth possible. So there's a lot of ignorance about money, which drives this. But there's also biblical prohibition. But the interesting thing about the biblical prohibition is this caveat to the biblical prohibition. It basically says, you cannot lend money at interest to your brother, but you can to a stranger, or at least that's the implication.
- 3.17** And the biblical prohibition is given to the Jews. It's in the Old Testament. And the Jews take it as, well, other Jews are my brothers, non-Jews are not brothers. So I can charge interest from non-Jews. Christianity, on the other hand, is a universal religion. So Christianity views my brother as everybody. So I cannot charge interest from anybody. So Christians are literally prohibited from charging interest on a loan. Again, according to Catholicism, at least until the late 19th century, until the Pope says no.
- 3.18** And they create this interesting—they decompose interest and they say, well, there are two types of interest. There's interest and there's usury. Usury means excessive interest. Defined by whom? Whoever. Excessive interest by the authorities. Excessive interest is wrong, but interest is OK. That was the solution the Catholic Church comes up with. Protestants don't care because, basically, Martin Luther says, in this world, as

horrible and crummy as it is, do what you need to do. If you charge interest, it doesn't matter. Do it.

- 3.19** In the next world, the perfect world, the ideal world, nobody charges interest. That doesn't exist. So there's a certain permission to do things in this world that are not right, are not good. And Martin Luther says interest is evil. But he says that's for the afterlife to sort out. It's not for this life. So there's a certain beginning of almost a separation of what happens in this world and church. So church and state, a little bit, there's a little separation there. We'll deal with all of that in the afterlife, and this life, do what you need to do.
- 3.20** But that means that forever the Christians are banned from charging interest and their banking is difficult. It's cumbersome. They have to come up with all these rules around it. And that a lot of the money lending is left to the Jews. And Jews do a lot of the money lending, and they lend money throughout Europe during the Dark and Middle Ages and into, of course, the 19th century and the big Jewish banking houses.
- 3.21** But they become oriented on finance, for one, because they're allowed to. Two, they're not allowed to do much else. They can't own land. So they can't be farmers. A lot of professions they're excluded from. So what's left? One of the areas that's left is money lending. But as a consequence of that, they're hated. Everybody loves to hate the bankers, generally. Bankers are not liked, generally. But think about an era in which charging interest on a loan is a mortal sin. These guys are committing the mortal sin. All of them, committing the mortal sin.
- 3.22** And since I need a loan, so I'm going to take out the loan, I'm paying them and facilitating their mortal sin. This is horrible. So a lot of anti-Semitism comes out and a lot of the anti-Semitic images, a lot of the anti-Semitic writing, stories, blood libels, come from this idea of the Jews engaged in a profession that is inherently evil, inherently bad. And that they're exploiting the Christians.
- 3.23** And the best illustration of this is a masterpiece, one of the great, great works of Western literature, which is the Merchant of Venice by Shakespeare, which I highly recommend if you get an opportunity to watch or to see it, because all of this is laid out in the Merchant of Venice. I mean, it's beautifully laid out. Shakespeare exhibits an understanding of the economic issues, as well as the moral and philosophical issues, just brilliantly.
- 3.24** Shylock is the Jewish money lender. There is a Christian money lender, but the Christian money lender never charges interest. Never charges interest. And the Christian money lender is always putting the Jewish money lender down. He's always discriminating against him. But there comes a day where the Christian money lender has his ships, he's invested in these ships. They're all out sailing, and he needs money, I think, for a wedding of his daughter, if I remember. And he needs money.

- 3.25** And the only choice he has is to go to the Jewish money lender, asking for money, for a loan, from the Jewish money lender. And he berates him, while he's asking for the loan, he berates him for charging interest. And the Jewish money lender has had enough. So he said, OK, I'll give you the loan, but if you cannot pay, I want in exchange a pound of flesh, which is pretty nasty, right? I mean, I basically want you dead, right? This is a period in which I don't think you can take a pound of flesh off of somebody and keep him alive.
- 3.26** And the Christian says yes, and in the end, he cannot pay. The ship sinks, and he cannot pay the loan. And the Jewish money lender demands the pound of flesh. And then there's a whole trial around that. The play is clearly anti-Semitic and horrific. Of course, Shakespeare had never met a Jew in his life because Jews were banned from England for 200 years around this period, so there were no Jews in England when Shakespeare wrote this. He had written it off of another play, a Venetian play. It's set in Venice.
- 3.27** Again, Venice, the trade place, where you would see this, and where there was an active Jewish community of money lenders who are lending money to facilitate all the trade that was going on across the Mediterranean by the Venetians. But I highly recommend it. There's a movie with, I think, Al Pacino plays Shylock. It's really good, and listen to the language. Listen how they talk about money. Listen to how they talk about finance.
- 3.28** I'm stunned by how good Shakespeare's understanding of what's going on is. He's on the wrong side, but he gets the issues, which is just stunning. So that's a good illustration. There's been lots of examples in the finance course that I've done for Peterson Academy. We go into more depth over the whole issue of money lending and the issue of anti-Semitism and some more examples. I'll just say this, that some of the pogroms against Jews, where a whole Jewish community is killed or big chunks of it are killed, are often motivated by the fact that the aristocrat in the area owes money to the Jewish lender.
- 3.29** Well, one way—and he can't pay, he's run out of money, he's got a war to finance—one way to get rid of your debts is to kill the person you owe the money to. And there's a famous massacre in York, a pogrom in York, England, where they kill all the Jews. And the first thing they burn are the records of the money lenders. And the aristocrat gets to walk away scot-free. And a little bit after that is when the English ban, kick all the Jews out, and Jews are no longer there for about 200 years. There are no Jews in England.
- 3.30** So money lending is definitely connected because through those origins to anti-Semitism. And to this day, when you have conspiracies around the Jews running the world, it's Jewish bankers. It's almost always Jewish bankers. Bankers are somehow involved. Banking, generally, is always mysterious. Money is always mysterious. Remember, it's the root of all evil, or the love of it is the root of all evil. So all of these

things are connected and connected into conspiracies and connected into a particular perception of Jews.

- 3.31 Jews did very well in banking in the 17th, 18th, 19th centuries. You got the Rothschilds, you got other banking houses. They ultimately did well on Wall Street, and that all connected in people into this general idea that they control things. And of course, that all connected back to the usury, the anti-usury laws.
- 3.32 Banks are developing and they're developing a credit. Again, in the Protestant world, there's a lot more tolerance towards banks and interest and a more diverse banking system. It's, I think, ultimately part of why, as you get into the 16th, 17th, 18th century, Italy fades, Spain fades, and where you get prosperity is in Northern Europe, which tends to be the more Protestant part of Europe. There's a lot more tolerance for economics and primarily for finance and banking and usury and things like that.
- 3.33 England and Amsterdam become the real hubs of commerce. These are two trading nations. They have ships that are now not just going to the Mediterranean, but going around the world. They're going to the New World. They're going to Asia. They are setting up empires. They're setting up colonial empires. Britain, at some point, controls vast swaths of the world. At that point, it's pretty much the world is known. And both Amsterdam and England become financial centers, as well as commercial centers. Those always go together.
- 3.34 When Venice was at its peak in terms of commerce, it was also at its peak in terms of finance. And in England during this period, what starts happening is people accumulating gold and silver. And it's inconvenient for, call it the nouveau riche, if you will, to carry around all this gold and silver, to keep it at home. There's a security risk, there's robberies, theft.
- 3.35 And what starts happening is, the local blacksmiths start offering the service of basically putting your gold and silver into a vault. And they keep your gold and silver, and they give you a piece of paper, a claim on the gold and silver, that you can any time come and redeem and get your gold and silver back. And you pay them a fee, a storage fee. They're storing your gold for you. And of course, they invest then in the security to make sure that it's not robbed, in proper vaults and in all of that.
- 3.36 Now, what slowly happens is, some of these goldsmiths realize that one of the things they can do is, instead of giving people, given the quality, they check the quality, one of the things they can do is, they can give you a bearer's note, a note that is redeemable in its gold. But that note could not have your name on it. It could just be a note. And then you can then use that note to go buy stuff. And then the merchant you bought it from could go to the blacksmith and get your gold, or gold.
- 3.37 And indeed, the blacksmiths, who are now becoming bankers really, now realize that they can actually mix all the gold up if it's of equal quality, and they can check all that. And when you deposit the gold, they can figure out the quality of the gold, and they

give you a receipt that is commensurate with the quality that you've deposited. And they can now mix it all up. Now, they still are willing to take your gold and store it for you. And then they'll put it in a bag, separate, sealed bag that's separate from the rest of their gold, and put it in a vault.

- 3.38** But if you don't care to get the exact same coins back, what they will do is, they will give you a note that just says, this is redeemable for X gold. And now what happens is, these notes start circulating. And people start using the notes as money. And of course, the bankers figure out how to make different denomination notes. So one pound, 10 pound, 100 pound. So now you can have different denominations as you're consuming out there.
- 3.39** But at any point in time, if whoever holds the note wants gold in its stead, they can go to the bank, give the note, they'll get the gold. Now, what you've done here is, this is probably the most important innovation in finance since the coining of money, because now you don't have to worry about quality, because the goldsmith is basically guaranteeing the quality. You don't have to worry about lugging around large quantities of a metal. You just have paper. And you now can transact freely with money, without worrying about the metal, the quality, whatever goes into that.
- 3.40** If you get coin from out of the state, from another country, again, you take it to the goldsmith, they'll do the calculation, they'll figure out what it's worth, and you'll get the paper money, the paper instead. Now, the paper in and of itself is not money, right? What is the paper? It's a claim. It's an IOU against money. The money is still gold. Gold is still the unit of account. Gold is still the store value. And gold is really still the medium of exchange. But now you've got a proxy for that gold, a substitute for the gold, but that is denominated in that gold to be able to facilitate trade.
- 3.41** Much more effective, much more efficient, easy to denominate. You don't have to think about different sizes of coins. Now you just have different numbers printed on the piece of paper. Modern money, right? This solves the whole debasement issues. At some point in Europe, in France, they tried to figure out all the different exchange rates, and they figured they had 38 different types of money that they had to figure out exchange rates between at a local bank. All of that is gone.
- 3.42** And while early on, very few people are using this system, it slowly grows and it really starts growing once bankers figure out this next step. They realize that they've got all this gold sitting in their vaults. And at any given point in time, very few people actually want to exchange the paper money for the gold. And they figure out, I could actually lend some of the gold in the meantime to other people and gain interest on that.
- 3.43** And yes, if all my customers come to me all at once to claim their gold, it won't be there because I've lent some of it out. But when is that going to happen? That's not ever going to happen. So I can take 20% of it and lend it out, charge an interest on it as

a loan, and start making money. And now I can induce people to bring their gold to me and to deposit it in my bank by paying them interest.

- 3.44** Now, you remember before, you had to pay a storage fee because there's no way for the banker to make money, unless you pay them a storage fee. But now there's a way for them to make money. They can use a portion of the money you're storing with them to make loans. And it's not like they can't back all the money that's being deposited. They can, it's just that there's a liquidity difference. You might want your money, and the loan has not been paid back yet. So the money might not be there.
- 3.45** But the money is there, as long as the loan pays back. And this is the talent of the bankers, to make loans that will actually pay back. If the loans fail, you've got a problem. So now you've got a modern banking business. You're bringing in deposits, you're paying interest on those deposits. Now, you're not lying to your customers, importantly. Now, if you told your customers, "No, it's in the vault. Don't worry, it's always going to be there," no, you say, "Look, I'm lending some of it out."
- 3.46** And there might be, if you look at your bank account, if you look at your contract, in the small type, it probably says there could be occurrences where you will not be able to get your money out of your bank account on demand. We hold the right to postpone, delay, in certain circumstances. And that's the reality because they don't have all the money that you put in there always available. That's the source of bank runs.
- 3.47** There's not a bank, certainly in the United States, there's not a bank in the United States that could survive a bank run, particularly today. We'll talk about that when we get to fiat money. But there's not a bank today that has the money that everybody's deposited in, available to pay out, even within 24, 48 hours. They just don't have it. So every single bank in the U.S. would go bust if there was a bank run on it. In those days, in the initial part, the amount that was lent was relatively small.
- 3.48** So, this modern banking starts developing. And you can see now, loans are being made by banks, productive loans to industry and business. Again, this is part of the development of the economies of these countries at this period of time. It's these entrepreneurial banks figuring out how to take, how to use money to, for productive purposes, to lend to businessmen, because how are they going to get the money back? It has to be a business that's profitable, that pays them back from their profits. So they're investing now in profitable businesses.
- 3.49** They're growing the economy by doing that. And you get England and Scotland, particular. Scotland is a little different than England, but in those days, they have a slightly different regulatory scheme. But both of these countries start developing these private banks, private money, because the money is being issued by the bank. It's not being issued by the government. It's not guaranteed by the state; it's guaranteed by the person who issued the money. There's no state guarantee. So all of this is private money now.

- 3.50** Not private coins. Don't need that anymore. Indeed, you don't need coins at all. You can have bars or you can have any kind of other way in which to store the gold. The banker could take the gold, the coins that you bring them, and melt it all down, and just create bars. It's much easier to store bars than it is to store coins. And now you get people using paper money. The paper money is redeemable 100% in gold. The money is a private money. So it says on the thing, bank of whatever the name of the bank is. It doesn't say England, Britain, that's not what. It's not being issued.
- 3.51** Now, later, the Bank of England very quickly takes over the monopoly. The state always tries to monopolize these things, and soon the money will all come from the Bank of England, but at least in the beginning, it is not. You get, as we said, checking accounts, banknotes. You get a deposit, you get interest, you get loans. You get all the characteristics of a modern banking system.
- 3.52** In American colonies, there was no paper money. There was no redeemable money. At the time of the Revolution, all we had were coins. Again, primarily those pieces of eight, those Spanish coins that came from Mexico. High-quality silver money was the dominant money. But with the establishment of banks in the United States and the first national bank in the United States, right after the Constitution, you get a flurry of commercial banknotes circulating within a few decades.
- 3.53** They start dominating the United States, the American economy. In Switzerland, coins were still about 87% of all the money held by the public as late as 1850. But by 1906, only 7% were coins. The rest was paper. So, country by country shifted to this model during the 18th and 19th century, primarily the 19th century. All right.
- 3.54** So, let's think about this fractional reserve banking system and how it works. At any given point in time, there are more claims on the gold in the vault than there is gold. Because think about this, when I take the 20% and lend it out, I'm not literally lending out the gold. What am I lending out? Claims on the gold. I'm giving them IOUs, I'm giving them banknotes. So if I have 100 units of gold that people have deposited with me, each one of the person who deposited gold with me got one of those units of a paper, a claim on that unit.
- 3.55** And then I went ahead and issued another 20 units of gold to somebody else. So if everybody comes to claim their gold, it's 120 for the 100. So of course, there's a risk there that everybody comes at the same time and I can't collect my loans fast enough to repay them. So there's a liquidity risk. Liquidity suggests, how quickly can I get access to money? Money being gold here.
- 3.56** The second risk is, as we said earlier, there's a risk that some of the loans will go bad. In banking, it's called credit risk, the risk that I won't get my money back. I'll get only 90% of my money back, instead of 100% of the money back. And then I take a real hit. I'd have to make profits elsewhere in order to make up for that. And that's why banks

have pretty diversified lending portfolios and they charge interest that takes into account risk, to compensate for the fact that some loans don't pay.

- 3.57** And there's the perception that this is going to be really unstable. There's something really wrong going on here, that fractional reserve banking is a lot of times put down, the source of economic instability and problematic. But the reality is that in a marketplace, it is not. There are competitive pressures to make sure the banks don't issue too much of the gold out as loans. What would happen if you suddenly realized, oh, that my bank has issued 50%, not 20%. What would be the first thing you demand?
- 3.58** You might want to take your money out, but you might also say, you're putting me at greater risk, right? So what would be the way to compensate you for the greater risk? More interest on your deposits. So I demand more interest on my deposits. So the cost for a bank to issue more loans on a fractional reserve basis is higher interest that I have to pay to depositors. So there's an equilibrium here in a competitive environment, where banks competing with one another, yeah, some banks will be more risky than other banks, but you'll be compensated for that risk.
- 3.59** And if you really want to be completely safe, you can ask to put the gold in the vault and pay a fee to have them protect that gold. You can choose how much risk you want to take as a customer, and you can see the relative risks of the different banks, based on how much interest they're paying, because the market works really fast here.
- 3.60** In the 1980s, during the S&L crisis, this was a crisis of, a lot of banks were going bankrupt in the United States, saving and loans banks in the U.S., all over. Hundreds of them went bust. Anyway, the Wall Street Journal used to publish every day a list of the banks that were offering the highest rates on, basically, deposits, on saving accounts, CDs. And there was almost 100% correlation between the banks offering the highest rates of interest and the banks that later would go bankrupt.
- 3.61** And the reason for that is, they were the riskiest banks. And nobody was going to give them money unless it was a lot of return. Now, why would people give them money anyway, if there was such a high probability these banks would go bankrupt? More interest, but if the bank goes bankrupt, you're not going to get the interest. I mean, in the United States, modern United States banking system, the reason is deposit insurance.
- 3.62** Because even if the bank went bankrupt, the government will bail you out, guaranteed. The government guaranteed it; the FDIC guaranteed it. It was just a matter of the inconvenience of getting the money. But you're going to get your higher interest. You're going to get your money back. There's no risk here, which is a perversion, but it's the way the system works.
- 3.63** But if somebody is offering you a way-above market interest rate on a deposit, there's got to be a catch, and the catch has to be more risk. It's just a question of figuring out where that risk is. But there's got to be more risk. They can't afford to give you a

dramatically higher than market rate unless they are doing something on the other side of the transaction, which is much riskier.

- 3.64** So there are market forces that, if you take away the deposit insurance or the government guarantees, that are constantly regulating these banks and regulating their behavior, to make sure that this doesn't get out of hand and is not a catastrophe. And indeed, what's called this era, the free banking era, there's good examples, particularly from Scotland, where they were really free. They were not really regulated at all. They issued their own money. They did pretty much what they wanted during the period where they had the free banking.
- 3.65** No bank runs, no bank failures. I mean, it worked like a charm. It worked really, really well. Same is true, surprisingly, I always find this surprising, that one of the freest banking system in the world that lasted for a very long time and did very, very well is the Canadian banking system, in the United States. There's actually a course here in the Peterson Academy by Charles Calomiris that describes the Canadian banking system and how well it functioned.
- 3.66** And Canadian banking system has never had a banking crisis. And that's because it was, for most of its history, left free, whereas American banking system has always been regulated, always been controlled. Canadians actually had a freer banking system than the United States, and they got a central bank later than the U.S. did.
- 3.67** So, the gold-backed, free market banking system worked. It provided capital to investors through lending, through fractional reserve banking. It provided interest to savers who put their money in the bank. It provided opportunities for investment, and it actually facilitated trade. And of course, what would happen if—how did this actually work?
- 3.68** So I get my dollars from my bank, and I go and spend them, and the merchant has a different bank, and he takes his dollars to his bank, and he wants the gold. Well, his bank will give him the gold, and then they will go to my bank and demand the gold be transferred to them. And of course, again, all this is done on ledgers. So any two banks have credits and debits relating to gold. And at the end of the week, or at the end of the day, or at the end of the month, they settle. And they figure out who owes what to whom, and the gold exchanges appropriately.
- 3.69** So that the system clears the gold in the background constantly, as an ongoing basis, between financial institutions, between the various banks, without the individual even knowing what's happened. So for you as an individual, it's seamless, as long as all these banks are considered credible banks, all these bank notes are considered truly redeemable into high-quality gold. And for a long time, this relatively free banking system worked.
- 3.70** The weakest player, probably in the free banking era, was the United States, because our banks were a lot less free than other countries' banks. But this gold clearing

facilitated a huge increase in industry, huge increase in economic growth, during the 19th century and early 20th century.

- 3.71** Of course, government set an eye on this because this was like a profitable industry. And as we said before, governments always want money. They need money. And a big part of that is to finance wars, and it always has been. So government were always eyeing the banking system and trying to figure out how they could get involved. Central banks, as a consequence, were government's response to all of this. It was a way for them to come in, in the name of supposedly standardizing things, come in and get a piece of the action, if you will, with regard to what the banks were doing.
- 3.72** Now, central banks initially were not there in order to issue currency or anything like that. One of the first central banks was the Bank of England. The Bank of England was established for the explicit purpose of helping fund the British government. That is, to buy, to facilitate the sale of their bonds, to facilitate a sale of government debt. It was established in 1694 to provide the government loans at low interest. That was it, so the government could borrow money.
- 3.73** Now, over time, as these kind of institutions are, its influence, its power grew. And at some point, basically, it became the only bank in England, Scotland was excluded, but in England, to be able to issue banknotes. That is, all the banks had to use Bank of England banknotes, and ultimately, all the gold had to sit at the Bank of England. So the banks could still do loans, take in deposits. But at the end, all of that was redeemable into Bank of England banknotes, which then became money.
- 3.74** So the money became now, again, centralized. Instead of printing coins, instead of minting coins, they minted banknotes, which were still redeemable into gold. So you could still get your gold and have it, but you could only get the gold and have it through the Bank of England. And that made it possible for the Bank of England, as it did on several occasions, to say at times of, let me think, war, that, "Hey, we can't redeem your notes right now. We need the money for something else, so we're not redeeming. Keep using the paper money as money, and one day, maybe it'll be redeemable again. But for now, there's no redemption."
- 3.75** And this happened during the Napoleonic Wars. It happened on a big scale during the First World War, and we'll get to that. But during this period, they could also issue more paper notes, in spite of the fact that the gold reserves were not increasing. It was much easier for them to cheat. There was no competitive pressure to hold the cheating back and to eliminate it. So whereas with private banks, there's always that competitive, you always have to make a profit. There's always competition. If your banknotes are not as good as somebody else's banknotes, they won't use them.
- 3.76** Once you monopolize, you can now debase the currency. So one way of debasing the currency is, let's say I have 100 units of gold and I issue 120 certificates on those 100, but not to loans that are then going to be paying me back, so that my gold grows. No,

I'm just issuing and hoping they never come and redeem them. Or when they do come and redeem them, say, "Well, they're not redeemable anymore." And they can't switch. They can't go anywhere else.

- 3.77** Indeed, the Scottish banking system in the mid-19th century was functioning fine, independent of the Bank of England, doing very, very well. No bank runs, no failures. And at some point, that made the Bank of England look bad. So what did they do? They grabbed them. They incorporated the Scottish banks under the same system as the Bank of England, and the Scottish Bank lost their independence. Although to this day, Scottish banks issue their own currency, now backed by pounds from the Bank of England. But they issue their own currency, so you can get a Scottish bank note that does not look like typical Bank of England banknotes.
- 3.78** So Bank of England, in a sense, became the banker for the banks. They deposited their gold with the Bank of England. It gave them the redeemable notes. They used those redeemable notes, passed them on to their customers instead of the gold. And if they wanted the gold back to their customers, they'd have to go back to the Bank of England to get the gold to give to their customers. So you could see they created an extra layer on top, in the name of stability and safety, but really for control and the ability to not redeem and to grow the money supply and to provide the government with additional resources when it needed it and wanted it.
- 3.79** And again, primarily during time of war, partially because—and wars are convenient because they ramp up the patriotism and then nobody's going to complain too much, if it's in the middle of the war, people are dying, and you can't redeem your money. Stop being greedy. Be patriotic. Stick with it. And the fact that your wealth has been eroded and you exit the war poorer than you did when you—again, nobody really cares.
- 3.80** So, during the 19th century, what we had was what is typically called the classical gold standard. And basically what this gold standard that was in England, that to some extent was in the United States, although the United States had, as we'll see, a bimetallic system, basically defined gold as the unit of account. A particular mass of gold was a unit of account for everyone. And of course, the pieces of paper were denominated in this unit of account. They translated into a particular mass of gold. Your dollar bill was equivalent to, I forget, 22 ounces and something of gold.
- 3.81** In other words, during this period, most of the 19th century, really until World War I, gold is money. Gold is money, and gold is standardized as money. It's a weight of gold. It's not now a particular coin. It's not different coins that you have to exchange. All the currencies, at least the British currency, the American currencies, and other currencies in Europe, are denominated in the same weight of gold. So they're all fixed against each other because they're all fixed against gold.
- 3.82** And this, of course, facilitates a boom in trade. This is a period, the first period of real globalization. While there was always global trade in the 19th century, it explodes. It

grows dramatically, partially because the British Navy is out there protecting the sea lanes, protecting the trade lanes, but also because you now have a unit of account that is truly global. Throughout the world, there's one unit of account, and that is a certain weight of gold as manifest in these bank notes.

- 3.83** So the dollar was actually 23.22 grains of pure gold, was a dollar in 1900. And that became the unit of account. There were still gold coins circulating, but people were using gold coins less and less, and they were slowly, basically phased out. It was translated to paper notes and currency notes, bank balances. Banks became the center and the hub for all financial transactions.
- 3.84** In the U.S., for a variety of reasons, silver still maintained a role, at least on and off, during the 19th century. A lot of the financial crises in the U.S. were caused by this fact that we had silver and gold working at the same time. There were certain interests that wanted silver because silver was deemed as inflationary. There was more silver than there was gold. If you flooded the market with silver, prices would go up. There were certain interest groups in the U.S. that liked inflation.
- 3.85** But there was a big populist movement that was around "we want silver." Because we want more money, not less money. We want prices to go up because our commodities are not, their prices are not going up. A lot of farmers held these views. So there was a lot of politics during the 19th century in the United States about, should we be in a pure gold standard? Should we be in this bimetallic? If we're on a bimetallic, gold and silver, to what extent are they interchangeable? What is the exchange rate between them? How much silver goes into gold? How do we do that?
- 3.86** And there was a lot of political fighting going on throughout the 19th century over this. And because the government kept getting involved in it, because now the government was determining whether silver would count as legal tender and how they would exchange with one another, because of that, you would have periodic bank crises in the United States, partially as a consequence of this. Generally, the U.S. government was particularly interventionist when it came to banks, both at the state level and at the federal level.
- 3.87** And again, the United States engaged in, what was it? What do governments do? Wars. And look, wars are sometimes justified. Sometimes you need to be in wars, but wars are expensive. So we had continentals, during the war of independence, which were basically paper money that lost all its value. We had paper money during the Civil War. We had the South's paper money versus the North's paper money. Of course, the South lost, so that money became worthless very, very quickly.
- 3.88** And they're both irredeemable, not redeemable into gold. And then when they made the dollar redeemable back into gold, I think in 1879, there was a question of how to do that and how to do that at the right exchange rate, and that's when the whole idea

was, should we include silver in that or shouldn't we include silver? So government is very involved, but nowhere near as involved as today.

- 3.89** Because it's still true that during this period, if you found gold in California, you took it, you could take it to a private mint. They could stamp coins for you, or you could take that gold to a bank, and that bank would give you a bank certificate with the bank's name on it, redeemable into gold. All of that was still true during this period. So the government was playing around more at the edges, and more at the level of bank regulations, than they were in actual monetary unit, again, with the exception of silver, on a regular basis.
- 3.90** So for the most part, there was relatively free monetary system based on gold. We lived in a relatively free market gold standard. And a consequence of that was stable prices. So we all know the term inflation. Inflation is used in different ways. But typically in the way most people use inflation, they talk about the fact that inflation is a loss of purchasing power. That is, your dollar buys less. Or in other words, prices have gone up. So when people report about inflation, they report CPI or they report some other measure of inflation; that means they're measuring prices going up.
- 3.91** Another way of thinking about that is, my dollar buys less. So somehow my dollar is being depreciated in value. During the gold standard, we went through very long periods—and next class we'll contrast that with fiat money and what's happened post-gold standard—we went through long periods of time with almost no inflation. 1821 to 1914, in the United Kingdom, prices went down 5%. But over the whole period, they went down a little tiny bit on average, throughout the period, and that means productivity grew faster.
- 3.92** The number of goods in the economy grew faster than the new gold entering the economy. So prices slowly decline, but it's minute, if over that entire period, it's just 5%. So prices are basically stable. In the United States, from 1879 to 1914, prices rose 0.2 %. So basically, they didn't. Prices were just flat. Now, there were periods where the value of the money appreciated, like it did in the UK, where your dollar could buy more and more stuff because productivity increased faster; the number of goods available in the economy increased faster than the amount of money increased.
- 3.93** And there were periods of slight inflation, like the California gold rush, where suddenly a lot of gold enters into the economy, and productivity hasn't caught up yet. So you have the same number of goods, more money chasing it, prices go up. Again, we'll talk about inflation more in a future class. Quantity of money doesn't increase much. As we said, quantity of gold doesn't increase much from year to year. So it's expensive to mine it. It's hard to find it. It's expensive to refine it, even when you find it.
- 3.94** So from 1807 to 1869, the gold stock grew at about 1.95 % a year. We talked about 1.5% before. That's across very long periods of time. But it's about consistent with the same range. From 1869 to 1919, because of the California gold rush, it grew by closer

to 2.5 %. But still, these are not massive numbers. And of course, this is also an era, and I don't think accidentally, of incredible economic success, long-range planning, people issuing 50-year bonds because there's no inflation.

- 3.95** So, remember we said interest is based on real rate, deferred; it's opportunity cost, risk, and inflation. But inflation is off, basically zero. So it's easier to project into the future and to price things way into the future. The British government actually issued, I think during the Napoleonic War, it issued a perpetuity. Basically, to this day, people who own that bond get interest. It never expires.
- 3.96** They're issuing 100-year bonds, 50-year bonds, because people had a very long-term perspective because they viewed money as stable. Money was gold. It didn't change. Quantity of money didn't change that much. And even if, when it did, productivity was increasing at such a rate, economic progress was creating at such a rate that basically, in terms of the purchasing power of the gold, that wasn't changing. Purchasing power of the gold was flat.
- 3.97** During this period, as we said before, we got a boom in international trade. Globalization 1.0 really happens and creates immense amount of wealth for everybody who's participating in it. Think about the fact that a gold standard creates fiscal discipline on a central government. The government cannot just print money and spend it. All the money is backed by gold. So there's a limit to how much is available, how much is accessible.
- 3.98** And if they take on a lot of debt, interest rates will reflect that. They'll have to pay higher and higher interest rates. They can't just wipe out the debt by printing more money. So the gold standard really placed a strong fiscal discipline. I mean, think about the fact that—and I don't think this is because of the gold standard. It's because of a completely different conception of the world of government back then.
- 3.99** But in the 19th century, the U.S. government, except for the Civil War, never spent more than 3.5 % of GDP a year, 3.5. Today, 21%. So 3.5 to 21%. So when they say they can't cut spending, you get a sense of what is actually possible. If you're willing to go back to the 19th century, which was pretty good, we grew at a much faster economic rate than we do today.
- 3.100** Now, the fact that there was fractional reserve banking on the gold standard, banks could expand the amount of credit they provided when there was a lot of demand for credit. When the economy is growing, where businesses wanted to borrow money, they could contract it. When the economy was not doing as well, they could limit how much loans they want. When risk went up, less lending. When risk went down, more lending. And you can see the whole system is very responsive to market conditions, very responsive to what's going on in the economy.
- 3.101** During the gold rush in 1849, 250,000 people moved to California during that period. Now, this population was pretty small back then. That's a lot of people moving to

California, from 1849 to 1857. Price level over those 26 years went up 1.3% a year, annually, which is 26% over 18 years, which is very little, considering this is a gold rush. This is a huge increase.

- 3.102** So again, stability across the board. Modern banking stabilizes, the business cycle stabilizes, lending and consumption, lending over time. And what you get is a thriving economy. Now, there were problems at the time, particularly with the American banking system, that, as I said, derived from a variety of different constraints placed on it by regulations. And again, I'd refer you to Charles Calomiris' course on the American banking system.
- 3.103** But one of the main ones was, American banks were not allowed to branch. So if you were a bank in America, you could have basically one branch in some states, only one branch. Other states allowed you to branch within the states, couldn't leave the state. So you weren't allowed to branch outside of the state. There was no, there was no diversification in banking. Banks were all concentrated in a particular region, particular state, and sometimes a particular town because they couldn't have more than one branch.
- 3.104** United States had, at the peak, something like 30,000 banks. Canada, same geography, approximately, right, a lot fewer people but same geography, had a dozen major banks, maybe 100 banks total, all well diversified across the entire Canada. Never had the kind of banking crises the United States had.
- 3.105** So, the gold standard during this period worked phenomenally well. It facilitated economic growth. It provided stability, no real inflation, and was an example of what happens in a free market when you leave it mostly, not completely, sadly, but mostly alone, and the kind of success and thriving that can result from that.
- 3.106** All right, next class, we are going to dive deeper into some of the pros and cons of a gold standard. We'll then talk about what happens when you combine central banking with a gold standard, which is what ultimately happened in the United States. And then in classes beyond that, we'll start talking about what happens when you go completely off a gold standard and embrace what's called a fiat standard, which is what we have today. And we'll spend the next few classes wrapping up the gold standard and then getting into the nature of a fiat currency and a fiat monetary system. Thank you.
- 3.107** This might be touching a little more into theology, but I'll try to, I'll ask away. You were talking about usury and interest and how, I think it was Martin Luther, how he just said usury was now known as excess interest.
- 3.108** Well, that was the Catholic Church. Usury is excess interest, and that was banned. Interest is OK. And that was the Catholic Church in the late 19th century.
- 3.109** So do you think that today, because today it seems like Christians, it's not necessarily— having interest isn't a sin, so to say, or it's not really a part of— how can I say this?

- 3.110** It doesn't bother Christians anymore.
- 3.111** No, they're taking part in it, no matter what.
- 3.112** Well, they always did. I mean, that's the point. They kind of always did. What did the Catholic Church do in the 19th century? They'd recognized the reality. It looked around and everybody was doing it. I would even argue, and there's plenty of evidence of this, the Catholic Church itself, for much of its history, was engaged in banking activities and taking interest and charging interest, in spite of the fact that it was condemning it at the same time.
- 3.113** And as I said, I think, in a previous class, there were places, I think in Genoa once, where they literally wanted to come in and take over the banking system there. They wanted to make their money. So they persecuted the bankers for committing the mortal sin and then came in and replaced them. So the church has always had, has had a certain theological position, but then haven't stuck to it, and often utilized their theological position in order to achieve its own aims.
- 3.114** And the reality is that today, nobody has any problem with interest. Most people don't even know the history about it. Most people don't even know that biblical passage, but it's there. But then there are a lot of biblical passages that Christians, Jews, Muslims prefer to ignore. But look, there still is hostility. I'll give you an example. Most states in the United States have anti-usury laws. And what do they mean by that? You can't charge excess interest. But what the hell does excess interest mean?
- 3.115** If you want a loan from me, and I want to charge you 35% annually, and you say, "I can't get a loan anywhere else, and I really need the money, and yeah, I'm willing to do it," and we shake on it or sign a contract on it, what business is it of the state to come in and say, "No, no, you're not. You're charging him excess interest, and we're going to put you in jail for that." So the remnant of the antagonism towards usury, towards interest, is still there.
- 3.116** I'll give you another example. We hate financiers. We still don't like financiers as a culture. It's not theological, but culturally. We blame every problem we ever have on financiers. Who caused the Great Depression? Wall Street. No. No economist believes that, but we all believe it, and we teach it in high school. Who caused the great financial crisis, 2008? Wall Street. Bankers. It's always bankers. No, no economist believes that. But who cares? So we still have the remnants of that theology still in the culture.
- 3.117** I think the anti-Semitism, the connecting of Jews with bankers and hating on Jews and bankers, all of that is in our culture. Whether we like it or not, it's there. And that all comes from that history and that theology, like it or not. So we tolerate, I'd say we tolerate finance, we tolerate interest, we accept it because we need it. We couldn't survive without it. Like, without banking, our civilization would collapse. But I don't get

a sense in our culture that we like it or we embrace it or we endorse it. We tolerate it. I think that's a theological remnant, if you will.

- 3.118** Thanks for helping me clear that. Get that out properly.
- 3.119** You mentioned that not growing the money supply is part of the things that leads to stability and financial caution, and with the government. Do you think if through maybe a constitutional amendment, they capped the amount of fiat currency they were able to create, that would lead to more financial responsibility or just more taking out more debt?
- 3.120** As we'll see when we talk about fiat currency, there's a lot to talk, there's a lot to unpack there. It's not even obvious today what you mean by money and what you mean by capping the amount of money growth. Is it M1? Is it M2? Is it M3? Is it MZ? Is it M0? Each one is a different measure of money that's out there. Is stablecoin money? If stablecoin is money, does that mean government debt is money? Because that's what all stablecoin is. It's just a fed off of—it just is, fiat money and the modern financial world has created such complexity, just in terms of even thinking about it, that it is challenging.
- 3.121** But look, there's no question that if you stabilized the money supply, whatever that happens to be, however we want to define money, if you stabilized the money supply, that has significant benefits to the economy. Would it stop the government from spending? No, not in and of itself, because they can now borrow as much as they want. Now, if you could guarantee they would never inflate, that would be interesting. If you could guarantee they would never inflate, then the risk premium on government debt would be default risk.
- 3.122** So right now, again, we'll talk about this, right now, when the government sells debt, you basically, again, the interest on it is composed of three things. It's composed of real rate, risk, and inflation. Today, risk is zero. Government can't default on its debt because it could print. So your main risk factor, if you will, is inflation. So really, government debt is priced as, in terms of interest, priced as the real rate, deferring consumption, and inflation.
- 3.123** Now, if you zero out inflation, what comes back? Risk. So you're trading off two different—how much of a difference would that make to interest rates? God, I don't know. It would be an interesting experiment. If we somehow could go like this and eliminate inflation from the equation, then people would have to start thinking about the U.S. government going bankrupt. And would that change the way they price U.S. government debt? And I don't know, because if they're rational, then the inflation is just as bad as the default, and it should be all priced anyway.
- 3.124** Now, maybe the inflation is softer somehow, so it's a lower price, and maybe we could get rid of that. Interest rates would go up a lot, and that would control government debt somewhat. That's possible. I mean, I'd have to think about it. But you see how you

trade those two? So today, we talk about government debt as riskless. Yeah, it's true. There's no risk. They'll always pay it back because they have a printing press.

- 3.125** Anybody who has a printing press can pay back their loans, but they pay back in, what did we call them? Debased currency, in currency that's worth less and less and less and less, because the more of it they issue, the less it's worth it. So it's a kind of a default. So if you think about it, Greece, you remember Greece in a financial crisis in 2013, something like that, 2012, in that period, Greece ran out of money. They couldn't pay back their debts. And because Greece doesn't have a central bank, it can't print money.
- 3.126** It's on the Euro. Only the European central bank can print money. So Greece actually has default risk—except, except if Germany steps in and bails them out. And now again, all your calculation is gone. So if I have to price Greek debt, I go, OK, real rate. Inflation is whatever the EU does, so that's all over Europe. And then default risk. But my default risk takes into account whether Germany can bail, or will Germany bail them out or not. And you can see how complicated this becomes.
- 3.127** So you sometimes look at Greek bonds and you go, Greek's a basket case. Why are they so cheap? Or why are they so expensive? Why is the interest rate so low? And you go, because everybody counts on Germany to bail them out. But take a country like, I don't know, trying to think of a country that—Argentina, before Milei. Argentina can print pesos, but as a consequence, nobody in the world is going to buy bonds, Argentinian bonds, denominated in pesos. So I'm only going to buy an Argentinian bond if it's denominated in dollars. Argentina cannot print dollars.
- 3.128** So what is the risk when I buy the Argentinian bond denominated in dollars? That they default. And how often have they defaulted on Argentinian bonds, the government, on Argentinian bonds denominated in dollars? Several times. It's a real risk. That's why interest rates in Argentina in dollars are so high. So it might work that that risk of bankruptcy is more real to people and will be priced more significantly than the risk of inflation. It's also zero-one, or closer to zero-one, you get some of your money back, not all of it, whereas inflation is more over time.
- 3.129** So you'd have to really think it through and model the two out and see if it would have an impact on it. But you're trading those two off. And you can see with Argentinian bonds, the ones in pesos and the ones in dollars, and how differently they trade.

Lecture 4: Fall of Gold

At a glance — The fall of the gold standard.

- 4.1 All right, so today we're going to talk about how the gold standard died. We're going to talk about how it died and why it died. But it died, right? So let's first recap what we mean by the gold standard, right? So the old system, the system that was probably in its purest form in the late 19th century in the United States, and in some form or another really was a global standard; most of Europe was on a gold standard during this period.
- 4.2 Think about it. People would go and mine for gold. You were part of the gold rush in California. You went out there, you bought your picks and your jeans, right? Levi jeans was basically established for the gold rush in that period of time. You know who made money during the gold rush? Just as an aside, like, who made money in the gold rush? Like, people who found gold?
- 4.3 No, people who sold the picks and axes and the jeans. They're the guys who made the money during the gold rush. The people who found the gold did OK, but they're not the ones who did really well. It's generally not a bad deal to sell the picks and shovels, or the equivalent, digitally, these days, to kind of the primary industry out there.
- 4.4 So anyway, you go out, you'd mine some gold, you'd find some gold dust, gold rocks or whatever it is. You would take that gold, typically; you could take that to a store, a local store, and buy the picks and ax, and picks and shovels, right? And they would weigh it and they would figure out how much gold there was. And then the store would then take that gold, maybe to a mint. And there were private mints in California where they would melt down the gold and give you back coins.
- 4.5 He should buy a private mint in California. And the store owner could then take those coins and put it in the bank. He might have even taken just the gold dust and put it in the bank, and the bank then dealt with minting. But in the end, the gold lands up in a bank, and the bank issues you currency. It issues you pieces of paper that are redeemable in gold. So you get these certificates, and at the end of the day, it's these certificates, redeemable in gold at the bank, which are used for commerce.
- 4.6 That is what we use as money, even though they represent what you would consider real money, which is the gold held at these banks. In the United States, at least, there was no central bank. The banks didn't do anything with the gold beyond that. Again, they lent a portion of it out. So they had fractional reserve, but that was controlled. There were limitations on that that were generated by market forces, by competition, by interest rates, by the riskiness of the bank, by the need of the bank to keep enough reserves to be able to deal with the regular flow of people redeeming gold.

- 4.7 But think about it. People didn't redeem gold that often. Right? But so where's the outflow from the bank? How does gold leave the bank? It leaves the bank when I take my certificates that bank A has issued me, and I go to bank, and I go to a merchant and use it to buy stuff. And then that merchant puts these certificates in bank B, and then bank B demands that bank A provide the gold that I had put in, in order to cover these certificates. So gold is moving around between the banks.
- 4.8 But it's very rare for individuals to actually go and collect the gold, unless they think the bank is shady or won't survive, and they want the actual asset. So that's kind of generally how the banking system, the gold, worked on a domestic basis. As the demand for money grew, because the economy's growing, as the economy expands and grows, demand for money grows; and as a consequence, the purchasing power increases.
- 4.9 So you only have certain amount of gold right now, but there's a lot of production going on. The economy is growing, so there are more goods to buy. You have a fixed, let's say right now it's fixed, we have 10 units of gold. But now the number of goods has increased from 100 to 110 goods. Well, the only way for that to adjust is for prices to go down, which means the purchasing power of gold goes up, which means the value, in a sense, of gold has gone up, which provides all of you an incentive to rush into the mountains and go find gold.
- 4.10 So it encourages finding more gold. It incentivizes more mining. Of course, as all of this is happening, as the economy grows, technology improves; mining technology improves, so we can extract more gold, given the technology. And what you get is a constant balance, right? If demand for money goes down, prices go up. In other words, the value of gold declines compared to prices.
- 4.11 Well, the incentive then to mine is reduced, because the value of what you're bringing out of the ground is less. So you mine less, and prices will adjust. So what you get is this constant rebalancing of prices, the purchasing power of gold, as determined by mining, the incentive to mine, the technology of mining, the cost of mining. Remember, you're only going to mine gold to the extent that the gold you extract is going to be profitable. That is, it's more valuable than the effort you put in in order to mine it.
- 4.12 So, one of the real beauties of the gold standard is this self-adjusted, self-correcting mechanism that keeps happening. And it's not self-correcting, self-adjusting based on some decisions somebody makes. It's just the way the market functions. It's all done through the price mechanism, right. Price, the value of gold, purchasing power of gold goes up, more gold comes into the market to bring it back down.
- 4.13 And we saw, we talked about last time, we talked about inflation, that is, the purchasing power of gold over long periods of time. Purchasing power of money over long periods of time was unbelievably stable during this period. There was really no inflation; there was no consistent, steady rise of prices, in particular as we'll see,

compared to the world we live in today, where the Fed is targeting 2% of inflation. So we've got 2% inflation built in.

- 4.14 So if you have a hundred dollar bill a year from now, if the Fed has its way, it's only worth \$98 and, a year after that, it's 2% less. So your \$100 is constantly declining in value if you hold it in your pocket. So a gold standard, yeah, sometimes it's worth a little bit more, sometimes it's worth a little bit less. But generally it's self-correcting over time. Now this also works internationally.
- 4.15 So, on an international scale, if during a particular period of time, let's say you're exporting a lot and you're importing a little—this is relevant to kind of balance of trade that people talk about today—you're exporting a lot, you're importing a little. So what does that mean? To export stuff, you're sending stuff overseas; what do you get in return? Well, gold. Gold is money. So you're getting gold in return. So the amount of gold in the U.S. goes up.
- 4.16 If the amount of gold goes up, what happens to prices? They will go up a little bit so the purchasing power of the gold will go down because there's more gold coming in. Prices will go up and, as a consequence, imports will now increase, and you get an adjustment of the relative price of gold. So if it's just import-exports, what you'd expect is, over time, those each balance out as gold flows in different directions. Gold comes in, prices go up, which causes gold to flow out, prices come down again, and you get these international trade balancing.
- 4.17 Now you still might have excess gold coming into the country, because people might want to be investing in the country, but that doesn't raise prices. Because what happens when people invest in the United States? Let's say they're sending gold to the U.S. not to buy stuff, but to invest. Well, that investment increases productivity, increases the number of goods to match the amount of gold that exists. So again, you get this constant balancing act between the amount of gold and prices.
- 4.18 And it's constantly adjusting naturally, without anybody guiding it, without managing trade or without managing the money supply. The money supply is constantly adjusting itself. Make sense? So if gold inflows to buy stuff, then prices go up. That disincentivizes people to buy stuff. Prices will then come down, and the other way around.
- 4.19 All right, the gold standard has this real equality that is independent of what, you know, really, of what the government does. In a true gold standard, it's banks are making decisions, investors are making decisions, consumers and producers are making decisions. And they're not dependent. Interest rates. How do interest rates set in a gold standard? Today we all wait for that meeting of the Fed Open Market Committee meeting. They meet eight times a year, and everybody's suspenseful.
- 4.20 And in the days of, I think it was Alan Greenspan, if the briefcase he took into the meeting was thick, then interest rates were going up. If it was thin, interest rates going down. I mean, there are all these speculations about what they would decide and

where interest rates were heading. But in a market with a gold standard, interest rates are not determined by government, it's not determined by a central bank, it's not determined by any central authority. Interest rates are basically determined by supply and demand.

- 4.21** Supply and demand for what? What is an interest rate, after all? Interest is kind of a price of money and a price on time. So it's a price for money over time. So a one-year interest rate, a 20-year interest rate, it's the price of, as we said, delaying consumption over that period of time. And what's going to determine what that is? Because yesterday we said, or last class we said, interest rates is composed of a real rate and an inflation premium. No inflation premium in a gold standard. So default risk, definitely default risk in a gold standard.
- 4.22** But people are going to have a different estimate of what their delayed consumption is going to be. We don't all agree on how much I need to be compensated in order to delay my consumption of ice cream for a year. Some of us might really value getting it now. Some of us might value it a little less. Some of us might have to be enticed to forego the consumption today or forego another investment today. Some people might not have to be that enticed.
- 4.23** So what actually is going to determine that interest rate is, in aggregate, how much we are willing to supply at a, you know, given, at a given rate; how much other people are willing to demand. That is, it's a supply and demand for loans at any given point in time. And if interest rates are too low, and I need a greater compensation or to give up my money, I don't give it up. So I don't participate in the market. So ultimately, supply and demand will find a price, and that price will be the interest rate.
- 4.24** But it's again a market phenomenon. And the interest rate for one year and the interest rate for five years and the interest rate 10 years are going to be somewhat different, but not hugely so, because the differences in inflation, the monetary unit is exactly the same today as it is in 10 years. That doesn't change. So it's just a question of, is delaying consumption over the long period of time different than delaying consumption over a shorter period of time? Remember, you're getting interest every single year.
- 4.25** And then, is there different risk levels over time? So interest rates are determined by the supply and demand for loanable funds. And who is demanding loans? Well, it's businesses, primarily. There's some consumption. People are borrowing money for consumption. Mortgages, car loans, things like that. And then of course, there's government. And government now has to, again, it is constrained by this supply and demand. It's going to offer its bonds.
- 4.26** And now it's a question of what people assess the default risk is going to be in government; typically, U.S. government, pretty small default risk. So it's really just going to be the real rate of return—what, at any given point in time, that real rate of return

tends to be. Real rate of returns, the amount of money the market typically demands for delayed consumption over a year is somewhere between 1 to 2%, historically.

4.27 So that's kind of an interest rate, where you start out, you start with 1 to 2%. And think about it. If your money holds constant, 1 to 2% is not great. You're not going to become rich over it, but it's OK. You're getting a return, a real return on your money.

4.28 All right, so what happens? So, as we said, between 1879 and 1914, you basically have a free market and a classical gold standard that is international. But already in countries like the UK and most of Europe, central banks are being established. And of course, the central bank in London was established in the 1690s, so it's got a long heritage. But the central bank keeps gaining more and more power. And part of that power is to aggregate the reserves, the gold, at the central bank.

4.29 So, gold all moves into the central bank and, at the same time, so banks, the banks out there that are working day to day, they don't actually hold gold. What they hold are central bank deposit slips. That's their assets. And they can, theoretically, they can go to central bank and demand the gold, and then give the gold to their customers when the customers redeem gold, but they don't hold it; now the central bank holds it. And now the central bank gets to decide how much money, paper money, how many claims on this gold, to issue.

4.30 We talked about this last time, that in a fractional reserve system of competing banks, there are lots of mechanisms to control how much fractional reserve is done, how much more redemption certificates are issued relative to the amount of gold they have in the vault. A central bank doesn't have anything to constrain it. So that's, remember that.

4.31 Second, it's important to note that even in the U.S., the system was far from really free and really a full-on gold standard. During the Civil War, the United States kind of went off a gold standard, issued a paper currency called the greenbacks, and then it had to kind of figure out what to do with those greenbacks after the war had ended up. And it had learned from the war of independence, when the money that was issued there was basically hyperinflated and people were very upset and very unhappy by that.

4.32 And the greenbacks, they dealt with a lot more responsibly and slowly took them out of the system without allowing for hyperinflation. But in the meantime, they suspended redeemability of banknotes into gold and they had to reinstate a gold standard in the United States. And that's 1874, is when they reinstate the gold standard. But at that point, they do a couple of things.

4.33 First, they make it illegal for state banks, banks that are chartered by states, which are most of the banks in the U.S. by a long shot, they make it impossible for those state banks to actually issue certificates of redeemability, in other words, issue banknotes. They can't issue banknotes themselves. The only entities that can issue banknotes are newly chartered national banks, so banks that have a charter; these are not national

banks in the sense that they have branches across the country. Those don't exist until the 1990s, so until 100 years later.

- 4.34 No, these are just, they're chartered by the federal authorities instead of, and regulated by, the federal authorities instead of by the state authorities. These banks can issue banknotes. These are the banknotes that circulate in the entire system. Of course that leads also—it means that they are also the ones who are actually redeeming the gold; they're actually holding the gold, because these banknotes are ultimately a claim against their gold. It gives these banks a lot more centrality in the system.
- 4.35 These banks also can only, in order to issue banknotes, have to also have a certain percentage of their assets be government bonds. So one of the ways the government is trying to get rid of its, trying to manage, call it, manage its debts during this period, the debts that it took on during war, because historically, that's again the main expenditure of government, the big expenditure of government was war, hadn't invented the welfare state yet.
- 4.36 And so it managed that through having the banks take on that liability and issue banknotes, based on the amount of the government debt that they held on their books. This landed up constraining the banking system, making it more complicated. What we described earlier, the simple, you know, gold standard and how it works and how it rebalances, this threw a wrench in there in all kinds of subtle ways that created crises that were then taken advantage of. We'll get to some of those in a little bit.
- 4.37 So, importantly, in Europe, gold is held by central banks who then are issuing these certificates, paper money. And of course, in 1914, World War I breaks out. And really, right off the bat, the Austrian-Hungarian empire, France, Germany, Russia, all leave the gold standard. Now why do they do that? What is it that the gold standard is restraining them from doing? Spending.
- 4.38 Well, it doesn't have to restrain them from spending because they could just tax their people, right? There's a war going on. So they could raise taxes significantly, take that revenue, and use that revenue for the sake of building up the war infrastructure and paying the soldiers and fighting the wars. But that is costly. It's costly politically. The people might turn against the war. It's one thing just to send men into battle, it's another thing also to tax their families.
- 4.39 Also remember that during war, and this is true of pretty much any war, economic activity goes down. Yes, you're manufacturing arms, but productive economic activity, economic activity that actually enhances human life, that actually is productive, goes down. And indeed, the quality of life, standard of living of people, go down—and now you're going to tax them more? I mean, there'd be a rebellion.
- 4.40 And particularly in World War I, where you could argue nobody actually knew why they were fighting the war, particularly soldiers in the trenches, what was this war about exactly? Nobody ever quite knew. So it wasn't a huge motivation, I think, among

particularly soldiers. Taxes were out of the question. So, of course, the way to fund the war was to print money. And to print money, you had to go off the gold standard. You have to make gold, the paper money, irredeemable.

- 4.41 Now, Britain does something very similar. It blocks redemption by the Bank of England and it blocks export of gold out of the country. In spite of this, gold flows out of the country, primarily to the United States. So gold reserves in the U.S. go way up as a result of the inflow from the United Kingdom. Generally, Europe views the United States during this period as a safe haven. U.S. hasn't entered the war yet. So a lot of gold is flowing to the U.S., not for any real productive purposes other than to shield it.
- 4.42 You don't know if you're going to get invaded; the gold gets stolen. It's faraway, it's protected. That way, you trust Americans to respect your, call it, property rights over it. So, gold is leaving Europe, money is being printed, prices are going up, because now we have more paper money chasing fewer goods, arguably, right, because you're destroying supply, you're destroying production, and inflation is picking up.
- 4.43 Now, the United States enters the war in 1917, and indeed the United States at this point bars the export of gold. So it doesn't want the gold leaving the United States; and the U.S., by this point, we have a Federal Reserve that is printing, you know, banknotes. And the U.S. experiences significant inflation during World War I. So you've got two factors here that are inflationary.
- 4.44 One is all the gold coming in, which is not really matched by anything going out. And second, you've got the Federal Reserve incentivized now to print money, in order to fund its own war effort. And there was a real effort during World War I to run everything very centralized and for the government to run the war effort in terms of war production and everything. So, it was very much centrally planned, as wars almost always are.
- 4.45 And indeed, by, from 1917, I think, for 1920, price level almost doubles. So you get real inflation in the U.S. Maybe that's from 1914 until 1920. Was funded by paper money, unlinked to gold throughout the world. Redeemability is out. And then the question is, what happens at the end of the war? So Britain is faced with this conundrum. It wants to go back to the gold standard, but there's a lot of paper and very little gold, because they printed so much more than actual gold. How do they go back?
- 4.46 By 1920, the U.S. is the only country in the world that has a semi-functioning gold standard. So it did not abandon redeemability and it continued to function. But the rest of Europe is struggling with how to manage kind of the post-war correction. So during this period, what you get is countries, particularly the UK is probably the dominant one, but the UK this time, this is the, the end of, you know, the British empire's heyday, but they are the second largest economy in the world after the United States at this point.
- 4.47 They have a thriving trade throughout the empire. They're still maybe not the dominant player anymore. The United States has become the dominant player. Up until this

point, the British pound was, in a sense, the currency around the world backed by gold. And so anything Britain did had disproportionate impact on everything else globally. It was the reserve currency. Today we talk a lot about the dollar being the reserve currency of the world. Well, in those days, the British pound was the reserve currency in the world.

- 4.48 It made sense, because Britain sat at the hub of international trade, primarily because of its empire. So the UK has too much paper money. So how do you reinstate convertibility? Now they could have devalued gold. They could have said, "Yeah, we printed a lot of money. So now it's going to take more of these paper notes to buy the same amount of gold as before." But they didn't want to do that. They wanted to go back to the same convertibility they had before the war.
- 4.49 So to do that they'd have to suck out all this, half of the paper notes that they'd printed, because there were too many paper notes. But that would have been, they feared that that would be disruptive to the economy, and that might be deflationary, and it might spiral the economy into a recession. So they tried to have their cake and eat it, too. And you know, when evaluating the legacy of Churchill, this is one of the big mistakes Churchill made, is, he was responsible for the decision to return to convertibility, return to the gold standard, at the old convertible ratio, even though there were way too many banknotes circulating.
- 4.50 Now this created a bunch of distortions in the market. Gold left the UK because it was undervalued, because there was so much, there were too many pounds to actually, if everybody redeemed, there wasn't enough gold. So gold was actually more valuable than represented by the conversion ratio. So nobody wanted to keep their gold in the UK. And it actually had ripple effects on the United States, on Europe, but primarily in the United States.
- 4.51 And generally, I'd say, if you look at the Great Depression and you look what happened in the Great Depression, there's a real sense in which the Great Depression is the price paid for the misbehavior in World War I. And really, the 20s are just a bridging period. But at some point, you had to pay for that inflation. And the way you paid for that inflation is massive deflation, which is what happened during the Great Depression.
- 4.52 Now, the Great Depression could have been avoided, but a lot of that avoidance would have required devaluation of gold, which the British weren't willing to do and the Americans weren't willing to do until 1933, I think, or '34, so when you get the devaluation of the ratio between gold and the dollar. And as a consequence, to a large extent, could argue, the consequence is the Great Depression.
- 4.53 So, if you will, what kills the gold standard at this point is its mismanagement by central banks. What kills the gold standard is the fact that central banks refuse to be disciplined by its mechanisms. They want to spend; they've got good stuff to spend it on. There's a war going on. And then not making the tough, tough politically decisions

on the other side of the war to fix it, to get everything back in balance, creates massive distortions and, ultimately, economic catastrophe.

- 4.54 And of course, note that a lot of people to this day blame the Great Depression on the existence of the gold standard. But it's the misapplication of the gold standard, it's the cheating on the gold standard, it's the not being willing to be disciplined by the gold standard, if anything, is what caused the Great Depression. So they've got it upside down. And a lot of the hostility today, a lot of hostility that followed after the Great Depression, to a gold standard was this misunderstanding of what actually happened, and what was cause and what was effect.
- 4.55 Governments get off the gold standard then, when they want to print more money than the standard will allow. They want to escape the constraint. They want to avoid raising taxes. And again, you can see this over and over again. I mean, it certainly was true in the American Civil War. Government engaged in a war. And again, I'm including here wars that are justified. They can't figure out how to pay for it through taxation, so they do away with redeemability. They print greenbacks in the case of the United States, and they go to fiat money.
- 4.56 And then they might, if it's a decent government and they think this through and they're more thoughtful, they might find a way back from that to get back on a healthy gold standard. But that requires a lot of government officials, particularly in a period where, arguably, the economic understanding is not that deep. There haven't been a lot of economists at this point.
- 4.57 We're talking about late 19th century. Let's say, you've got Adam Smith, you've got other economists during the 19th century. But there's no monetary, there's no deep understanding of monetary theory. There's no real analysis of this. I mean, I think that's an excuse; at the end, I think it's mainly just the irresponsibility of our political class and those running things. But it's complicated; it's not trivial and it's not simple.
- 4.58 And remember, there are massive political pressures here. There are real political pressures on trying to do things, trying to do things differently. So as we'll see when we ultimately completely go off the gold standard, it's government spending which is driving it. It's government spending and the inability to basically raise enough tax revenue to cover that government spending, or the unwillingness to do so because of the political costs. So politics drives this.
- 4.59 Now you have to also say that, for some politicians, certainly going off the gold standard also involved this idea of, we want more control over the economy. We want to be able to control things. Right now, everything adjusts automatically. We don't get a say. And there was this idea that if we give economists, experts, control over the supply of money, we could smooth things out. We could get rid of the business cycle.
- 4.60 Business cycle is, sometimes you get economic growth, other times you get economic slowdown, maybe even recessions, maybe even depressions, and then economic

growth again. And again, there wasn't a lot of understanding on how this happens and why it happens. But maybe, and this is Keynes' contribution, of course, to economics, is his idea is that we can smooth this out.

- 4.61 In times of economic growth, we can save money, the government can save money. In times of recession, we can spend money and even borrow money, deficit spending, in order to cover the difference. So we smooth it out, and then once things recover nicely, then we pay back all the debt and we do this nicely. It never quite works out like that. Once you unleash the government's ability to do all that, they never quite ever pay it back, it seems.
- 4.62 But that was the idea. The idea was things are cyclical, will be countercyclical, to offset the cyclical. And if we can be countercyclical, we can smooth things out. Now gold does that. But their perception, and again, primarily the business cycle, particularly in the United States, was always a consequence of the fact that we didn't have a real gold standard. It was always being manipulated. There were always being controls. The government always wanted to tread in it; they always wanted to play around with it.
- 4.63 So there wasn't a real understanding of what was causing this. But if we can give it to experts, then they will get it right. This is a period in the early 20th century where there's a lot of belief in what's called the science, managerial science. We can scientifically manage things. Look at these big businesses; they are scientifically managing steel, railroads, I don't know, oil. And we could do even better by bringing in principles of accounting and finance into these businesses and making them even better.
- 4.64 Well, why can't we do that with government, and why can't we do that with the whole economy, right? If we can do it with a business, and the business is pretty big, why can't we do it with the whole government? And now we can bring in these ideas of scientific management into government and money and the economy, and centrally plan much of this.
- 4.65 Much of the New Deal, FDR's New Deal, was motivated by this idea that we can do a good job, we can do a better job, we're scientists; and we'll manage it all, and we'll smooth this out, and we'll get us out of Great Depression, and things will be better. Didn't quite work out like that, but that was the idea. The Progressive movement was a movement that dominated intellectual life in America in the late 19th century, early 20th century. And it was convinced around this scientific management of the economy, that scientific management of the economy was possible.
- 4.66 It was part, we'll get to, part of the motivation to establish a Federal Reserve was to bring this scientific management to the idea of money, into regulating and managing money. And ultimately, they wanted this across the board in the economy. So, the gold standard is slowly abandoned, and it's always abandoned unilaterally by government. That is, you never find a market, a market that's just left alone on a gold standard or

some other kind of solid standard, that abandons it and goes to a non-standard monetary system, a monetary system based on paper.

- 4.67** It just doesn't happen as an evolutionary process. It's only going to happen as a result of top-down dictate by a government. This is what we're going to do, this is how it's going to function. This is kind of, there's a sense in which, a real sense in which this is sad; and it's sad because the gold standard was good. We talked about its self-correcting nature. We also talked about, we talked about last time about, kind of gold is almost an ideal money in a sense of, it's the scarcest money. You only get about 1.5%, 2%, inflow of new money every year.
- 4.68** The economy, a good economy in real terms, is probably only growing at about 2% a year in real terms. So you get a nice match between the amount of money coming in and the growth of the economy, so that you don't get inflationary pressures. Gold has a utility, so its value is not completely determined by its use of money. It's also used as predominantly as jewelry.
- 4.69** And the value is determined in free exchange. It's not dictated from above. And it seems all the values in the economy, all the economic values in the economy, are determined by free exchange. They're determined by individuals' supply and demand; that means individuals' valuing of things. So it's a system that embraces the idea of freedom, the idea of individualism, the idea of individuals deciding what they will buy, when they will buy, how they will buy it, with a unit of exchange.
- 4.70** That's what money is, right? A unit of exchange that keeps its value. They don't have to rush to sell it, right? They don't have to rush to use it to buy something because they're afraid it's going to diminish in value. You know what happens when you get hyperinflation? You spend money immediately, because tomorrow it's worth less, a lot less.
- 4.71** So if you live in a place that has, I don't know, 1000% inflation or several hundred percent inflation, you get your paycheck, and ideally you get paid every day or at least once a week, not once a month. And you immediately go out to the store and you buy stuff. So people tend to have three refrigerators, not because they need three refrigerators, because the refrigerator is going to maintain its value more than your currency, than the money. So you can later sell the refrigerator after the inflation's gone, the money's gone; so, stable.
- 4.72** And while that's hyperinflation, with a little bit of inflation, we're still all tempted to do that again. \$100 in my pocket, if I keep it there for a year, it's only worth 98. So I'm not going to keep it in my pocket, even if that would be my preference. I'd like to keep some money at home, just for day-to-day stuff and just in case. Why would I keep it now? So it changes the way people behave. The gold standard allows us to behave, in a sense, optimally, to make economic decisions optimally because the value of our money is constant.

- 4.73 And of course we've talked about the self-stabilizing, self-stabilizing aspect of it. Gold allows us to preserve our wealth at no risk. Today if you want to preserve your wealth, you have to find an investment for that hundred dollars that earns at least 2%. And there's always some risk when you make such an investment. If you buy a government bond, maybe there's no default risk. But what happens if inflation lands up being 4%, not 2%? Then you still lose 2%. Or if you take on, if you buy stocks, stocks are risky; they're very volatile. They could go down just as much as they could go up.
- 4.74 So today, in order to preserve your wealth, in order for money to be a store of value, you have to invest it. You have to use an intermediary to actually do something with the money to get you a return, to just get you back to where you started. Just to get you back to where inflation has taken you, away from you, get that inflation, if you will, back. Gold, you could just bury in the garden. I mean, in a gold standard, right? And it won't make money, it's not going to grow anything, right?
- 4.75 But if you bury it, it's going to be worth approximately, in a gold standard, not today, but in a gold standard, approximately the same as when you buried it. You preserved your wealth. It's a store value, you've stored the value, and it kept up with that value. So to a large extent, losing that is losing a lot. It's losing a lot. So let's look a little bit at kind of how this happened, and then we're going to talk about, in more detail, we'll talk about in next class kind of the, how fiat money system works, and what the consequences of fiat money system are.
- 4.76 And we'll probably spend, you know, probably two classes on that, and talk about how the money system works today. How does, how, how does money get created today? What happens today? How do interest rates get set today? What is it that the Fed actually does? I mean, as we'll see, it's super complicated, as compared to the gold standard, which was really pretty simple and pretty straightforward.
- 4.77 I mean, really, all that's happened is, as we centralize, centralization and control ultimately are going to kill something like a gold standard, which is decentralized and uncontrolled. So the fate of the gold standard was really set as soon as you established a central bank. As soon as the central bank of England was established, 1694, it was just a matter of time. Slowly, legislation gave the central bank more power and more power and more power, and it kept cheating. And sometimes, as we said, the cheating was fixed and sometimes it wasn't.
- 4.78 During Napoleonic Wars, it eliminated redeemability, printed money, in order to fund the war, then had to somehow find a way to get back onto redeemable currency. In those days, the economy was really speeding ahead. This is the early part of the 19th century. UK was experiencing the beginnings of the industrial revolution, and it was the empire in the world. It could afford, kind of in a sense, to play around with these things and survive. By the time we get into the 1920s, not so much. It's not in a good state where it can afford to make mistakes in terms of monetary policy.

- 4.79 By 1913, the Fed is created, and by the early 1930s, the Fed has basically taken over the banking system. So between 1913 and 1930, banks still have a role. There's still nominally a gold standard, but as we'll see, by the early 1930s, by the Great Depression, that's all out the window; it's gone. So just the entry of a central bank, the giving of that kind of centralized power, that kind of power never shrinks. It always expands.
- 4.80 And the reason to establish a central bank is to be able to smooth out the business cycle, to make banks safer, whatever; and therefore, means more and more and more power to be able to achieve those things, particularly when you fail initially, which the central banks certainly did. Central bank notes become the dominant notes in the economy, even if there are parallel systems of banks issuing notes. Central banks, again, can issue notes. They're not constrained by any market forces, so they can issue large amounts of notes.
- 4.81 And ultimately, government suspends their redemption. And as we've described, things go off the rails from there. So let's look a little bit more closely at the Federal Reserve in the United States, its creation, and then kind of how it evolves, particularly in those early years. So in 1907 in the U.S., there's a major financial crisis.
- 4.82 The financial crisis is primarily caused by the structure of American banking, by the fact that American banks are not diversified across state lines, that you have little farm banks in different places, and then you have regional banks that don't have branches in these other places but sitting in the region. Then you have these larger New York banks. But no bank has branches in more than one state because that is not allowed.
- 4.83 And it's also caused by a shortage of currency, a shortage of paper money, because of this idea that the money has to be backed by these government bonds; and the government bonds are being paid off, and there are not a lot of them. So there's a shortage of currency. There's a problem of diversification. Banks are collapsing; banks in New York are collapsing because the banks that they have deposits of from other regions in the country are collapsing. There's like a domino effect.
- 4.84 And basically what happens, ultimately, is J.P. Morgan, the legendary banker of the period, gets a bunch of the big New York bankers into a room. Legend has it at least, he locks the door and he basically said, "We're going to figure this out, and we're not leaving this room until we figure this out." And they basically figure it out. They decide which banks are going to go bankrupt. They decide which banks they're going to save. They decide how it's all going to be structured after the fact and who owns what, and they do it.
- 4.85 And they basically save the U.S. economy from a devastating potential depression, certainly a sharp recession, and the economy recovers. But the consequence of this, politically, are pretty harsh. I mean, suddenly, people in Washington, D.C., don't like the fact that a private banker has as much power as it turns out J.P. Morgan has. He has

just single-handedly saved the U.S. economy. And a lot of people view him as a hero, but a lot of people are also looking at him with suspicion.

- 4.86 We talked about in previous classes, about the fact that we generally don't like bankers. We view them with a lot of suspicion. And J.P. Morgan was the most powerful of all bankers, and we certainly didn't like the idea that he had the kind of power that could control, in a sense, the fates of the nation. That's the job of politicians, not of business people. So you start having hearings after this crisis in Congress. J.P. Morgan is brought in front of them; he's attacked viciously by Congress.
- 4.87 Might be familiar to those of you who might have watched some of the hearings where businessmen come in front of Congress over the last 10, 15 years or during the financial crisis. And basically, the outcome of this is the decision by Congress to establish a new institution called the Federal Reserve that would take on, basically, what J.P. Morgan did. It would be now responsible for getting us out of these financial crises.
- 4.88 It would be now responsible for making sure that there was enough money in the system, because part of the problem was, of course, there was not enough money in the system in 1907. It would be responsible for making sure there's enough money in the system. They could have done away with the restrictions on bank branching. They could have created a national system of banking.
- 4.89 They could have done away with the requirement of banks to hold government securities in order to issue currency and taken us onto a pure gold standard. They could have, in other words, liberated banks and turned the United States into a true gold standard. By the way, at the same time as this is happening, Canada has a gold standard, has a banking system that doesn't have the constraints that the United States has, and has no 1907 crisis, and does fine.
- 4.90 And indeed, over the history of banking in the United States, we've had 13, 14 banking crises. Canada's had exactly zero. So, the structure matters. And instead of moving towards the structure that seems to be working, they experimented with something new, or something that they took from Europe, which is more central control, more central planning, and the establishment of a Federal Reserve.
- 4.91 And the establishment of a Federal Reserve is, even here, it's one big compromise, political compromise. On the one hand, this is a government institution that they want to structure. On the other hand, their interest to say, this is America. We shouldn't have these big government institutions. We don't need a central bank. So they kind of make it pseudo-private, so it's like a public-private partnership, the Federal Reserve. It's not really; it's really a government institution, but it's pretend private.
- 4.92 So, the Federal Reserve technically is owned by commercial banks. They even get dividends. But all the profits of the Federal Reserve go to the treasury. And of course, it's government that appoints the governing body of the Federal Reserve. So from 1913 on, the United States has a central bank gold standard. The banks no longer are now

holding gold. They are shifting that gold over time to the Federal Reserve. Federal Reserve is issuing notes, our dollar bills, issued by the central bank rather than by banks, rather by national banks.

- 4.93 Gold gets sucked into the Federal Reserve over time, and we are all left with the paper currency, the dollar bills that we have, similar to the ones we have today. The central bank holds two assets, gold that it got from the other banks, and what's the other asset? Treasury bills. So it's, of course, funding the government. Why do we need a central bank, really, at the end of the day? Why was there—I mean, the good thing about the Bank of England is, they were honest about it when they founded it. It was to provide cheap loans to the government.
- 4.94 Well, that's exactly what the Fed was there for. So the main asset that it has are cheap loans to the government. It issues banknotes and buys government securities. So those are the two assets the Federal Reserve had in those days. And banks then hold reserves at the Fed that are redeemable into gold. So the banks can still get the gold from the Fed and then give it to the customers, as we described earlier.
- 4.95 Now, we're not going to get into all the details of the Great Depression, because that's a whole course in and of itself, but it's pretty common knowledge today among economists that the Great Depression was caused by the mismanagement of money by the Federal Reserve. For example, there was a period in the very early 1930s where gold reserves, the amount of gold at the Federal Reserves, is actually increasing. And they could have issued more currency as a consequence, and they didn't.
- 4.96 So generally, what you see in the early part of the Great Depression, starting in 1929, 1930, is a shrinkage of the amount of money in the economy. Massive deflation, prices plummeting, companies not being able to be profitable, laying people off. Unemployment reached 25%. A quarter of people employed in the United States were fired. I mean, that's unimaginable today. Today, we get jittery if unemployment goes to 7, 8%, as we should, 25%, I mean, it was crushing.
- 4.97 And much of that, and again, this is not, not controversial anymore, though not what they teach you in school—they teach you in school is still that the Great Depression was caused by 1929 stock market crash and by the gold standard, by capitalism or whatever. But the reality is that it was caused by this new institution, the Federal Reserve, not knowing what it was doing, and functioning under a variety of different monetary theories that had nothing to do with reality. They were just plain wrong.
- 4.98 And the consequence of them being wrong was a Great Depression, was a massive blow to the U.S. economy. So what you got was 1933, 1932, FDR gets elected. Banks, at that point in 1932, there's a massive run on the banks. People have lost all confidence in the banking system. Gold is flowing out of the banks, banks are shutting down. Hundreds of banks are going out of business, all over the United States. And FDR comes into office in 1933.

- 4.99** And the first thing he does is, declares a bank holiday to stabilize the banking system, to stop the bank runs. He basically takes the United States off of the gold standard, so banknotes become, stop being redeemable into gold. And he does more than that. He basically confiscates all the gold in the United States. So it becomes, people don't know this, but it becomes illegal for Americans to own gold. You literally can own certain small quantities of gold for jewelry. But if you have bars or coins or large jewelry, you have to take them to bank, get banknotes for them. And that jewelry gets sent to the treasury.
- 4.100** Just to make it clear that we're off the gold standard, the Federal Reserve no longer has any gold. All the gold gets shipped to the treasury. And that's what Fort Knox is for. It gets stored somewhere. And the Federal Reserve no longer carries gold as an asset. The only asset at the Federal Reserve is basically government debt. So, literally, 1933-34, all the gold flows from the American people, from American banks, from your wherever you have it.
- 4.101** Now it was hard to enforce. So I'm sure people kept some gold tucked away in different places. But it was illegal. You could go to jail. You'd go to jail, so this was real. Indeed, it doesn't become legal again for Americans to own gold until 1971. Think about that America. Pretty un-American thing to do, but it was done at the same time, or a little bit later, the dollar then is depreciated. Up until this point, a dollar is worth 22 ounces of gold. Or it costs \$22 to buy an ounce of gold. Now it costs \$35 to buy an ounce of gold.
- 4.102** And the U.S., for the most part, there's a tiny sliver of a gold standard still left. We'll talk about that next class. But for the most part, the United States is off the gold standard. And by this point, as we're heading into the 1930s and we're heading towards World War II, really, so is the rest of the world. The gold standard is dead. Its remnant, what's called the Bretton Woods system, which is a partial gold standard, we will talk about that next class.
- 4.103** With interest rates being determined by supply and demand purely, it seems like, unlike today, where there's an interest rate, it's documented, it's in stone, it seems like interest rates would be floating or variable by a great degree. Not that that's a problem, but it just seems like they're based on geography, need. There'd be a lot of variability in interest rates.
- 4.104** And that's absolutely right. But even today that's the case, right. So if you look at the 10-year bond rate, which is the government bond, every day it's different; within the day, it's constantly moving, right, because supply and demand is still determining the 10-year bond. And as we'll talk about, the way the Federal Reserve sets interest rates is by buying and selling. So it is in the market to manipulate. But that means that there's constant volatility that it's trying to smooth out, or it's trying to manipulate in some ways.

- 4.105** But for example, the 10-year, the 30-year bond are determined by the markets and they're constantly moving, they're constantly changing. And I would say even on a regional basis, I mean, again, we talk today about an interest rate, but there is no such thing. Because if I go to the bank and ask for a mortgage, is there going to be a variation between a mortgage rate in California and a mortgage rate in Texas and a mortgage rate? Maybe, maybe not today, because some of that would be arbitrated by Internet, like, mortgage facilitators.
- 4.106** But my guess is, before the Internet, banks in Texas might have given different mortgages than banks in California, given different in Nebraska. Suddenly risk is different in different regions. Like, you might have a declining house market, and banks are very reluctant to give mortgages there and might demand a higher risk premium because prices might go down, versus a booming market in which banks might be more eager to give out mortgages and might be willing to give it a lower interest rate.
- 4.107** So I think even today there's a regional aspect to this. Farm loans are probably different than industrial loans, different than home loans, are different than all of these. And the Fed, again we'll talk about what the Fed actually, which interest rate the Fed actually sets. But the Fed really, today, is only trying to control one industry. Now when it does QE, when it does quantitative easing, which it did in the 20-teens, it was trying to control more than one. But generally, it's trying to control right now one interest rate, and then once in a while it dabbles with some others.
- 4.108** All the other interest rates are floating and indeed determined by supply and demand of loanable funds in the marketplace. And exhibit everything you describe in terms of variability and can be different in different places. I know, you know, I live in Puerto Rico, and money costs differently in Puerto Rico than it does in mainland U.S. There's a Puerto Rican premium. We pay a little higher interest rates on the loans that we take out in Puerto Rico.
- 4.109** I'm confused on, in Keynesian economics, they say consumption drives the economic growth. And how do they respond to F.A. Hayek's knowledge problem, and the idea that we believe, for example, production drives economic growth?
- 4.110** You know, basically, the way they view it, particularly because they view it in terms of aggregates, is they look at the economy and they view the whole relation between producer and consumer the opposite of the way I typically view it, right. They view it as, we are consumers, we demand stuff. We demand stuff, and then businessmen go and they create the stuff that we demand. And if we demand a lot, businessmen will create a lot. If we demand a little, businessmen will create a little.
- 4.111** So you really want to monitor demand. That's the really key in the economy. And you have to figure out how to do that, how to, in a sense, manipulate demand, so that we're always demanding a lot, so the businessmen are producing a lot, so the economy's growing. And I think that is upside down. That is the opposite of the way it

actually, the world actually functions. But we're told constantly, 70% of the economy is consumption.

- 4.112** And therefore, if you want to control the economy, what you want to control is the incentive to consume. And money is one way in which you control the incentive to consume. You give people more money. Think about COVID, we'll talk about COVID later, but think about COVID. You grant people, everybody, money. They go and consume; they go buy stuff. And that, yeah, during the great, during the, during the COVID period, it actually does create some economic activity. It also creates some distortions and perversions, inflation, for example.
- 4.113** But the reality is that the way an economy works is, you have to produce in order to consume. So first of all, what does it mean to be a consumer? There is no such thing as being a consumer. You have to produce. You have to first make stuff. Think about a barter economy. You know, if you always, if you're confused about economics, always go back to a barter economy. For me to trade with, with you, I like the spears that you have. I want one of those cool spears that you've just created. How am I going to consume that spear?
- 4.114** The only way for me to consume that spear is to go produce something of value to you, and then trade. Nothing changes because you have money. I still have to produce something of value now to somebody. It doesn't have to be particularly to you, because now that I have money, I could produce something valuable to him. He gives me money, and I can then buy the spear from you. But nothing has changed. In order for me to be a consumer, I have to be a producer.
- 4.115** Now, we can deal with the welfare state as a separate issue, right? But fundamentally, somebody has to produce, in order for even a welfare recipient to consume. So, every act of consumption assumes that you as a consumer have already produced. So every act of consumption is really the flip side of an act of production. The only reason I have money is so I can consume, right? Or invest, but consume, right? But where do I get the money? By working.
- 4.116** And then of course, there's another act of production that's involved, and that is the production that the producer, you making the spear, so spear has to be produced for me to want it, and then I have to go produce in order to get it. So two acts of production to one act of consumption. And it's also, kind of reverses, at least to a large extent, reverses cause and effect in terms of why I consume what I consume, right?
- 4.117** It's not that I had a demand for an iPhone before there was an iPhone. I had no clue. I didn't want an iPhone, because I didn't know it existed. And even if somebody had told me, I would have said nah, you know, because I did this, right, when cell phones first came out. I said, "Nah, I don't want one of those. Who the hell needs a cell phone? I have a cell, have a phone at home, have a phone at work. Why do I need a phone for

the in-between? You know, it's ridiculous." It didn't take me long to figure it out and to have lots of cell phones, right?

- 4.118** But it's, you know, when McCaw cellular brought forward cell phones into the marketplace, he had to educate us about the value cell phones represented for us. So we became consumers, because the producer produced something that was of value to us. I didn't know I wanted that spear until I saw you had one, and it looked really cool, and I want one of those cool ones. And that created my need. Today everybody talks about need. My need is created by the fact that a producer has produced something that now I see that I really do want and I really do desire.
- 4.119** So, consumption is a response to production, and it's a consequence of production. It's production all the way. So the fundamental in economics should be, and very few economists, sadly, view it this way. So most of the economists you'll talk to don't even think in these terms. It's all about production. And if you want to think about how do we spur economic growth, well, why is production not increasing? What's wrong with production?
- 4.120** Because production means, if somebody's going out there and building a new factory and producing new stuff, it means they're hiring people. That's creating new consumers. Instead of giving people money, create new consumers by giving them jobs. And not giving them jobs, hiring them because they're going to produce and they're going to benefit whoever runs the business. So I don't know what their response to that argument is.
- 4.121** First of all, they see things in terms of, everything's in aggregate. I think, to really understand economics, you have to go to the particulars. You have to do the micro before you do the macro. So you have to understand how businesses work, what incentivizes businesses, how consumers and businesses interact. What is a consumer, the fact that he has to work, and all of that has to happen. This is why it's really useful to break down economic concepts to things like barter, to things like very simple.
- 4.122** A great way to think about economics is, how would it work on Robinson Crusoe's island? If I can understand how it works on Robinson Crusoe's island, I can then slowly start to expand my understanding to a modern economy. Modern economy, this is why central planning is impossible, is unbelievably complex and involves, today, billions of people and billions of people with different value systems, with different ideas, with different motivations, with different passions, with different interests.
- 4.123** Integrating all that, that's what we have, the beauty of the market and the price system. And if we try to centrally plan that, it's impossible because we can't get inside people's heads and evaluate what their values actually are, what they like and what they don't like, what they're willing to pay for and what they're not willing to pay for. We have to make estimates about that. We have to make assumptions about that, which are almost always going to be detached from reality.

- 4.124 So you mentioned the unemployment rate during the Great Depression was 25% and that we freak out now when it gets to 7 or 8. Didn't we change, the government change the way they reported the percentage? So isn't 7 or 8 actually like real inflation? Isn't it a little bit higher?
- 4.125 A little bit, but it's not that big of a difference. That is, the scale is completely different. The Great Depression, even as late as 1938, unemployment was 17, 18%. And that would, maybe that would be more, you know, that would be a lot more than 7, 8% we have today. Maybe it would be more like 10 to 12%, but we don't even reach that. When was the last time we had double-digit unemployment rates? Can't remember what it got to in the 1970s. It got pretty bad in the '70s, but nowhere near. It's a reason why that's a depression, and what we usually have are recessions.
- 4.126 I mean, in the 2008 Great Recession, not just a simple recession, a Great Recession, what did unemployment get to? 8%. So it's not in the same category, even though they do change the measures periodically; they change the inflation measures. It's not enough of a difference to change the nature of what was going on back there. I mean, it was the worst it's ever been in American history.
- 4.127 And as we'll talk about, one of the things central banks have actually learned is how to prevent that from happening. Depression's unlikely. We might get other stuff, but depression's unlikely because they figured out how to prevent that, at least that kind of depression. There might be other forms, and there might be other forms of decline that are not represented by a massive contraction of credit and contraction of money, which is what happened in the Great Depression, when the Fed could easily have expanded the amount of money available.
- 4.128 And if we had a gold standard, a flexible international gold standard at the time, that would have exactly been what would happen. Money would have expanded, banks would have expanded. But here, everything collapsed because the Federal Reserve screwed it up and allowed it all to collapse. And again, because they had printed up more money, they ultimately had to devalue the gold standard. You have to go off the gold standard in order to devalue gold. So it becomes kind of this cycle that keeps getting worse and worse and worse.

Lecture 5: Rise of Fiat

At a glance — The rise of fiat money.

- 5.1 Last class, we talked about the demise of the gold standard. And really, for practical reasons, in terms of the population, in terms of Americans, in terms of individuals, the gold standard is completely dead by the early 1930s. By 1933, 1934, the gold standard is finished. Redeemability is gone. The Federal Reserve is no longer constrained by gold in terms of issuing currency.
- 5.2 But there is a challenge, and that is that in the post World War II era, really, this is being thought of during World War II—what are we going to do in terms of international trade? And what are we going to do in terms of international capital flows? And how are we going to manage this process once the war is over? Because the gold standard was a very effective, very efficient tool to regulate naturally, through the market, the value of goods and, and imports and exports and money flows.
- 5.3 And there was a global currency, and the currency was gold. You could denominate it in dollars, you can denominate it in pound sterling, you denominate it in francs or any other currency, but it didn't really matter. The currency of the world was gold. And exchange rates between every other currency in the world that was on a gold standard were fixed by gold.
- 5.4 But now gold is gone. So what do we do? What do we replace this? How do we reestablish a trading system, a global financial system without any gold? There's a sense in which this had never been done before, certainly not in the era of the nation state.
- 5.5 So in 1944, a bunch of economists and politicians met together in Bretton Woods in New Hampshire—it's a nice resort out in New Hampshire—from all different countries, all the allied countries that were fighting in the war to try to figure out what is going to be the monetary system of the world for the purposes of international trade moving forward. And what came out of that is what's called the Bretton Woods monetary system.
- 5.6 And what was really decided during that conference was that the dollar would basically—the American dollar, the American money would basically become the world currency. That it would serve the purpose that gold once had served, that everything would be denominated in terms of gold. All other currencies would be fixed—sorry, in terms of dollars—so all other currencies would be fixed to the dollar. And that the dollar would be redeemable into gold at \$35 an ounce.
- 5.7 Now, not redeemable into gold for you and me. Individuals would still be completely out. All the individuals would have dollar bills, but foreign central banks, bank of England, the bank of France, they could redeem any dollars they had and they would

have dollars. If they, for example, exported stuff to America, they would send us our stuff and we would send them dollars. They could take those dollars and instead of buying American goods with them or making investment in America with them, they could send them to the Federal Reserve and demand gold, redeem them into gold.

- 5.8** And the understanding was that they wouldn't. The understanding was that there wouldn't be any reason to, that the dollars would now basically circulate to facilitate trade, but that everybody knew—and this would anchor the dollar as the reserve currency—everybody knew that if they needed to, if they wanted to, if America had misbehaved from a monetary perspective, if it devalued the dollar, they could always get their gold at \$35. And this was supposed to be a mechanism to discipline the Fed, to make the Fed honest on an international scale.
- 5.9** We can't discipline the Fed, that was taken away from us, that ability. We couldn't redeem our dollar bills for gold, but foreigners could, foreign banks, central banks, could. And this system basically functioned from 1944 or 45, the end of the war, through 1971. And it is a system that elements of it still remain today in the sense that the dollar is still the reserve currency of the world. So things are denominated in dollars. Things are not fixed to the dollar. Exchange rates are not fixed anymore; they're floating today. We'll talk about that. But in those days they were fixed. Every country had a fixed exchange rate.
- 5.10** Now, of course, this only worked, this would only work long term if the Fed behaved itself in a sense that it didn't print more money, more dollars than the gold that it had, if every country behaved itself and didn't print more of its particular units relative to the dollar, so that its exchange rate could stay fixed. Of course, none of that happened. Nobody behaved themselves and in particular, the Federal Reserve didn't behave themselves. Everything was fine, seemingly going into the early 1960s.
- 5.11** Really what happened after World War II in many respects is truly amazing. Government spending after World War II plummeted. The government stopped spending a huge amount of money because all the war efforts were gone. Indeed, most economists believe that right after World War II we would go into a Great Depression. We would re-enter the Depression because money had just been sucked out of the economy. The government was not spending it. Who was going to replace it. All those jobs would be lost. Everything would be gone.
- 5.12** And in addition, if you think about the complication here, money's going away to pay salaries, to start new factories, to do all that. And at the same time, millions of soldiers are coming back expecting jobs. The expectation was that that would generate a depression. There was a lot of angst and there was a lot of concern. The economists came up with all kinds of schemes to figure out ways to soften the blow. Luckily, nobody—the Truman administration and ultimately the Eisenhower administration paid no attention to all of that, and they just allowed it to happen.

- 5.13** What filled the gap, unsurprising, was private investment. Private investment came in and sucked up all those jobs, started producing. Now, a lot of that private investment was to build stuff that would facilitate products for the rest of the world. Remember, the rest of the world had just been blown to smithereens. There was no productive capacity in Europe. There was very little capacity, productive capacity in Asia. And the United States was the only place that hadn't been bombed and therefore still had its productive capacity.
- 5.14** So private investment rushed in to build the factories, to convert factories that were making tanks and everything else to make stuff that could go to Europe and to go to Asia to rebuild those places. And there is a huge boom in industrial production in the United States. But when people point at that and say, "We don't produce stuff anymore," that's not true, we produce more than we did back then. But that was artificial. We had just blown the world up. We needed to blow it up. I'm not against blowing the world up when it's necessary, but it was gone. And the people needed—had real needs that only the United States, in a sense, could supply.
- 5.15** So the economy had done very well in the late 1940s and going into the 1950s and pushing into the 1960s. Prices were stable. Government ran—I know this is like science fiction, but government ran surpluses. They actually paid back the debt that they took during World War II, and things looked like this was working. The system was great.
- 5.16** And of course, Europe was recovering not only because of the Marshall Plan—the Marshall Plan was a lot of money, American money, being invested in Europe, but it was a very small factor of European recovery. A lot of the European recovery happened because the economies there were freed up and entrepreneurial spirits took hold and people started building, creating and making stuff in order to recover their own lives. They had the incentive to work.
- 5.17** Productivity went up, production went up, and the global economy looked, at least the global economy of Western Europe and the United States and even Japan—there's not much of a global economy at that point. The rest of either. Soviet Union, Asia, everything we think today of Asia was nothing. I mean, China was one of the poorest countries in the world. Per capita GDP was maybe really one of the poorest countries in the world. Korea Peninsula, there was nothing. There was very little anywhere in the world except for Western Europe, the United States and Latin America, again, had been saved somewhat from World War II, although never really capitalized on that in any meaningful way because of its own corruptions.
- 5.18** So generally, the world economy seems to be doing pretty well, but there's problems underneath. For example, there's significant outflow of gold from the United States to Europe. In spite of all these banks saying, other countries saying, "We're not going to redeem our dollars into gold," they're doing that. Not at a large scale yet, a small scale, but money or gold is flowing out.

- 5.19 And of course, the United States enters the 1960s in this relatively robust condition, and then it decides within a matter of what—I don't know, eight years or something like that—less than eight years, to get involved in a major war in Southeast Asia, the Vietnam War, to launch really the largest expansion of government since the New Deal. In some ways, over the long run, a much greater expansion even than the New Deal.
- 5.20 I mean, people forget that before the 1960s, there's no war on poverty. The welfare state was really established at the federal level during the 1960s. There was no Medicare—Medicare today is one of the largest items on the federal budget and will become the largest. Medicaid—there was no Medicaid. None of these things existed. And suddenly all of them were put into place, and the budget projections for all of them turned out to be way, way, way, by a factor of 10—no, I think by a factor of 100—they underestimated how much they would ultimately spend on things like Medicare.
- 5.21 So Suddenly, in the 1960s, the United States is spending huge amounts of money on a war and on a welfare state. And to do that, it is basically creating new money, it is printing money, printing paper money, which the Fed can do—we'll talk about how it does it later on. The amount of base money, the amount of basic money, doubles during this period. Of course, if you think about it—by the way, broad money, a different definition, we'll get into definitions in a minute—a broader measure of money that includes more of your accounts at the bank actually triples during this period. So money, generally, the amount of money increases significantly in the economy.
- 5.22 Of course, the Europeans are watching this. We've got more money. Gold is in the account. Their dollars, the dollars they have and the dollars they get, are redeemable at \$35 an ounce. But given how many dollars the Fed has just printed, they gold is much more valuable than that; it should be maybe \$70 an ounce. So the temptation to go buy the gold or redeem the dollars for gold is very high.
- 5.23 Indeed, in spite of significant diplomatic efforts on the part of the United States to try and convince our allies to not to redeem gold, it's the French, not surprisingly, who, under de Gaulle, who is very independent minded, remember, he actually leaves NATO, he establishes his own nuclear umbrella and goes off on this independent streak, he's not going to be a minion to the, to the Americans, he is going to redeem his dollars into gold. And in the late 1960s, you have ships coming from France to the United States, loading up in gold and sailing back to France.
- 5.24 The amount of reserves gold in Fort Knox or wherever they keep the gold goes to something from like 20,000 metric tons in 1950 to 9,000 metric tons. So less than half by 1970. And the pace of gold leaving the United States is increasing. Those are the French in the late 1960s, coming into the 70s. There's no end. The war has not ended yet. The welfare state is only going to grow from this point, there's no indication otherwise.

- 5.25** What do you do? You're Richard Nixon. You're the President of the United States. Within a few years, there's going to be no more gold in the United States if you keep this up. There's no way to shrink government spending at this point, you're overly committed. So in 1971, Richard Nixon announces that Bretton Woods is over, that the gold will no longer be redeemable into gold. Think about it as reneging on a promise, on a contract. But of course, the real reneging on the contract was the printing of all the money that happened before that.
- 5.26** Dollar was no longer exchangeable at fixed— well, it continued to be at fixed rates, but all those broke because at that point, you don't have any anchor, which is gold. So now the relative value of different currencies is going to be determined by what? By the relative purchasing power of those currencies. And that's going to be determined by how much every government prints of its own currency. And that can change over time.
- 5.27** That's why today we have wildly, relatively wildly, fluctuating exchange rates. You look over the last—just this year, the dollar has lost significant value. I think it's 13% last time I looked to the euro and to the Brits, because nobody wants the whole dollars for a variety of political reasons we can get into and for economic reasons. There are reasons why they don't want to hold dollars right now. As a consequence, the value of the dollar has declined across the board, including vis a vis the Japanese yen and pretty much any other currency that it fluctuates against.
- 5.28** So Bretton woods provided some stability in the post World War II era, but it was never going to last. It was never going to last because as we know, central banks always cheat on a gold standard. The incentive not to cheat is greater when it's your own citizens who are redeeming the dollars into gold, and they cheat when that happens. When it's foreign governments, yeah, they're going to cheat even more. They care a lot less about what happens to foreign governments than they care about their own citizens. So they're going to cheat even more. Indeed, that's exactly what happened.
- 5.29** You can look at Richard Nixon and say, "What a horrible thing to do," but of course, by 1971, what option did he have? He could have undone the welfare state. He could have undone Medicare, Medicaid and got us out of the war, which he did ultimately. But given the politics, that was, of course, impossible. So it was much easier to get rid of the gold standard completely.
- 5.30** Of course, what happens to price of gold? It takes off within a decade. I think by the late 1970s, early 1980s, gold is priced at 20x the \$35 it was priced originally. It makes it up to 700, \$800 an ounce. Anybody knows what it is today? Let's round it up to 3,500. So 35 to 3,500, that's a big increase. Now, gold is not serving as money, but you could argue that today, what gold serves as is a proxy for money. People still keep gold as, in a sense, a monetary unit. They keep gold just in case the world ends. They can always barter with it, they can always trade with it. Gold is still shiny and pretty. We still like it. So it's still going to be something that in a time of crisis you can trade.

- 5.31** I have a few gold coins tucked away somewhere. So if the civilized state collapses, I could trade with a few people. Not enough, I'm afraid. So I don't know how much I could trade. But it serves today, or is believed to serve today as a proxy for a hedge against inflation, as an alternative to the dollar. And if that's true, if it represents the value of the dollar, then the dollar's lost value by not 10x but 100x just from 1971 to today. It's not that long of a period.
- 5.32** That is, of course, the end of Bretton Woods we'll get to. At that point—we'll get to what happens in terms of inflation. Of course, inflation picks up in 1970s as a consequence of this. Everything is now unanchored. There's no limitation on what the Fed can do. We're now on 100% fully fiat currency.
- 5.33** What does that mean? It means a currency that is not redeemable. People think of money as an IOU, that somebody's going to give them something for that. Well, not necessarily. The government's not going to give you anything other than it's legal tender, so you can pay your taxes with it, but there's no gold over there.
- 5.34** Or I don't know, people talk about American government's assets as if you can exchange your dollars for the American government assets. You can exchange your dollar bill for taxes and you can exchange—it's a medium of exchange within the United States because of its status as legal tender. And we'll talk about why fiat currencies continue to exist or what maintains their value. But it's unmoored from anything. It's unbacked by anything.
- 5.35** For the first time really in human history, on a global scale, do we have money, a medium of exchange, the common medium of exchange that is just paper? There have been occasions where this has happened in different places around the world. It happened in China a very long time ago. They shifted to paper money.
- 5.36** It happened in France in the early 18th century. There's a famous episode where John, a guy named John Law, Scotsman, comes in and tries to revamp the French monetary system and makes a complete and utter mess and disaster of it. Although you have to say that if you go back and study what he did in France between 1715 and 1720, it very much is similar to what everybody does later on when they switch to fiat currency. That is, he invented, in many respects, he invents the concept and it completely blows up in his face. And later on, people consider him a genius and they mimic what he did.
- 5.37** So for the first time on a global scale, we now have a currency that is not backed by anything, unmoored. And its quantity, its value, in other words, its value, its purchasing power is completely now dependent on a central bank and on the incentives, motivations of a central bank. We'll get to this. I mean, this is huge. It's huge in a sense of the importance of money to our lives and how important money is to our lives, to our saving, to production, to our ability to plan for the future.
- 5.38** Suddenly we've gone from a system in the late 19th century, early 20th century, where this was backed by something real, where you could understand the mechanism by

which it worked and you could count on it being worth the same in the future as it is today, to a system where there's some black box inside the Federal Reserve that nobody really understands and nobody knows exactly how it works, that prints out money when they think it's necessary, sucks in money when they think it's necessary. We're just supposed to accept it and not understand it.

- 5.39** These days, the only seeming alternative to that seems to be that instead of having the Federal Reserve, headed by Jerome Powell, make these decisions in their black box, that we shift the responsibility to Donald Trump and his even darker box to make decisions about what the monetary system is. We'll talk about Fed independence and the push right now to eliminate Fed independence and to bring the decision about money to the government, to politicians, and let them decide. It's bad. I mean, I would argue it's bad enough where it is now, but it could get quite a bit worse.
- 5.40** All right, so let's think about how this works to fine tune it a little bit. So dollars are the unit of account. We get paid when we go to work, we get paid in dollars. Basically nobody actually gets the cash, or most of us don't get the cash. It's basically transferred into our bank account. The money in a bank account is—we can convert it into cash and we can convert it into stuff we buy using a credit card or using an ATM. We can go and buy stuff with that and then it'll move around the banking system.
- 5.41** If we buy from a merchant, the money that is sitting in our bank account moves into that other merchant's account and basically it's a pretty simple transaction. There is no physical asset that moves. Everything today really, with exception of the cash you have in your pocket, everything today becomes electronic. Everything today is just zero ones. And there's no constraint on how much of those zero ones exist out there in the world. That is a product of how much money the Fed creates.
- 5.42** And there's a sense in which banks can create some money through the fractional reserve banking system, as we've seen. A bank can issue a loan. When it issues that loan, it creates a deposit. So if I give you a loan, I'm a bank, I give you a loan, I create a bank account on your name in my bank with money in it—that's the loan I just gave you. So I have an asset, I have a liability on my account. I've just created some money. You can then go and spend it.
- 5.43** Now ultimately you will spend it and those other vendors will come back to me and I'll have to give them back money. So it's not that I've created this out of thin air. Ultimately my bank reserves, the reserves that I have, are going to have to shift to this other bank that now holds the money. So banks hold a certain percentage of their assets at the Federal Reserve. They don't hold gold.
- 5.44** They have basically, if you look at a balance sheet of a bank, they have loans that they make as an asset because it's going to pay them. They have securities—most of the securities are government bonds. They're not allowed to speculate on stock market or anything like that. So almost everything they buy are government bonds. There might

be mortgage backed securities. There might be certain securities that they're allowed to own through regulations. And the rest is, in a sense, money that they have at the Fed.

- 5.45** They have a checking account at the Fed and that's called Federal Reserves. Reserves the banks have at the Fed. The Fed could release those reserves to them or they could pull those reserves. They could take those reserves and make loans with them. They could pull those reserves—I mean constantly, the reserves are moving around because that's how banks settle between each other. Every single bank of the U.S. has reserves at the Fed. If I owe you money, I just move some of my reserves to you. So reserves are constantly shifting around. Money literally isn't moving other than reserves really shifting around at the Federal Reserve level.
- 5.46** The dollar that we have, which used to be an IOU, the dollar literally was redeemable for something. It's no longer redeemable into anything, even though it's technically a liability on the Federal Reserve's books. But it's not redeemable into anything. Again, taxes—you can buy a government bond with it, but you're not going to get anything tangible. The Federal Reserve is not going to give you—show up at the Fed and say, "Here's my dollar. I want something." You don't get anything. Nobody's going to give you a thing.
- 5.47** All governments, as we said—all currency is government issued. There's really no market to govern the supply of fiat currency. The supply doesn't adjust to anything happening in the marketplace. It can shift, and it does shift all the time, but it shifts horizontally based on a Fed decision, "We're increasing the money supply." It's not responsive to markets.
- 5.48** Now, one hopes, and they will claim they do, the Fed is responding to market signals. The Fed is responding to what they think is optimal for the economy to have in terms of money at any given point in time. So it's completely at the Fed's discretion. Again, the conversation right now happening between the Fed and the Trump administration is: should the Fed be independent? That is, should it be at the Fed's discretion or should it be at the executive's discretion? Does the President of the United States, should he have more power, should he have discretion over the amount of money out there?
- 5.49** The Fed claims to be independent of politics, it claims to make decisions around the amount of money based on economic considerations. The Federal Reserve System employs more PhDs in economics than any institution in the world. It's the largest employer of PhDs in economics in the world. They want to be scientific. Whether they are or not is a question, but they want to be scientific.
- 5.50** They claim not to be political, although there's plenty of evidence to suggest that they are political. Not as political as if politicians ran the Fed, but at the margin they make decisions that are positive to the incumbent. They like the incumbent. An example: it's pretty rare that interest rates are raised during an election year. So in the election year,

the Fed tries to be neutral, certainly not raise interest rates, maybe reduce interest rates, but not in any overly dramatic fashion.

- 5.51** Arguably there have been cases where you could argue that the Fed's behavior swung the election one way or the other, because it could have decreased interest rates faster, or it should have raised interest rates, and it didn't. But it can't avoid politics. Ultimately, as we'll see, the President appoints the chairman of the Federal Reserve, they meet regularly—there's clearly influencing. Not enough to satisfy Trump, but there's clearly influence.
- 5.52** So the question is: what gives this paper value? Why do we use fiat money? Why do we assume that \$20 is \$20? Where does it get its value? There's supply and demand for this medium of exchange, but how does it hold up as a medium of exchange? When we talked about barter, we talked about the fact that in order for something to arise as money, it had to be saleable. It had to be something we want, and a lot of us want, so that we could actually use it to trade among each other.
- 5.53** Nobody wants paper, even if it has pretty pictures on it. So what makes it desirable enough to facilitate exchange? I'd say there's really two things. One is the power of the sovereign, the power of government. This is the main thing. If the government tomorrow said, "Nah, we don't care about the dollar. We're not going to take—if you want to pay your taxes with it, we're not going to accept it. We want, I don't know, shells. If you want to pay your taxes, you're going to have to pay with shells." Dollars are going to disappear as a medium of exchange. Everybody's going to rush and find shells or whatever it happens to be.
- 5.54** So to a large extent, it's the fact that we can pay our taxes with it. It is legal tender laws. Legal tender laws mean that if we have a debt agreement, then you have to accept dollars in repayment of the debt. So even if the debt agreement puts the repayment in some other currency or, let's say, in gold or something else, if I say, "No, no, no, I can't pay you in that. I want to pay you with dollars," you have to accept dollars. That's what legal tender means. It's the way in which we settle debts.
- 5.55** That's force. That's government force. That's coercion. We have no choice but to, but to abide by it. You go to jail if you don't. And of course, we have to keep enough money aside to be able to pay our taxes. So we have to at least have enough of this money in order to pay the taxes. In a world in which our taxes are anywhere from, I don't know, 0 to 60% of our income, maybe more if you take into account all the different taxes you pay— sales taxes, property taxes, everything else—in some states of the union and above certain levels of income, you might be paying more than 60% of your income as taxes. So a significant amount of money that is out there is going to taxes.
- 5.56** Indeed, if you look at total government expenditure, the federal government spends about 21, 22% of GDP and then the states spend another 10— or more than 10—15% of GDP, and maybe even more than that. So overall it's getting about, 40% of GDP is

spent by the government. So on average, 40% of your money ultimately lands up in the hands of government, on average, across the entire economy.

- 5.57** So government is a big part of it. And government only accepts, only deals with this form of money. They're gonna pay their contractors with this money. They pay the wages of the millions and millions of people who work for the government with this money. It becomes the thing that everybody wants because it's the thing that if you interact with government, you need. And we, for better, for worse, we are constantly interacting with government. We have no choice about the matter.
- 5.58** The other thing is that there's a certain—something has to fill this gap of a medium of exchange, something has to facilitate it. There's a certain inertia, a certain acceptability—we've all gotten used to it, we're all using it. There's a certain path, dependency, we got here in a certain path. This is what we have for a variety of historical reasons we've not talked about and what else are we going to use?
- 5.59** You find, I mean it's rare, but you find societies where everything has collapsed, central bank has collapsed, everything has collapsed, but there's still paper money out there. They're still using the paper money to transact with one another because it's convenient, it's just convenient. Keep everything as the same unit of exchange.
- 5.60** Somewhere crazy like Somalia, the Somali shilling, for years after Somalia government basically evaporated and you basically had no authority, people were still using this paper money that was worthless to transact. Now, I think they were getting rid of it as quickly as possible—you don't want to be saving in Somali shillings, because you didn't know when it was going to go to zero— but for a while there was a certain inertia about doing it, a certain convenience and efficiency. But it's unbelievably fragile. It depends on our willingness to keep using it, which is not anchored. And of course, the only anchor it has is the government authority over it.
- 5.61** All right, so we today live under a fiat money regime, as the does, I believe, every country in the world. Not every country in the world has a central bank, and not every country in the world has its own currency. For example, a country like Ecuador has no central bank and has no currency; they use dollars, which is very convenient for them.
- 5.62** Milei has said that he would like to dollarize his economy, that is, get rid of his central bank, get rid of the peso and go to dollars. There's still a central bank; it's just not sensitive to your politics. It's sensitive to somebody else's politics, which is a problem. I mean, I don't know why Ecuador has to suffer from the particular conflicts that happen in the United States around monetary policy, but if the United States, I don't know, raises interest rates because it believes there's inflation in the United States, something like that, it's not clear how that reflects towards Ecuador.
- 5.63** But I think it's a superior—for Ecuador in particular— it's a superior alternative to having their own central bank because the incentive is, when you have your own central bank, to inflate, which has always been the temptation of central banks. So the

Federal Reserve is not—I mean, by my standard, doing a terrible job—but by international, global standards, doing an amazing job at managing the money supply. So everybody attaching themselves to dollar is not a bad idea.

- 5.64** The problem is if you link your currency to dollar but you maintain your own currency in your own central bank, what do you do? You cheat. You print too much money and ultimately you have to devalue the whole thing. But if you don't have a central bank, if you eliminate that, you can't cheat. You just use dollars.
- 5.65** You've got Ecuador, you've got Panama and I think El Salvador, all, for example, use American dollars as their currency. All of those countries had at some point hyperinflation. And this was the way to get rid of the hyperinflation, hyperinflation caused by their own banks. We'll see how central banks do that.
- 5.66** European Union has a central bank, but that means that the member states within the European Union don't. So when Greece got into trouble in 2012, 2011, 13, it couldn't print itself out, it couldn't inflate itself out because it doesn't have a currency. Its currency is the European Union's currency and it has a central bank that by constitution—when the EU was established, they were worried about all kinds of misbehaviors of the central bank. So they created a constitution for the central bank. Guess what happened to that constitution when the first crisis happened? It was flushed down the toilet.
- 5.67** The constitution of the central bank, for example, said that you can't buy sovereign debt of the member countries. What did they do with Greek financial crisis? They bought stuff and debt. But it's removed somewhat, at least, from Greek politics.
- 5.68** And indeed, I would argue Greece right now economically is doing quite well since that crisis. It's recovered really nicely. Wages are up, income is up, GDP per capita is up. But every measure they've done well. And I would argue that the reason they've done well is because they didn't have a central bank. It's because they were dependent on the euro. Now, they got bailed out by the Germans and everybody else. But since then they've done well because they had to reduce some reforms internally which the currency—which the fact that they were in default caused them to do.
- 5.69** The same thing happened in Spain. And Spain, in spite of having a socialist government, now economically is doing decently well because of the kind of reforms they had to do because they were stuck with this currency that they couldn't inflate. For example, they had a lower some people's wages and they had to fire certain people and they had to restructure the economy in order to cope with the fact that they couldn't just print money to pay the bills. They actually had to figure out how to pay the bills.
- 5.70** So not all, as I said, not all countries have central banks, although most do. Even the European Union, they have one central bank for all the members of the European Union. And that group is—well, that's not true. Not everybody in the European Union is on the euro. So those are not—that's not 100% 1 to 1. There's some countries that are

members of the EU, that is, they have the open, free movement of labor, capital and goods, but they don't have the common currency.

- 5.71** The common currency is a separate thing to the EU. They're not exactly the same thing, although there's a lot of overlap. So there's a number of countries that have their own currencies. Denmark, for example, have their own currency, but they're part of the EU, so there are exceptions.
- 5.72** Let's talk about the Fed. The Fed today has what's called a dual mandate. It is responsible economically for two things and it's supposed to manage the money supply and manage interest rates and everything else, all the levers that it has. It has a big lever in bank regulations as well to manage these two. The first is price stability. No inflation.
- 5.73** The Fed defines no inflation as 2% average inflation over the long run. So no inflation doesn't mean what it would mean to us, I think, zero. "No" means usually zero. It means 2%, and they've decided 2% is the right number. It gives them some flexibility. What they hate, what they really fear—this is modern economics generally—really fear deflation, dropping prices. This is the one lesson from the Great Depression; dropping prices, that's bad, that leads to depression.
- 5.74** That's completely wrong. It depends why prices are going down. Prices are going down because there's no money and because credit is contracting and because the, the, the, the, the banking system is seized up, then yeah, that's bad. But if prices are dropping because productivity is really high, then that's actually not a bad thing.
- 5.75** But they hate, they hate deflation. So in order to avoid even a little bit of deflation, they would rather inflate a little bit. So the target is 2%. Right. And, and that's the mandate. And it used to be the target was 2%. And then a few, a couple of years ago they changed it to average of 2% over long run, which gives them even more ability to be volatile. They, they weren't making the 2%. So an average seemed like a better way to do it.
- 5.76** The second mandate is to maximize long-term employment in the United States. So it's an employment mandate, not a monetary mandate, with the idea, again, somewhat derived from the Great Depression, but generally from Keynesian and modern economic theory, that employment is a monetary phenomenon. If you want more jobs, you flood the market with more money, and then more jobs will come into being. And if there's not enough money, then, then there's not enough jobs, and people, people lose their jobs. Right.
- 5.77** Instead of, again, money being something that kind of balances all these things out and, and makes them all work, you get to somehow manipulate it. You hand businesses money, and they create jobs. Doesn't quite work that way. But that's kind of the Fed doing that. And often they believe that these dual mandates are in contradiction to one

another. So if I keep prices under control, I might cause unemployment, and if I cause employment, I might cause inflation. So they're constantly balancing these two.

- 5.78** And of course, the means by which they do this is central planning; that is, it's kind of fascinating to me that, in modern economics, in almost every realm in the economy, we understand that central planning doesn't work. The idea that central planning can calculate, can figure out what is the right amount of bread or the right price of bread, or how many cars we should manufacture, what price you can manufacture, that that can be done has been dismissed by economists quite thoroughly and shown that it's not possible.
- 5.79** And yet, when it comes to money and interest rates and employment, we give one institution the ability to centrally plan all of that or to try to centrally plan all of that. So the most important price in the economy, in my view, is the price of money. The most important quantity in the economy is the quantity of money, because that drives everything else; that, we centrally plan. Wouldn't do that to bread, wouldn't do that to automobiles or computers. But that is OK. And we suffer the consequences of all of that.
- 5.80** Now, really they have more mandates than that. That's the two official mandates. They're written in the law. This is a law the Congress passed. They used to have only one, used to be just price stability. And then in the 1970s, they added the mandate for full employment, or to, or for long-term maximizing employment. But they also have a mandate generally perceived to be, keep long-term interest rates low.
- 5.81** Now that could, is related somewhat to the employment mandate. Long-term interest rates low means companies are comfortable to make investments long-term and that will generate employment, so those are related. They also have a mandate of keeping the financial industry stable, out of crises. Ultimately, the Federal Reserve is one of seven regulators of banks, but you know, the seven agencies that regulate banks. But it's probably the most powerful in many respects, most, most significant of those.
- 5.82** It has a whole regulatory division. There is a vice chairman of the Fed responsible for regulation. So it's a big chunk of it is to regulate banks into stability. We can talk about how well they've done that, right, given all the banking crises we've had in the U.S. just recently, right. Just in 2023, we had Silicon Valley Bank go bankrupt and almost a lot of other banks go bankrupt. So, one could ask whether the Fed is doing a good job in that sense.
- 5.83** And then it also has the mandate of being, call it the lender of last resort in the economy. It used to be perceived that the Fed was a lender of last resort vis-à-vis other banks. So if a bank needed money, needed to borrow money and couldn't find anybody to lend it money, it could go to the Fed. There was something called the discount window. They could go to the Fed and they could go to the discount window and

borrow money from the Federal Reserve to keep themselves liquid, to keep themselves going. And the Fed was the lender of last resorts to banks.

- 5.84** And then the financial crisis happened. And in spite of the fact that this was completely illegal at the time, the Federal Reserve became the lender of last resort to pretty much anybody. Corporations, businesses all over the country came to the discount window and got loans. Foreign companies had the ability to come to the discount window and get loans. Now they all paid them back, so theoretically, we didn't lose anything. Right?
- 5.85** But the reality is that the Fed is now responsible for not allowing another Great Depression to happen, not allowing a Great Recession to happen. Bailing everybody out when they need to be bailed out. Bailing the ones that want to be bailed out, in other words, and not others. They bailed out some of the investment banks; again, there was no, there's nothing in the law that allows them to do a lot of what they did in 2008. 2008, they made stuff as they went along, in order to— the whole mindset was no Great Depression. We do whatever is necessary to avoid a Great Depression.
- 5.86** And if that's the standard by which you want to measure life, they did a good job. We didn't get a Great Depression, so they succeeded. But in order to do that, Congress and the executive basically turned a, not turned a blind eye because they actually encouraged us, encouraged the Fed to basically go way outside the scope of what it's authorized to do and do whatever's necessary in order to get the economy going again. And this, as I said, happened on steroids in 2008.
- 5.87** So what the Fed controls is the quantity of fiat money that is out there in our economy or, you know, almost all of it. It's true that, you know, banks through fractional reserve can create some money, but not endlessly, not without limit, and even that the Fed can somewhat control through its interest rate policies. But the base fiat money, the base on which everything else is measured, the amount of reserves the bank has at the Federal Reserve, that is what the Fed is responsible for.
- 5.88** That's what the Fed can manipulate. That's what the Fed can, can target. So when it's targeting price stability, it's really targeting a certain growth in the money supply, although money supply is an ambiguous term these days, because money is an ambiguous term. What is, what is money now, right? It used to be, money's gold. That was easy. But what is money now? Well, there's M1 and M2, and M this and M that, and base money and, and, and, and, you know, broad money.
- 5.89** And nobody can exactly agree on which one matters and which one we should care about because, you know, is your money in a saving account? Is that money? Is it a medium of exchange? Well, if it's locked up and you can't literally spend it right now, then it's not really money. You can't use it as a medium of exchange right now. But is that really the case? But your checking account, is that money? Yeah, it seems like that's money because I can withdraw it anytime.

- 5.90** Cash is clearly money. My mutual fund, is that money? Well, some mutual funds, you can actually get an ATM and draw money off of them. So maybe that is money. But other mutual funds, where they're locked up, not so much. So it's complicated, to actually really figure out, what am I looking at? What am I measuring?
- 5.91** This is why they have, you know, a base money, broad money. But then even broad money is not quite as broad as, let's say, all the wealth that we have, everything we denominate in money, all the financial assets we have, which is a much broader category. And now we're going to get stablecoins. We'll talk about that in a future class. Is that money? How does that relate to money? And in what way is it money, and what backs stablecoins? Government debt. So maybe government debt is money. It's a mess. It's a real mess.
- 5.92** Anyway, what the Fed directly control is what we call, you know, basic, you know, the basic money, the reserves that banks have and the money we have in our pockets. Indeed, the Fed determines every year how much cash it'll actually print; that is, the actual cash that you have. It determines how much is going to be available. It tries to evaluate how much money you took out of your ATM machine over the last year. There's some kind of estimate and that's the money that's supplied to the banks, so that they can—treasury prints it up. But the Fed is the one determining that.
- 5.93** So, if you remember, you remember Y2K? Anybody remember Y2K? The hysteria over Y2K? Everything was going to collapse. Everything was going to, the world was going to end. This is when the clocks shifted from 1999 to 2000, supposedly there was something in the software of every single computer on planet Earth that was going to collapse it all. Planes were going to drop out of the sky. I mean, I knew people who 100% bought into this, right?
- 5.94** Nothing happened. But the Fed wasn't that sure. It was worried. So they actually stocked up on cash, fed that cash into ATMs because what they expected was that there'd be a mass— all of us would go to ATMs and load up on cash because we'd be worried that the next day we couldn't go to the ATM. I mean, it didn't happen. There was a slight spike in cash withdrawals, but nothing close to what the Fed expected.
- 5.95** But they expanded the amount of cash in the economy dramatically just before Y2K, in order to, just in case people panicked and they wanted to withdraw cash. So that's one of the ways in which they control. But the main way in which they control the money supply has been by buying government bonds. So, you know, historically the Fed would go into the marketplace. It actually had certain banks that were, that were, in a sense, market makers for them, and only certain banks.
- 5.96** And they would, the government would issue bonds, these banks would buy the bonds, and then the Fed would buy the bonds from the bank. So think about what's happening there. You've got a, a piece of paper, promissory note, an IOU. The Federal Reserve is

buying that IOU. So the contract gets put into the balance sheet of the Federal Reserve. And what does the bank get in its place? Money. Money.

- 5.97** Now the money is in the form of a reserve. It's a liability on the Fed's balance sheet. So the asset is IOU. The liability is the bank reserve, money that the bank has on reserve at the Fed, and theoretically can withdraw their money and then lend it out or consume it, pay wages, whatever it needs to be. But it's new money into the economy, didn't exist before.
- 5.98** And there's a sense in which, as long as that money stays on the reserve at the Fed and doesn't, it hasn't really entered into the economy; so it hasn't increased the money supply because it's still sitting on, it's just a potential for the increase in the money supply. Until the bank actually, in a sense, uses that reserve to lend money out or to do something with it, nothing has really happened. All you've done is move money from one side of the balance, from one place to another place, but it's still static. Nothing has happened. No effect, no economic effect is actually manifest.
- 5.99** So, mostly, money is being created by the Fed buying securities from the government. In 2008, 2008 is a big year, at least for the Fed, it started to do a few things. One, for the first time in American history, it started buying non-government debt. So we have a policy in this country to really support housing. We really believe people should own their own homes. That is a bipartisan, bipartisan belief system. Whether it makes any sense or not, probably when it's bipartisan, it doesn't. Anything bipartisan is suspect.
- 5.100** But we want people to have homes. And when the housing crisis happened in 2008, the first thing was to, was to, you know, support the housing market. And one way to support the housing market is to make sure the mortgages continue to be made. And one way to make sure mortgages continue to be made is not allow all those old mortgages to go bust, for people to lose money off of them.
- 5.101** So the Fed stepped in and bought them all, or a bunch of them, not all of them, a bunch of them, and to this day has large quantities of mortgage-backed securities on its balance sheet, thus supporting, giving a kind of support for the mortgage industry. Basically telling people, yeah, don't worry about the risks of mortgages, you'll be bailed out. If nobody else will buy the mortgage-backed security, we will.
- 5.102** But you know, again, par for the course, for our government is bailing out, bailing us out constantly, and thus incentivizing us to take the kind of stupid risks that get us into trouble to begin with. So they started doing that. So today the Federal Reserve has a lot of government debts, treasuries, and a lot of mortgage-backed securities on its balance sheet, right? Those are its, those are its assets.
- 5.103** The other thing that in 2008 the Federal Reserve started doing, before 2008, the reserves banks had at the Fed did not pay interest. So there's money the banks put there and didn't, they didn't get anything for it. They put it there because the Federal Reserve demanded that they put it there. It was expected. And because they needed to

keep a reserve, because they're fractional reserve banks; you need some reserve to be able to have some money, to be able to distribute if people run on the bank.

- 5.104** But reserve requirements are very low, and they kept shrinking because it's much more profitable for a bank to lend the money out than to hold it in reserve and not get anything for it. So they kept shrinking. The Fed tried to have reserve requirements, but those kept shrinking because they kept being an incentive to make the banks more profitable by shrinking their reserves and allowing them to lend more.
- 5.105** So, if in a gold standard free banking system, banks might have had, I don't know, 50% reserves, maybe if it was a particularly aggressive bank, 40% reserve, some banks 100% reserves, some bank 80, 70, a variety of different reserves, in a fiat-based system in which, in which the reserve requirement is set by law, in a sense by the regulatory agency, there is no incentive to have more reserves, then otherwise you can always borrow more money from the Fed.
- 5.106** The Fed can always facilitate funds if you need them. You can get liquidity from them; you can go borrow them from them. So the incentive is constantly to lower your reserves because that's, you're going to be more profitable in a competitive market, because of this entity that is led to always lend you money. You know, reserve requirements only go down, down, down, down, down. And the Fed has ultimately no incentive to stop that.
- 5.107** So when they realized they couldn't really control that, what they started doing in 2008 is, they started paying interest on those reserves. And what they, what they do is they try to pay enough interest on reserves to incentivize banks to keep reserves, not to lend it out, not even to buy treasuries with them—that's what the Fed does; Fed buys treasuries—to disincentivize the banks from buying treasuries. So they have to pay them an interest rate that is equal to or higher than the treasuries they would get, and on a risk, on a risk basis, higher than what they could get if they made out a loan.
- 5.108** So today, when the Federal Reserve says we're raising interest rates, they used to talk about the Fed funds rate. That's the rate they used to raise. That is the rate at which banks lend and borrow from each other. There is no Fed funds rate anymore, not really. The Fed funds rate is, banks don't borrow and lend from each other. They don't need to. They just go to the, to the Fed and to the Reserve.
- 5.109** Now, when the Fed says they're raising rates, they're raising the rate they pay on the reserves at the Fed. When they say they're lowering rates, they're lowering the rates they pay on the reserves at the Fed. That's what they mean when they change. They're not controlling the 10-year interest rate. They don't control the 30-year, 1-year, 2-year, 5-year, although sometimes they try; they don't. The only rate the Fed actually directly controls is the rate they pay of interest on the reserves the banks hold at the Fed.
- 5.110** The smaller those reserves, the more money is going to be out in the economy. The larger those reserves, the less money is going to be out there in the economy. And you

know, assume everything else held constant. Right, because if they pay very little, then the banks have an incentive to pull their money and give out loans or buy securities themselves. If they pay a lot, I don't have to, you know, that's hard work to go out and make loans. I'd rather just leave it at the Fed.

- 5.111** So it's become the number one lever that they have over, over, you know, over the money supply, if you will. All right, so, so the Fed is, so the Fed's main tool to control the amount of money in the economy is its purchase of government securities and the interest rate it sets on those reserves, and therefore how much it allows banks, you know, in a sense, to lend out money into the economy and have that money circulate in the economy. Those are the two tools that it is using on a regular basis.
- 5.112** The Fed can, therefore, increase or decrease the reserves, increase or decrease its balance sheet. You know, a lot of times people talk about the Fed's balance sheet. What they're really talking about is all those assets it has, all those government bonds, all those mortgage-backed securities. Now imagine if the Fed decided to sell government bonds, says, "I have too many government bonds on my balance sheet, I'm going to sell some." What happens to money? Does it go up or down? It goes down.
- 5.113** Security enters into the banking system, and money goes back to the Fed. If it was physical money, they burn it. It's not physical money. It's just an electronic thing. They basically take down the reserve that the bank has on the, on the reserve accounts. In a sense, everything is reserves now. So, money is reserves that the bank has and the cash that we have. So when they sell a security, the amount of reserves decline. When they purchase security, the amount of reserves increase.
- 5.114** And the assumption is that when that happens, that has an impact on the rest of the economy through bank lending, through bank lending and banks using the money for their own needs. But those are the ways in which it changes. We will demand money. The demand curve is still a regular demand curve, demand for money. We want money. When we think the purchasing power of money is strong, when we think the purchasing power of money is declining, when we've got inflation, what do we do? We want to get rid of our money, right?
- 5.115** It's like that guy who gets paid in hyperinflation every week and goes out and spends it. We'd rather spend the money or invest the money or put the money away, but we don't want to actually hold money in a time where the money itself is losing value, when it can purchase less and less and less over time. And we'll see kind of the consequence of that.
- 5.116** So in the end, you know, the Federal Reserve central banks are responsible for the price level, they're responsible for the supply of money. They can't, they, they indirectly can control demand by how much inflation there is, but they directly control the supply, and therefore control inflation. And basically, that's what they play at all.

5.117 Right, so let's stop there and when we get back, we're going to talk a little bit about, we're going to talk a lot about inflation. So we'll talk about inflation and what it means. We'll talk about why inflation is as damaging and as bad as it is, what the damage that inflation does. And we'll talk about the importance of having a money that can sustain its value over time, and how what we have today is anything but.

Lecture 6: Inflation

At a glance — Inflation.

- 6.1 We have discussed the transition in our monetary system from a gold standard to a fiat standard, and discussed some of how the Federal Reserve works this and, in a sense, tries to manage the supply of money and the supply of reserves.
- 6.2 Today, we're going to talk about, primarily, the big consequence of that, or one of the big consequences of that, which is inflation, which is a decline in the purchasing power of the unit of account, of our medium of exchange, of the dollar, and what consequences that has, both economically and what consequences that has to our ability to save, and everything else.
- 6.3 Before we do that, I want to step back a little bit and get a big picture for a minute in terms of, just to give you a real sense of the importance of time in life. Should be pretty obvious, but what the hell. Human beings survive by producing. Beyond the very basic hunter-gatherer societies, and even there, there's production happening, we need to plan into the future.
- 6.4 We need to know when to sow the seeds and harvest the crops, and we need to be able to keep some of those seeds and not consume them, so that we have seeds to plant in the future. We need to be able to plan long-term about our lives, our production. We need to know that what we save is going to be worth something in the future. It's essential, in a sense, for human life and human productivity, this ability to save.
- 6.5 We save across time. We live across time. We produce across time. Farmers can get away with planning one season in advance, maybe. And even there, they have to think about when is the soil depleted, and they have to have seven-year cycles and all kinds of stuff like that. But if you're in an industry, if you're in tech, if you're in any productive activity, you have to plan over not just one season, but over a career, over a lifetime, over a whole productive period.
- 6.6 And to do that, one of the things, one of the great innovations, one of the great innovations in all of human history, was money, which allowed you now not only to be a medium of exchange, to dramatically make your exchanges with other people, your barter system, more efficient and more productive, and you can consume much easier.
- 6.7 But money now made it possible for you to save over time. You weren't just saving seeds to plant in the future. Now you were saving money, which could then be constantly invested and constantly generate the kind of results that you anticipated into the future. I mean, think about some of the things that companies are investing in right now.

- 6.8** They're building these massive farms of servers for AI that are going to take a year at least to build, and then to put all the servers in there, with the anticipation of someday down the road, this paying off. They're investing trillions of dollars right now, all based on plans well into the future. Business, generally, is a long-term endeavor. It's not about tomorrow. It's not about one season. It's about years.
- 6.9** And investing capital and money, and we as individuals do the same thing. This allows us to produce into the future. It buys us time, it buys us life, and it makes us richer, wealthier, which is, again, buying time and buying life. The ability to save means we can consume when we want, we can produce when we want, we can invest when we want. We're not dependent on hand-to-mouth. It changes your perspective on life.
- 6.10** Think about living hand-to-mouth. With your money, hand-to-mouth, with no ability to plan into the future, any of these aspects. Think about how you put money aside for your kid's education or to buy a house or to buy a car. They all depend on your capacity to save. It all depends on the capacity of the unit of account to maintain its value.
- 6.11** And when you start depreciating that, when now I have to take into account the fact that the money is not worth what it is, that prices are going up constantly, that I don't know what prices are going to be in the future, that I don't know what the value of the money that I have is going to be in the future—what can it buy? What it won't be able to buy? The kind of burden that places on each one of us as individuals. Now, we've internalized it because we live with it constantly, so we don't even think about it. But it's a real burden. It's a wasted burden. It's a waste of time. And time is all we have.
- 6.12** And it creates massive uncertainty, that we need to find ways to protect ourselves and hedge against. And sometimes we get blindsided by it, and sometimes prices go up, and we don't have the funds to deal with that, and we have to change our plans. So, money is an incredible time-saver. It's an incredible fuel for production. It makes production possible. And it is a means by which we save, in order for all of that to happen.
- 6.13** When you destroy money, when you undermine money, its value and its ability to save, you're really destroying lives. You're destroying production. In that sense, inflation is really evil, in that it is destructive of human life. It's a cost in human life. It's a price that we have to pay every day. For what? We're not getting anything in return. There's no value we get in return for the fact that our dollars diminish in value 2% every year, if the Fed gets its way, more likely 3% or 4% this year. It's just gone into thin air.
- 6.14** I think that's important just to recognize, money, common medium of exchange, it's a unit of account, but it's also a means of saving. It's also a way to preserve value. It's also, because it's a unit of account, if it can preserve value, a way for us to plan into the distant future. And when you take that away from us, you take a piece of life away from us. And fiat money has done that. It's taken that away from us.

- 6.15** Because at the end of the day, what the Fed does by controlling the money supply, and since it has no natural way in which to control the money supply, the money supply is not adapting to what is actually going on in the economy, it is being adapted by central planners, trying to estimate and figure out what's going on in the economy. There's no natural way to do it. The money supply tends to expand and tends to, therefore, reduce the purchasing power of every unit on a regular, systematic basis.
- 6.16** We call that inflation. We call that decline in the purchasing power of our dollars, inflation. There are lots of different measures of inflation, and they're not all consistent. You hear about the CPI, consumer price index, but there are other measures of inflation that might be more accurate or less accurate. And of course, each one of us experiences inflation completely differently, because each one of us has a different basket of goods.
- 6.17** So when they say the CPI has gone up by 3%, and a lot of that is, I don't know, food and energy, but for some people, food and energy is most of what they buy. For other people, food and energy is a fraction of what they buy. They're buying computers and they're buying vacations and they're doing all kinds of other things. How inflation affects each one of us is going to be completely different. There's no one inflation, in that sense. Each one of us has his own inflation.
- 6.18** How much do the goods I purchase, how much are they increasing in price? How much are my dollars depreciating in value, given what I want to buy, given what I tend to buy? Inflation is always caused by the same thing. Fundamentally, at the end of the day, there's a process. It can be multiple different processes. At the end of the day, it's an increase in the amount of money in the economy.
- 6.19** It could be that prices go up because of what they call a supply shock. There was a lot of talk about this after COVID. For some artificial reason, there's less stuff being produced, and yet the money supply hasn't changed. Then what happens to the prices of those goods? Still a lot of money chasing fewer goods, the prices will go up. But then once the factory starts churning stuff out again, what will happen to prices? If the money supply stays the same, they'll come back down. This is why they consider it a temporary supply shock.
- 6.20** And COVID was a temporary supply shock. And you'd expect that prices would go up and they'd come down, but they didn't come down. They never came down. They went up and then they went up even more, and then they went up a little bit less, but they kept going up. So they don't come down. Supply shocks tend to be temporary— a hurricane, a pandemic, a war. But sustained inflation, the sustained depreciation of your purchasing power, comes from an increase in the amount of money that is in the economy.
- 6.21** It means there's more of the stuff you use as a medium of exchange, and therefore, there's more money chasing the same number of goods. We bid the goods up. We bid

prices up. So just to be clear, it's sad that this even has to be said, but in the world in which we live, you need to say it— inflation is not caused by greedy businesses, and that's not because businesses are not greedy. It's because businesses are always greedy. The greedy industry does not change suddenly, and they raise prices.

- 6.22** If they could have raised prices yesterday, they would have done it yesterday. The reason they can raise it today is because there's more money chasing those goods. And there's nothing wrong with businesses being greedy. This is how they make a profit. This is how they make their business function. Work. But that's not inflation. Indeed, it's the greed of businessmen mostly that lowers prices, because in a competitive environment, the only way to stay in business is to outcompete your competitor. One of the ways you outcompete a competitor is by lowering prices, lower than the other one.
- 6.23** Unless, in a free market, in the market we have today, you really can identify a so-called monopolist, somebody who dominates a market and has no competitors, it's very difficult to raise prices, because your competitor won't. Gain more market share from you. And even if, let's say, you dominate an industry and you raise prices over here, OK, so the relative price went up. People still have the same amount of money. They have to spend more on your stuff. What happens to all the other goods in the economy?
- 6.24** They have less money to spend on them. So prices there will have to come down. So on average, what happens to prices? They don't change. So Elizabeth Warren can talk all she wants about "greedflation" by grocery companies. Even if that were true, which it's not, complete nonsense, what about computer companies? What about energy companies? What about all these other companies? They must have to lower prices if the supply of money is the same.
- 6.25** It's just a deep, deep, deep ignorance of economics and, of course, of human nature and how business works and everything else that drives this attitude. But it's a prevalent attitude on, sadly, both sides of the political aisle. This is not unique to any one side. It's always much easier to blame business. We always, I mean, this is just a general principle in life, I think. Whenever there's a crisis, we blame business. We never blame government.
- 6.26** Great Depression was caused by Wall Street, not by the Fed, not by FDR, not by, I don't know, tariffs, not by Hoover, not by anybody like that. It's always the great financial crisis was caused by business, was caused by the mortgage industry, was caused by bankers, it was caused by all that, not by Dodd-Frank, two politicians, not by the Federal Reserve, not by the people who actually caused these things. So we always blame business.
- 6.27** And whenever you hear people blaming business for some problem in economics, always be suspicious. Always be suspicious. The people really almost always to blame

for systemic risk—what's the most systemic institution we have? Government. It affects all of us. It passes a law, we all have to abide by it. It passes regulation, everybody has to abide by it. If the Federal Reserve increases the money supply, it affects all of us.

6.28 A bank can screw up. A business can do stupid things. But for that stupid thing to affect all of us, and for all businesses to become stupid all at once, is very unlikely. Very unlikely. So when they talk about systemic risk and they talk about systemic institutions, they're talking about government and they're talking about the Federal Reserve. There are no systemic institutions among businesses, not really. So again, be suspicious, suspicious generally of what you hear from politicians.

6.29 So, as we discussed, the Fed started inflating in the 1960s. Started inflating in the 1960s because it had all these expenses and it didn't want to raise taxes. Sound familiar? Didn't want to raise taxes, but it wanted to spend money. And at the time, there was a gold reserve, and obviously, that had to implode. And once that imploded, then all constraints on the ability of the Fed to produce money went away. They didn't have to redeem for anybody. Not a farm bank, not a domestic, nobody.

6.30 And indeed, starting in the 1970s, starting in 1971, the Fed expands the money supply dramatically. And most of you are too young to have lived through it, but it was a time of rapid inflation. 1974, inflation was 12.2%, 1980, it was 14.6%. I mean, we get upset now when inflation is 5%. 12.2%, and it's the same definition now as back then, and inflation was significantly higher than it's been recently.

6.31 And interestingly, a lot of people believed—economists—believed that inflation was related. So if you create inflation, you also create economic activity. The economy also grows. And yet in the 1970s, inflation, the idea was, there's more money, more money circulates. We're buying more stuff. Goes back to buying, consumers. We're buying more stuff. Producers are producing more stuff. Economic activity rises. We hire more people because economic activity is rising.

6.32 And yet, in the 1970s, we got high inflation and no economic growth and high unemployment. Unemployment reached almost double digits. So we coined a new phrase, which Keynesian economists or some economists believed could never happen, which was stagflation, stagnation and inflation. Stagnation and inflation. That's what you had during much of the 1970s, stagnation. It was from '71 to 2021, so this is until recently, 50 years.

6.33 The CPI, the consumer price index, rose 569% over 50 years, 569%. I know because, you see, interest compounds, inflation compounds. You have inflation one year of 3%, and then the next year it's 3%. Well, it's 3% over the base, plus the 3%. So it's really high. So that translates into an average 3.9% a year. But it comes to 568%. So think about what a dollar was worth in 1971 and what a dollar is worth today—a fraction, a tiny fraction, of what it was.

- 6.34** You can't save 50 years. Think about 50-year horizons. I'm going to put my money aside. And think about how I'm going to invest it into production over the next 50 years. I can't because I don't know what the money is going to be worth in one year, never mind 50 years. It might be worth 569% less, which is a lot. Well, you'd only go 100% less, but you get what I mean by 569 %.
- 6.35** And look, in the Fed, it's not that bad. Other countries had it much worse. Argentina was running a triple-digit inflation annually, not over 50 years, annually. I lived in Israel in the 1980s, and we had triple-digit inflation on a regular basis. There was actually some talk of maybe dollarizing the economy, which never happened, but triple-digit inflation. Again, this is the kind of inflation where you spend the money as soon as you get it. You don't hold on to it because the next day it's worth a lot less.
- 6.36** Stores, in those days, things weren't electronic. Stores had to change prices every single day. And in some hyperinflation countries, they change them every hour. So literally every evening, if you owned a grocery store, you went and you pasted the new prices on every single good, every day. Think about, again, the waste of effort, time, life that that consumes, that that takes up. It's just horrific.
- 6.37** So the Fed Reserve, it's not that bad. But it's really horrific when you compare that inflation rate to the gold standard. From 1879 to 1913, inflation in the United States on average was 0.13%, basically zero. You had a piece of gold in 1879 and a piece of gold in 1913, they were worth about the same. They could buy about the same. Compare that to 569 %, or 3.9 a year. 3.9 a year sounds like low, but then you compound it and you see how big that number gets to.
- 6.38** So inflation goes up, went up during wars, World War I, World War II, because again, the government's spending money in order to fund the military. And it goes up when the government starts spending without control, because the government needs to get the money from somewhere. And it could raise taxes. The government has the option of raising taxes, balancing its budget, spending all it wants, but making us pay for it directly, honestly. At least that's honest.
- 6.39** Instead, what it does is, it doesn't ask us to pay for it now. It asks us to pay for it over time, with a depreciating currency, with a depreciating money, with a money that's worth less and less. Still taxing us, still a tax. There's a 2% tax every year on all of us, from the fact that our money is worth less. But they don't admit it. They don't tell you it's a tax. And most people don't realize it. Most people don't know. All they know is taxes. "Oh, I pay that at the office," right?
- 6.40** And yet, the fact that we lose our purchasing power of the money is going directly to make it possible for them to run the deficits that they run, because what happens? Why is the Fed buying all this government debt? Because the government's issuing a lot of debt. Because it has massive, massive deficits that are only growing. Over the next—this budget bill that just passed is basically—our debt is going to grow by \$30 trillion

over the next 10 years. \$30 trillion. Trillion. And that's grow above what we already have. Payback, none of it. Only grow. Only take on more debt.

- 6.41** Well, where's that going to come from? Where's the government going to get the money? Because it just cut taxes. It didn't increase taxes. So in an act of unbelievable dishonesty, it cut our taxes. "No, you get to keep all of your money, and then we'll depreciate it even faster." I mean, that's unbelievably dishonest. And yet that's what's going on. And it's been going on for a long time. It's nothing new, running deficit after deficit.
- 6.42** The last time the United States had a surplus, for a very short period of time, I think for one year, was in 2000. Bill Clinton's the last president to run a surplus. And most of that is luck because, in 1999, the Nasdaq, Nasdaq stock exchange, Nasdaq index went up over 90% in one year. A lot of capital gains. A lot of taxes collected off of those capital gains. Guess how much it lost in the next two, three years? 75%. So a lot of people paid a lot of taxes in 2000 and then lost all the money on which the taxes were based.
- 6.43** Pretty horrific situation to be in. They probably paid more taxes than what they were worth a year later. We don't run, we don't run surpluses. Again, we haven't run surpluses in a long time. And before 1999, I can't remember what the last year, or 2000, I can't remember what the last year was that we ran a surplus, probably going back to the '50s or the early '60s.
- 6.44** In more recent times, of course, we've had a financial crisis. And coming out of the financial crisis, a lot of people thought we would have a lot of inflation. Because after all, the government bailed out everybody. It issued a lot of money. Then we got something new called QE, quantitative easing, where the Fed bought up huge quantities of bonds, mortgage-backed securities, government treasuries, government long-term bonds, not just short-term bonds; typically, the Fed buys short-term bonds.
- 6.45** It seemed like there was this flood of new money coming into the economy and to be used. What was interesting is, we didn't get any inflation. Inflation was fairly low. Now, again, when I say we didn't get any inflation, still the inflation was 1, 2%, so we're still getting taxed. We're still getting the value of our money being depreciated, just not the massive inflation people expected. A lot of economists were really puzzled by this.
- 6.46** But if you really look at what was going on, if you look at the balance sheet, yes, the Fed was buying a lot of these treasuries and a lot of these mortgage-backed securities. And the money was becoming reserves at the Fed in the name of these banks that were selling these securities to the Federal Reserve. And then the Fed was paying interest to these banks, to keep it as a reserve at the Fed— in other words, not to lend it out, not to introduce it into the economy.
- 6.47** So the Fed seemed to have found a way to manipulate interest rates by buying and selling treasuries, to support the housing market by buying mortgage-backed securities, without creating inflation, which was kind of a cool parlor trick. And it surprised a lot of

economists. It probably didn't surprise the Fed. I think they knew what they were doing.

- 6.48** They were basically immunizing. Immunizing is doing the opposite, to offset the effect, the effect of introducing more money into the economy by, in a sense, paying higher interest rates than otherwise on the reserves, to suck money out of the economy. So on net, the amount of money in the economy did not increase that much. And that's why you did not get inflation during those periods of time.
- 6.49** On the other hand, in 2020 and 2021, the Fed did not insist, did not increase interest rates on reserves, did not encourage reserves to stay where they are. On the contrary, they wanted people to go out and spend the money. So if you remember the big stimulus packages, there were two or three of them. The first one was under Trump in 2020, and then there was the big one in January 2021, the first bill that Biden passed. These were trillions of dollars, both of them, and these were supposed to be stimuluses.
- 6.50** So they were basically checks into the bank accounts of Americans. And what happened there is, the Fed directly bought treasuries. The treasury took that money and sent it to all of us. We all got checks. And we went and spent it. The banks didn't take that money that now appeared on their balance sheet as a deposit and put it in reserve and keep it there with a high interest rate. They didn't offer us a high interest rate to keep it in our checking account. Quite the contrary, they wanted us to go out and spend it. They wanted to stimulate the economy.
- 6.51** So we did. We went and spent the trillions of dollars they gave us. They gave us a payroll protection program. There was just a stimulus where we each got a check in the mail. We all got money, and we all ultimately went to spend it. What happens when you suddenly have a bunch of money and you go spend it? And the number of goods in the economy hasn't increased; it's actually shrunk because it's a supply shock. You get inflation, and prices went up.
- 6.52** In 2022-23, it takes a while for this thing to filter through the economy. And even though the supply came back, prices continued to go up, because they had distributed so much money into our hands that it had that effect, and they did not suck the money back out. Now, they tried. They started doing that in late 2023 and into 2024. They started sucking money out of the economy. If you look at the Fed's balance sheet, the Fed was actually selling assets, selling bonds, which means money was going into the Fed, trying to get money out of the economy, trying to get it off the reserve accounts of the banks, so they would lend less.
- 6.53** And of course, the Fed started increasing interest rates. What does it mean, the increase in interest rates? They're trying to incentivize the banks to keep the reserves, so that they didn't enter the economy and cause consumers to go out there and raise prices. Raising interest rates is indirectly a means to reduce the circulation of money,

reduce the amount of money we spend, which will at least reduce the amount of inflation out there.

- 6.54** And we're still in the middle of that experiment. Inflation is still somewhere between 2 and 3%, depending on the month. They still haven't got it down to where they want it, which is 2%. Now there's a battle about, should they reduce interest rates? How much should they reduce interest rates? If we head into a recession, do we want to then stimulate things by reducing interest rates so money circulates or we get out of a recession? Again, all the wonderful fantasies of a central planner, thinking about how they can manipulate things.
- 6.55** This is the reality of central planning, and this is how it works. If you think about a gold standard, if you think about the old system, nobody raises interest rates, nobody lowers interest rates, nobody pumps money into the economy, nobody pumps money out of the economy. It adjusts. I know it seems like fantasy. It seems like, it seems magic, but that's how markets work.
- 6.56** Markets work that way. They adjust the supply and demand, without the need for this centralized intervention. In 2022, we actually had, if you remember, we had 9% inflation, which was the highest inflation rate we'd had since the 1970s, which is high. Today, again, it's running at somewhere between 2.5% and 3%.
- 6.57** What is the cost of inflation? Now, if you think about it, there are two types of inflation or two ways in which inflation manifests itself. One is what we call anticipated inflation. If I know inflation is going to be 2% every year for the next 10 years, OK, I can price that. We talked about interest rates, real rate, plus an inflation premium. I know what that inflation premium should be, 2% for the next 10 years. That's easy. And I can price everything in. I can include it in my calculations.
- 6.58** So it has, some people think it has no downside. I'm not going to argue that, but it has less downside. Because now I know what it's going to be. It's constant. But even there, it's eroding my purchasing power, and it's basically transferring wealth to the government away from all of us. Every year, the dollars that I have are just worth less, and the government benefits from that.
- 6.59** How do they benefit from that? How does the government benefit from inflation? Well, its debts are worth less. Its debts are being depreciated. Its debts are denominated in dollars; and a debt is worth less every year because of inflation. I can't, I mean, generally, who benefits from inflation? Imagine you've got a mortgage. I mean, this actually happened to me.
- 6.60** I had a mortgage on a small apartment, a small mortgage, fixed rate, 30-year mortgage or 20-year mortgage, I can't even remember. And I can't remember the amount. Let's say \$100,000 mortgage. And I was paying, I don't know, 7% a year. And inflation goes up to 200%. What happens to that mortgage? It becomes nothing. If my home value is increasing with inflation, then my home value just doubled. The size of the mortgage is

still the same. I'm still paying 7%. The amount of money that the mortgage constitutes within a few years is going to be close to zero, because of the hyperinflation.

- 6.61 Who loses there? I gain a lot. I love to have fixed debt in a hyperinflationary environment. If you're the debtor, you're golden. Well, the banks, they hold the mortgages. They're getting crushed. They thought they were going to get X. They're getting zero, or something very close to zero, because of it, because the money they are getting is worthless, given hyperinflation.
- 6.62 Also, renters.
- 6.63 Yeah, renters are going to pay a lot more, but their rent is going to adjust to the inflation. So are new purchases of homes. Now that we know that inflation is higher, we'll adjust the mortgage rates. Now you won't pay 7%, you'll pay 100% a year interest to adjust. So whether anybody would take out a mortgage if interest rates are that high, probably not, so people who want to sell their homes can't sell their homes.
- 6.64 It doesn't cost anything to produce more money. Government uses the additional units that it is producing in order to spend, so it can increase. Our purchasing problem is being eroded. So even when it's anticipated, there's a price to pay and a benefit to the government. On top of that, we now have to spend time and effort to think about how to protect ourselves from this 2% inflation or whatever the number happens to be.
- 6.65 There's time and effort expended by individuals and business people and everybody else, to try to deal with whatever inflation there is, even if it's anticipated. Time and effort needs to be put into it. This is time and effort that you could be using to be more productive, to creating more, rather than just preserving what you have. It could also be time and effort that you spend on leisure or whatever you want to spend it on. Your money.
- 6.66 Remember, life is time. And it's the one finite, precious thing you have. Can't get any second back, can't relive, can't rewind. Life is one way. Every minute that's gone is gone. Any minute you spend on something useless is a minute less of life that you have. You should be pissed off when that happens. Sitting and waiting for the bureaucrat to stamp your whatever, that should infuriate you. Because that's all you have, is time.
- 6.67 And of course, we also then keep less money, less money in our pockets, less money in our checking account. Like, today, I'm not going to keep a lot of money in my checking account. I'm going to immediately roll it into a saving account, because a saving account, I can get, I don't know, 4%. My checking account pays me nothing. I'm losing value. Every dollar that sits in my checking account loses value every day, even today with a low inflation. It literally loses value every day.
- 6.68 So it doesn't make any sense to keep money in a checking account. But you need it, because we need to transact. So again, we have smaller pools, more dealing with all of

that. And of course, inflation just creates more noise out there. It just creates more uncertainty, even if it's stable, even if it's knowable. We don't know exactly.

- 6.69** Let's say it's going to be 2% on average over the next 10 years. We don't know. One year, it could be 4%, another year, it could be zero. It could be fluctuating throughout that time. It just creates uncertainty, reduces efficiency, and reduces our ability and our efficiency and our productivity of allocating resources. So it's just a drag on the economy. Even anticipated inflation is a massive drag on the economy.
- 6.70** Now, take unanticipated inflation, inflation we don't expect, like the inflation we had in '22, '23, where people, '21, '22, not '22, '23, people didn't expect it. Well, now I took out my mortgage at 3%, inflation is at 9%, I'm happy. My bank is taking a beating. Take Silicon Valley Bank. You remember Silicon Valley Bank went bankrupt in 2023? During COVID, Silicon Valley Bank, Silicon Valley Bank traditionally had not made a lot of loans.
- 6.71** It was in the business of taking in deposits from startups and venture capital companies, and then selling them a lot of services. And then they would take that money, they'd do some loans to startups, but startups don't borrow that much money. Most of the money they'd invest in securities. During the early part of COVID, interest rates were very low. I mean, the Fed drove interest rates very low, basically zero, and they couldn't make money on all these investments that they were doing.
- 6.72** So what they did was, instead of buying short-term securities, they started buying longer-term securities that paid a little bit of higher interest rates, so they could make some money to run the bank. That's their profit. The profit is between the money they spend on interest payments. I don't think they were paying depositors any interest, but still they need to make a margin. So they bought longer-term securities under the assumption that they could make money off of the gap.
- 6.73** Well, what happened to those longer-term securities when interest rates went up because of inflation? The value of those longer-term securities go down. Interest rates and value are inversely related. So suddenly, they had this asset on the balance sheet that they had a mark-to-market down. Suddenly, their net worth was negative.
- 6.74** Now, the reality is that if you'd done the calculation for every bank in the United States, I'm guessing, but it's not a bad guess because I think I know this, well over 60% of American banks were bankrupt on paper at that point because they all were writing down their securities. They all had done basically the same thing, in one way or another. And even their loans needed to be written down because now there was— because of inflation, because interest rates were up, the value of whatever they were receiving from these loans was lower.
- 6.75** So if they'd written everything down, mark-to-market, every bank, almost every bank in the U.S., would have been deemed bankrupt, negative net worth. Silicon Valley Bank had the disadvantage of— they tried to raise capital in order to beef up their capital position, to beef up their net worth. It became public that Goldman Sachs had turned

them down for capital, or they couldn't cut a deal. And then the community of venture capitalists discovered this, figured out, "Oh, this bank's in trouble," and started running the bank.

- 6.76** And these are sophisticated investors, so they didn't wait till the morning to go stand in line. They all got on their phones and they moved their money out of the bank. By the time the bank opened in the morning, their deposits were gone. And they were shut down that day, because no bank in the United States can withstand a bank run. They just don't have enough reserves available, liquid reserves. And if Silicon Valley Bank had to sell all the securities at these lower valuations, even then they couldn't pay off all their deposits.
- 6.77** But that's the circumstances of unanticipated inflation. Inflation went up, the value of the bonds went down. If they'd known inflation was going to go up, they never would have bought the bonds. So real harm, real damage gets done. And in this case, a number of banks went out of business. Two banks were closed, and a lot of banks got massive hits to the valuations. They lost a lot of money during the weeks that followed the closure of Silicon Valley Bank because everybody suddenly realized, "Whoops, all these banks are really underwater."
- 6.78** Now, the government, of course, bailed everybody out and secured everybody's deposits. So the bank run stopped because nobody has an incentive to run on a bank if the bank is guaranteed by the government anyway, so I'm not going to lose anything, which is what the government did. Not good for the long run, because now everybody knows that, no matter what the bank does, the government is going to bail them out.
- 6.79** And Silicon Valley was not like, what did they call it? A systemically important bank. It was a big bank, but not that big. And outside of Silicon Valley, it had almost no presence. It just wasn't that important, and yet they got bailed out. So the idea is now, everybody's going to get bailed out, which is standard practice. So, when inflation is higher than expected, it harms creditors. It harms the people who make loans and it helps debtors.
- 6.80** Indeed, I think I mentioned in a previous class that during the 19th century, farmers were really interested in introducing silver into the economy as money in order to inflate the economy because they wanted inflation, because all their debts would have depreciated in value and they would have benefited and basically screwed New York bankers who they were happy to screw. Generally, we don't like New York bankers, so they would have lost money. Creditors lose money, debtors benefit. Like my mortgage, I benefited, the bank lost.
- 6.81** So high unexpected, higher than expected, harms creditors. If lower than expected, if inflation is lower than expected, then creditors love it. I charged you 8% on the loan, but it turned out that inflation was only one. So I made off really well. If inflation had been one when we issued the loan, and I'd known that, we'd probably only negotiated

a 7% interest rate. Now I got a 9% interest rate, so I got 2%. So creditors like lower than expected inflation, debtors like higher than expected inflation.

- 6.82** So again, somebody wins and somebody loses, not because of themselves, not because of anything in the transaction, not because of anything in their business, not because of anything in their control. But because the government decided to do this, not that, because the Fed decided to inflate or not inflate or whatever. Volatile inflation or an uncertain inflation makes planning very difficult. As we I said, can't tell what the money is going to be worth in the future.
- 6.83** It deters borrowers and lenders because they don't know. Am I going to be the one who loses or are you going to be the one that loses? We know one of us is going to lose because we don't know what inflation is going to be. We can guess, but it's just a guess. So let's not do the deal. Sometimes it's easier not to do the deal than risk the loss. Again, it's not in your control. It takes control away from you.
- 6.84** And as a consequence, financial markets shrink, economic activity shrinks. Inflation risk now, in a sense, is added on to default risk. You've got another risk to take into account when you're calculating all these things. So again, added complexity. And the consequence of that is shorter time horizon. I think I mentioned in a previous class that railroads, American railroads used to issue 50-year bonds, 50-year bonds. Nobody would issue a 50-year bond today because you have no idea what the money is going to be worth in 50 years.
- 6.85** You can estimate, you can say, oh, 2% inflation. But what if they decide to change the target at 4%? Or what if they miss the target, which is likely? No way to do that. So generally, maturities of corporate bonds have shrunk from, typically, 50 years ago, maturities were about 20 years, they've shrunk to 10 years. And again, in the railroad times, there were even 50 years. That's gone. That you need a gold standard for.
- 6.86** And then there's always the risk of hyperinflation. And hyperinflation, when you're talking about hyperinflation, now you're talking about destruction of money. Zimbabwe, at some point, I had a bill. I had a Zimbabwe bill. I used to travel with it. I should have brought one with me. \$100 trillion. And you could buy a loaf of bread with that, maybe. It was worthless. And of course, it completely collapsed. And that's completely a consequence of their central bank just printing money. You can't get hyperinflation without massive printing of money, massive increase in the money supply.
- 6.87** And in the end, they had to go away from that, and they started using foreign currencies, so the South African rand and others. Venezuela has gone through the same process. And any country that has triple-digit inflation, it's always unanticipated. Nobody actually predicts that. And it's always devastating to standard of living, quality of life, production. Economies just are crushed by it. It makes the Great Depression look like child's play, given what happens with hyperinflation.

- 6.88** Inflation is devastating. We have learned to live with it, to our own detriment. We don't get angry about it as much, to our own detriment. But even, even here, I was a little surprised, and pleasantly surprised, that the inflation of 2023, 2024, there was a real political cost. I don't think Biden, Kamala, whoever it is, loses, if not for that inflation. Out of the '70s, American politicians learned something. They learned that people don't like inflation.
- 6.89** And there was a real commitment not to have inflation or to minimize it. Can't not have it, given fiat money, but to minimize it. And then they forgot or they didn't realize what they were doing, and they got the inflation, and they suffered the political cost of that. And that's a good thing because maybe politicians will learn again not to do it, because the one thing they care about, they don't care about us, they don't care about a standard of living, they don't care about the quality of life, they don't care about the economy, they care about getting reelected.
- 6.90** And maybe this will hold them back a little bit, but maybe not. We will see. All right. So inflation, really, really damaging. Let's look at some other aspects of fiat money quickly before we wrap up on this topic. So we talked a little bit about government borrowing under the gold standard versus government borrowing under a fiat money. Just to remind you, on a gold standard, government faces real default risk.
- 6.91** If the government can't get its hands on the gold to repay, and the only way, excuse me, the only way for it to get its hands on the gold is to either tax it or confiscate it, or arguably, that's the same thing; but they have to get their hands on gold. Otherwise, they can't repay their debts. So when people buy the debt, they include a default risk premium. It's in the interest rate. As the government's debts increase, default risk increases, and interest rates increase. It's pretty straightforward.
- 6.92** And as such, I think history has shown that a gold standard places constraints on government spending, constraints on government debt, at least. If it wants to spend more, if it's going to stick to the gold standard, it has to tax more. It's a political cost. It puts in question whether they really want to spend. So there's a real physical constraint on government when you're on a gold standard.
- 6.93** When you're on a fiat money standard, governments cannot go bankrupt. There is no default risk. What can they do? Print money. And we add an inflation risk premium to their debt, but we don't know how much to put there. We don't know how high inflation is going to be, how low inflation is going to be. It's much more difficult, much more complex. Here you can see government increasing debt dramatically and not suffering quite as much of a premium or cost to doing it as I think you would under a gold standard.
- 6.94** Now, we'll see this latest round of government going crazy and increasing its deficits. We'll see what kind of cost that has in interest rates and the interest rates they have to pay. And think about what happens when the government has to pay more interest on

the debt that it accumulates. That now is part of its spending. So today, the second most money that the government spends after social security, second highest amount is interest on the debt.

- 6.95** And that's only going to increase, because the debt is skyrocketing and interest rates are not going to plummet that much, particularly not long-term rates or the rates at which the Fed borrows. One of the reasons the Fed borrows very short-term is because the rates are lower and it's trying to game that, even though, arguably, it should raise money at long-term rates. That gives us stability over time. But politicians can only think short-term, so they do the short-term rates. And when those go up, suddenly the cost of funds goes up.
- 6.96** A big reason why Trump is hounding Paulson, not Paulson, Powell, I knew it was a P, Powell right now about lowering interest rates is that he thinks that if he lowers interest rates, then government debt, he'll be able to refinance the government debt at short-term, at lower interest rates, and lower the cost of interest for the government. And that may be true in the very short term, but it's not going to be true in the long run because lowering interest rates could raise inflation. If inflation goes up, interest rates will come up, and it will just make it worse, not better.
- 6.97** So fiat money, because there's no default risk, you've got inflation risk, more complicated, more complicated price, less market discipline on government not to grow its deficits, not to grow its debt. And of course, the deficit, the larger the deficit, the bigger the incentive to inflate. How are you going to get rid of this deficit? I mean, if you run the numbers, the amount of taxes they would have to raise, even if tomorrow they balance the budget somehow, which they're not going to do, they would have to raise taxes dramatically in order to fill in the gap. And they're not balancing the budget tomorrow.
- 6.98** So the only way they can get rid of this debt is by inflating it away. So we know we're going to get high rates of inflation in the future. We just don't know when, we don't know how high, but we know it's going to happen. And it's going to be interesting to see how interest rates adjust to that knowledge. And again, it's very hard for short-term interest rates to adjust to that knowledge. We could expect ongoing deficits into the future and no mechanism to bring them down and no real incentive anybody seems to have to bring them down. Politicians, it's very easy to spend money; it's very difficult to stop spending money.
- 6.99** Now, one of the reasons the Fed was created, one of the reasons to bring in fiat money, was the idea that, by the Fed having this flexibility of being able to print money, suck money out, and do all this stuff, play central planner, it could smooth out the business cycles, get rid of recessions, get rid of downturns. And yet, if you look at pre-Fed, post-Fed, post-Fed is much more volatile, a lot deeper recessions, a lot more of them than pre-Fed.

- 6.100** Even if you take out the first, what would it be? You'd have to take out the first 30 years of the Fed and say, "OK, they were babies then. They didn't really know what they were doing. They were still learning the task. So we'll take the Great Depression out. We won't count that," it's still more volatile. So it's not clear what the advantages are. Well, we know what the advantages are. They're political. The advantages are the ability of government to manipulate all these things and to control all these things.
- 6.101** A big issue these days is Fed independence. It's right now July of 2025, and the President is putting a lot of pressure on the Fed to do what he wants them to do or he's threatening to fire and to replace the chairman of the Fed with somebody he wants, who would do what he says. By the way, Powell was first chosen by Trump in 2018. And the reason Powell got the job, because the other guy, there were two people running, and it was Powell and John Taylor.
- 6.102** John Taylor is an economist out of Stanford University, very well-known economist, who believes that the Fed should be ruled by a rule; that is, that money supply shouldn't be determined by people, by theories, by whim, in a sense, but by a rule. And he has a Taylor rule. There's an equation that bears his name. It's called the Taylor rule. He was going to go into the Fed and implement the Taylor rule, which would have taken away any discretion from how much money, how the money supply changes. It would have been dictated by the rule, by the equation.
- 6.103** Donald Trump, I think, didn't choose him on purpose because he was not manipulable, couldn't influence him, because there was a rule. He chose Powell because Powell is the guy he thought he could manipulate. Once you give a president that kind of power and that kind of authority, do you really want the Fed, as badly as it's run, and it's run really badly, to now be completely at the whim of politicians, completely controlled by politics? And politics is at most a four-year cycle, often a two-year cycle, so no long-term thinking, and all about manipulating the next election.
- 6.104** Is that what we want? Because that's what we're going to get if we lose the independence of the Fed. So it's not like an independent Fed is some miraculous organization that's going to do the right thing. They can't; they're central planners; they're doomed to fail no matter what. But their failure is within specific bounds. It's run by economists. They know something about what's going on, and they're not incentivized by the next election cycle. You want to replace that with people who are completely dependent on politics, would be an absolutely unmitigated disaster.
- 6.105** And one of the reasons it's not happening right now, that Powell is not being fired, is that the market, stock market, bond market, knows that. Every time Trump says, "I'm going to fire him," stocks tumble. And he changes his mind, the stocks go up. So it's very clear what the stock market thinks about this. And they're right. Independence, as bad as an independent Federal Reserve is, it's better than a non-independent Federal Reserve. This is the system we have today. Like it or not, it is indeed what we have.

- 6.106** And the fact that we're running these massive deficits, the kind of size of deficits that we have, suggests that we're not going to move away from this system anytime soon or anytime not so soon even, because there's too much entrenched interest in this. The only way to fund these deficits is going to be through inflation, and everybody knows it, and nobody's going to move away from it. It is the system that we have.
- 6.107** And the question is, can you make a fiat system better? Can you make it work? Can you make a fiat system work? And I'd argue, the answer is probably not. Not if you, because it's unmoored from markets. It is central planning, no matter what you do. It is unmoored from supply and demand, from our values, from what we actually want to consume, want to produce, want to do at any given point in time. So the question really is, how do we take a central bank and make it as least damaging as possible?
- 6.108** And I think if we look at it that way, I think something like a Taylor rule probably makes the most sense. Figure out a rule with economic parameters in it that in 99.9% of the time will give you some kind of predictable rate of inflation and won't mess up on a grand scale. And even if it does mess up, it's within particular bounds. Right now, it's again, it's a discretion of the people who run the Fed, who are appointed by politicians. And even though we think of them as independent, they're only partially independent.
- 6.109** So a rule probably makes sense. Of course, rules can be changed. We know central banks cheat all the time. As I told you, the European central bank had a constitution. They were supposed to limit what it could do and only do certain things. And that was thrown out as soon as the Greek financial crisis happened, and then it was thrown out again during COVID, and it just doesn't exist anymore. There's no constitution. European central bank can basically do whatever they feel like doing. There's no limits on what they do.
- 6.110** So, can fiat money work? It's never really worked in the past because of its separation from fundamentals, from markets, from prices. It sets interest rates, which are market phenomena, and should be set by the market, supply and demand of lower level funds. That in and of itself is distortive. I don't see how it can work. I mean, it works now. We still have a high standard of living. We still do pretty well. We're still fairly wealthy. We don't know how wealthy we could be, how rich we could be, how much of a standard of living, how high it could be.
- 6.111** There's alternative universe somewhere where we never got a Fed and we're still on a gold standard, and we're a lot richer than we are right now, and a lot more productive than we are right now. But this is the system we have. It's probably not going away. But people are thinking about alternatives. People are thinking about, could we return to a gold standard? Not a lot of thinking going into there, but it's definitely a possibility.
- 6.112** Some economists have thought about it. There's a way to make that work, to convert the current system to gold standard system with minimal damage, with minimum costs. There are variations of it. Some want a central banking gold standard. Others want a

true free banking gold standard. But then of course, there's the big alternative that's out there that people are presenting today, that people are enthusiastic and excited about today. And that is crypto, bitcoin as money to replace ultimately a fiat money. We'll talk about that next class.

- 6.113** A question just to validate a possible observation. It seems like the collectivist, the more centralized planning, it's the same thing like the neo-Marxists. They're unmoored from reality, essentially. It's like they're not constrained by reality, versus the gold standard would be more in connection to actual—
- 6.114** I mean, the gold standard anchors us to reality. It is based on reality. It's based on markets. It's based on individual preferences and individual decision-making. Yes, I think generally collectivists are unmoored by reality. But remember, when you say that, then almost everybody's a collectivist. The number of people who believe that fiat money is a bad idea, me and five other people, I don't know, I mean, it's not a lot of people. It's a very small group, even within the economics field.
- 6.115** 90% of economists, no, I'm probably underestimating, 95% of economists, 99% of economists, are pro-fiat money central banking and can't even imagine an alternative, and even some decent economists. So while I think this is true, and the facts are on my side, our side, it's a minority view within economics. It's certainly minority, there's nobody in politics, nobody, zero in politics.
- 6.116** I mean, Ron Paul at some point says, "Audit the Fed. Just audit it." I don't know what help that would do, to audit it. And he got lambasted for that, and that never happened. We don't even audit the Fed, never mind end the Fed. So no, there's just, the vast majority of economists, the overwhelming majority of intellectuals, and the overwhelming majority of the American public support this system.
- 6.117** Now, to the public's, in defense of the public, they're mostly ignorant. Hopefully this course will change that a little bit. But they're mostly just ignorant. They don't know. This is what they were born into. This is what they know. This is what happens. They don't know there's an alternative.
- 6.118** There are a few that play around with bitcoin, who think they know what the alternative is, and think they come up with a better system, and they have a better understanding of the evils of the current system than anybody else does, even if they might not necessarily have the solution, but at least they have an understanding. But again, 99.9% of the people, 99.9% of the intellectuals and people out there, have no clue and are part of this, if you will, detached-from-reality perspective.
- 6.119** So I've played around with compound interest calculators, and if you just keep going at a rate of 4% inflation, you're going to be dealing with millions and millions of dollars. So is that just going to be the case, or do you think at some point we're going to say, hey, we're planning, 200 million now is the equivalent to a million today?

- 6.120** I told you I lived in Israel in the '70s and '80s, and we took five zeros off the currency—two once, and the inflation continued, so we had to take another three off. They changed the name twice. We used to have lira, then it became the shekel, then it became the new shekel. Five zeros total off the currency. And look, ultimately, it stabilized. You get 2% inflation and all that.
- 6.121** But this is the other thing you have to take into account is, so yes, at some point, you either take zeros off or it doesn't really matter anymore because you never carry currency. There's a huge, we'll talk about this in a future class. There's a huge incentive to go all electronic, so you never actually have currency. So you never see the number of zeros in that sense. You're going to use some digital currency in the future. The central bank would like you to use theirs, if they can get away with it.
- 6.122** I think what the powers that be will try to do is divorce you from the touch and feel of it as much as they can. But yeah, it's inevitable. I think we'll probably have some really high inflation every period in the United States, and then maybe we'll have to restructure stuff, and maybe we'll get some sanity at the federal level. But as long as they're running these deficits, high inflation, not yet hyperinflation, but high inflation is a must. There's just no alternative to it.
- 6.123** And if you compound it—the nice thing about compounding is it also works the other way around. Take your income and compound it at 4% a year for 40 years, and you suddenly realize how rich you're going to be. And take a poor person's income. This is just an aside, not related here. Take somebody making \$30,000 a year today; compound that at 4, 5%. Let's say we could achieve 4, 5% GDP per capita growth annually, which I think is doable. I don't think it's going to happen, but I think it's doable.
- 6.124** Suddenly, there are no poor people in the world. Everybody's making six figures in real terms, because I'm talking about 4% growth, real 4% growth, not nominal. And so the solution of poverty has always been growth, economic growth. The solution of poverty is not welfare. The solution of poverty is higher economic growth, so that nobody is poor anymore.
- 6.125** Now, there are relatively always going to be people who are poorer than others. But if everybody's making six figures, nobody's poor. And arguably, there are very few people today in America who are poor, particularly if you look at other countries and how much people make in other countries. Very, very few people. But yes, it's insane once you compound stuff.
- 6.126** So, if we were to go back to a gold standard, and given most of the world's on fiat currency, how do we avoid the problem that we had post-World War II, where we're exchanging actual backed currency for, essentially, funny money, and then they're turning it in for the gold?

- 6.127** It's going to be very difficult. And this is why, if you go back to a gold standard, it's probably possible to do, but then you'd have to have some kind of capital controls, where their money is not redeemable. But ideally, when you go into a gold standard, you have some international alliance and you create some international agreement where everybody goes on a gold standard one way or another. But you could do it.
- 6.128** America is a big enough economy that you could just do America. If you're a small country, if you're, I don't know, El Salvador, you're not going to go on a gold standard all by yourself. You could go on a dollar standard because you're sharing now a currency with, money with America, but you're not going to go on a gold standard. America could probably get away with it and then make other currencies non-redeemable into gold.
- 6.129** And if they are, if they're not fixed to the dollar, if the currencies are moving, then they would adjust based on their own inflation rate vis-à-vis our inflation rate. So it would work. But it would have to be a large economy to start it off. And it's much better if it was global, because then you get international trade going as well.
- 6.130** Before we get to that, then, how would you distinguish between money, currencies, and assets? And why is bitcoin and gold not immune to inflation while under a fiat currency system?
- 6.131** Why is what not immune to inflation?
- 6.132** Bitcoin and gold, they're not really immune, in our current fiat system today, versus if we were on a gold standard.
- 6.133** How do I differentiate? So money is basically the medium of exchange, the thing that you use in order to buy and sell, not literally, physically, but the unit of account that is measuring what you're buying and what you're selling. So even though we had paper currency, it was all denominated in gold. Gold was the money. The paper was the currency. So paper is what we transact with. But since it's redeemable in gold, gold is what is actually covering our purchases. Nobody's taking the paper for the sake of the paper.
- 6.134** Now, in the world in which we live today, if you have money, currency and money, money is broader because money includes, for example, reserves at the Fed, whereas currency is just the paper money we have in our pocket. Even our checking account, I don't think it's currency. It only becomes currency when we take it out in the ATM. Our credit card is not currency. So money are the dollar bills that are available to us to transact.
- 6.135** An asset is something—it's not a medium of exchange. It could be used as a medium of exchange on occasion. But an asset is something we invest in. It's something that we purchase with the expectation that it will appreciate in value. Usually, assets pay some kind of cash flow. They pay a dividend or an interest payment, something that

appreciates, or rent. A home can be an asset. Your home is not an asset. In spite of what you're taught, your home is not an asset. Your home is a consumer good. And in a rational free market, its price would go down over time, not up over time.

- 6.136** The fact that your home value is appreciating is a complete artifact of a statist economy that does not allow for supply to match demand by building more housing. Now, in particular areas, it might go up because demand for those areas might be lucrative and prices go up. But on average, home prices should go down, just like car prices. As soon as you take it off the lot, price goes down. You used it. You're using your home. It's worth less now than it was new. It's less nice now than it was when it was new. So it should be a depreciating asset, but it's not. So it's not an asset. It's a consumer good.
- 6.137** But rent, if you buy a home to rent to people, then that's an asset because you're getting—now, you do have assets that don't have cash flows, and whose value is determined by speculation, call it? Greater fool theory? By the assumption that somebody will buy it from you for a higher price sometime in the future. Or the idea that, at some point, there'll be cash flow.
- 6.138** For example, for many, many years, a lot of the tech companies didn't give out dividends. There were no dividends, so there was no cash flow coming into you. But the assumption was, they would at some point. They're profitable, there's money coming in, and at some point, they'll start paying out dividends or in some way providing cash flow to investors. When an asset is an asset that does not pay cash flows and is not clear where its value comes from, I call that a speculative asset. So it's basically, its valuation is driven by speculation.
- 6.139** And as we'll discuss next class, bitcoin is, in a sense, a speculative asset. I can get where the value comes from, but it's very speculative because the probability of that outcome actually happening is very low. So that's why I consider it a speculative asset. It's certainly not, again, we'll talk about it next class, it's certainly not money because it's not a common medium of exchange. You could argue it's a store value, only because it's a speculative asset and it's gone up. But speculative assets can also go down, and then it's a store of disvalue. It destroys value.
- 6.140** One thing that is a common principle in macroeconomics is this concept of the crowding-out theory that when governments are running high deficits and have a lot of debt, that therefore all the money that gets used that would normally go towards investment and growing the economy is instead spent on just funding the government debt through the purchase of treasuries. One of the ways, at least in theory, that the U.S. has avoided this is just by increasing the money supply a lot by having the Fed just print more money.
- 6.141** If we were to either keep the money supply stable or slightly reduce it, but still keep running our massive deficits, does that increase the risk of potentially having other

assets be lowered in value by the fact that suddenly all the money would just be going towards the debt?

- 6.142** I'm not sure that inflation really has that effect, because it's crowding out no matter what. So no matter what, it's crowding out because every dollar that is going to the government could go for productive activities. Remember, the government doesn't produce anything. It only consumes. So government dollars could be going to productive activities, and they're not. So even if you're inflating, those inflated dollars could go to a productive activity. So it doesn't really matter. There's crowding out, no matter how you look at it.
- 6.143** And indeed, arguably, look, it's not arguably. The U.S. government has been bailed out. We all, not just the U.S. government, we all have been bailed out by the fact that there are foreigners who are savers. We don't save. The United States hasn't been saving for a number of decades now. But the Japanese save, the Chinese save, the Koreans save, many cultures out there save, even some of the European countries, they save. And one of the things that they do with their savings is, they worry about their own local economies.
- 6.144** They worry about their own local currencies, their own central banks; is they convert their local money into dollars. And they do that, by the way, by selling us stuff. That's how they get dollars. So the whole trade deficit is, to a large extent, driven by this fact. They want dollars in order to invest in the United States. And their primary investment is safe investments. They buy U.S. treasuries. So they are constantly buying American stuff because they are savers and we're not.
- 6.145** And that, to a large extent, has made it possible for us to run the kind of deficits and everything that we do. If we had a balance, if we actually got rid of the trade deficit, if we got rid, if we actually started, if we stopped importing, we stopped importing, stopped sending dollars overseas, who would buy our government debt? We won't. We consume it. We just spend it. We don't save. So luckily, the Chinese and Japanese save, and they save in dollars, and they save by buying our debt.
- 6.146** So again, the whole conception of the trade deficit is completely wrong. The whole way in which people think about trade is completely wrong. The fact is, the trade deficit has massively benefited the United States. Now, we're funding it by carrying these massive deficits, which do matter, which are the deficits the government runs. That's what matters. Trade deficit doesn't matter. It's actually good.
- 6.147** What matters is our spending deficit. And that's the disaster, and that's what nobody will touch. So instead of touching that, they focus us on the trade deficit. We get really upset about the trade deficit when that's not the problem. That's just a shiny object out there that we can focus on and get all excited about, when the real problem nobody wants to address.

Lecture 7: Crypto as Currency

At a glance — Crypto as currency.

- 7.1 The last couple of classes we've talked about the existing monetary system that we live under, that really the whole world lives under, which is the fiat monetary system. We have fiat money, money that is not redeemable into anything else, money that is controlled by the central bank, which we can pretend is mostly independent, but has strong political influences. So really, money that is very much influenced by politics, by our government.
- 7.2 And we've talked about some of the limitations of this money, maybe most importantly, the fact that fiat money inflates, it loses purchasing power over time. And when a government is acting responsibly, when a central bank is acting responsibly, or you could say when a central bank knows what it's doing, then that inflation might be moderate, might be fairly predictable and stable.
- 7.3 But we know from the history of fiat monies now for quite a bit of time that that is not sustainable, that typically you get bouts of inflation every few decades or even more often. And in some cases, indeed in many cases in parts of the world, you can get significant bouts of inflation, hyperinflation, even triple-digit inflation, where the value of the money gets wiped out. And this creates, we talked about this, real problems, problems for planning, problems for savings, problems for investing. This is not ideal economically to have both elements of predictable inflation and unpredictable inflation all the time.
- 7.4 And of course, inflation is often a product of politics. It's almost impossible to divorce fiat money from politics. So a big challenge with fiat money is inflation. It's also this idea that it's controlled by a central government and therefore politics enters it. And now you could have influences on money, on its purchasing power that are not economic, that have to do with politics, with favoring particular groups, creditors over debtors or debtors over creditors. And that could influence the way money is manipulated.
- 7.5 As we know right now, we're in July 2025, Donald Trump is putting a lot of pressure on the central bank to lower interest rates dramatically. Does that make economic sense? No, but it makes some political sense and certainly makes short-term political sense. And this could be a disaster if it's followed through. It might not be. It's really hard to tell. But now it's politics; it's not economics.
- 7.6 And indeed, we've taken the market out of the equation. So in many regards, we've divorced our money from real economics in terms of market pressures. And much of money, again, is politics. For several years, the president of Turkey, Erdogan, tried to keep interest rates low. He believed in low interest rates with a passion, and inflation spiraled out of control. He kept those interest rates low no matter what. And

economists kept telling him, "No, no, no, you have to raise interest rates. This is how—" He didn't listen to them. It was an economic disaster for the Turks.

- 7.7 But of course, again, this is what happens when politicians control. And it's not like I'm saying economists—economists would probably do a better job, but they can't do a perfect job either because you've taken away markets, you've taken away supply, real supply and demand, and you've enforced central planning. Central planning generally doesn't work on anything, never mind on money.
- 7.8 The fact that we have fiat money and that everything therefore ultimately goes through the Federal Reserve, everything ultimately is controlled by the Fed: how much currency we have, but really, more importantly, our bank accounts are monitored. They know, they have access to, what we have there. There are lots of laws on the books, particularly since 9/11 and the Patriot Act. They can decide that you might be a national security threat. They could decide that you're money laundering. A lot of different reasons give them access directly to your bank accounts, to your transactions, to everything that you do.
- 7.9 It's very easy to do because most of what we do is electronic. And the Fed really sits at the heart of the electronic system. Everything goes through the Fed. They're the ones clearing all their bank reserves that we talked about. So they know what you are doing financially unless you use cash. If you use cash, they don't know. That's the one anonymous way in which we can transact.
- 7.10 I don't know if you've noticed, but if you've ever had the opportunity to try to deposit at your bank more than \$10,000, suddenly you have to fill in a bunch of forms and you have to explain what it is. If you withdraw more than \$10,000, fill out a bunch of forms. If you bring into the country, one of the few things they ask you when you come into the country is, "Are you carrying more than \$10,000 in currency?" And there's a lot of pressure to lower the threshold to start doing that with much lower numbers.
- 7.11 When the \$10,000 was put in the legislation, that was decades ago, it wasn't adjusted for inflation. So \$10,000 is pretty low for what they intended back then. But now, again, this allows for much more surveillance of what we do by the state. And again, if the state is mostly good and is not bothering us, then unless you're a criminal, you've got nothing to worry about. But we know from history that states can change, and privacy is something that is valued. How much it's valued is a question we'll talk about, but it has some value. So again, some of these challenges with fiat money are in the current system.
- 7.12 And then finally, one of the challenges with the current system is speed. Up until recently, it was very difficult to transfer money to other people in terms of electronic transfers. I don't know how often you've done wire transfers or ACH or all these things. You have to go in, fill out a form, bunch of security behind it, and then it takes time. Try to make a wire transfer outside of the country. It could take days, and they clear it on

both sides. When you deposit a check, for example, they don't—you don't have access to all the money necessarily. They keep some of the money. Why can't—

7.13 Why isn't everything moving at the speed of light? It's all electrons now, but it doesn't. Everything is going through the Fed. The system is slow, clearing is slow. Now it's getting better. Right now, most ACHs go within the same day. Most wire transfers go within the same day, but it's still clunky, slow and cumbersome. I still get wire transfer fees from my bank—15 bucks for a wire transfer, which is not a lot of money, but it's something. And again, what does it cost them? It's nothing.

7.14 On the other hand, we're at the point where we can do PayPal or Venmo or Zelle transfers, peer-to-peer, that are instantaneous and cost zero. So why can't I do it with larger sums? I once tried to do a PayPal transfer for a large amount of money. It took weeks. PayPal kept calling, "Are you sure this is the right amount of money? Are you sure it's going to the right person?" They called the person who was supposed to get it, "Please confirm your identity." It was a disaster. So it's great if you're transferring small amounts, but once you go above a certain threshold, it becomes really difficult. Everything has to clear. In spite of the fact that we live in the Internet age, in the electronic age of instant gratification, transferring money, particularly in size, is cumbersome and difficult.

7.15 While the transaction we do, like with Amazon or something like that, is instantaneous, the settling that goes behind the scenes is complicated. We love using credit cards. Credit cards are great. They're instantaneous. They're easy for us to use, buy stuff, but the fact is, the merchant has to pay a 2.5% fee, which means who's paying that? You think the merchant's paying it? You're paying it. So goods generally in the economy are about 2.5% more expensive than they would otherwise be if we all paid cash, because they have to make it up somewhere.

7.16 So transactions are costly. Credit cards are costly. It's not free. The fact that you get points and miles and all this stuff on your credit card just suggests that they're charging somebody, somewhere, something to compensate them for giving you the points. So again, the massive amounts of fees and costs embedded into the system make generally everything more expensive. That's the consequence of that. So it's slow, settlement is difficult, and particularly international is very difficult and very expensive.

7.17 When you have immigrants to the United States come and they make money and they send some of the money home—which is huge, that's a huge business, sending money home from the United States—it's costly, it's expensive. And these are typically people who can't necessarily afford that. And yet a portion of what they're sending home gets taken in the process, the Western Union or whatever the mechanisms used to be to transfer it.

- 7.18** The current system has real challenges, particularly given how sophisticated the world in which we live is and how sophisticated our telecommunications are and how sophisticated the Internet is. You'd expect better.
- 7.19** And indeed, a lot of people over the last, I don't know, 40 years, really, since the 1980s, since the Internet started—the Internet was born sometime during the 1980s, really became the Internet as we know it today, or similar to what we know today in the mid-1990s. But even in the 1980s, people already starting to think, "Oh, we can connect computers. How do we create digital money, some form of digital money? Using cryptography so that it was safe, so that it couldn't be stolen, so that it maintained privacy. How do we do that?"
- 7.20** There were a number of different attempts at this. As early as 1983, a guy named Chaum, C-H-A-U-M, invented something called eCash, which allowed for blind signatures, in a sense, digital contracts, private electronic transactions, all based on mathematical proofs, a proof of identity that was determined mathematically, all through cryptography. He actually started a company based on this idea called DigiCash in 1989, that early, 1989.
- 7.21** 1989 is, I think, when I discovered email. And that was early. The only reason I could do that is because I was connected to a university system, that was connected to other university systems and people—the killer app, by the way, the killer app that made the Internet was email. So that's about when we discovered email. He was already thinking about DigiCash. These people are pretty amazing.
- 7.22** Anyway, he tried to do this. He struggled for about 10 years. In 1989, ultimately, DigiCash went bankrupt. And what he basically discovered was that privacy is not that important to people. Convenience, speed, ease of use, just doing what you're used to doing—much more important than privacy. And generally, I think what you see over and over and over again is, and we'll see this with Bitcoin as well—privacy is not that important to most people. To some people, it is. To a subgroup, it is. But most people don't care.
- 7.23** We talk a lot about privacy. We say we value privacy. In surveys, we say we value privacy. But the real test of these things is how we actually behave. And most people, they'll pull out the credit card and they'll use that even though, pretty much anything, even though somebody knows exactly what you've bought every second of the day, all the time, and people don't care. And I sympathize. I don't care. I care very little about this kind of privacy.
- 7.24** In 1996, somebody invented something called e-gold. E-gold was—you had online accounts. They were denominated in gold, in grams of gold. You could transfer the gold. You could transfer. And there was literally a reserve of gold. There was a depository where there was gold. So these accounts, these tokens that you got, were redeemable into gold. And what happened was we could transfer the gold between us,

and as we transferred the gold, they would transfer ownership of the physical gold in the back on a ledger between people, and you could get the gold if you wanted to. You could redeem it.

- 7.25 This was quite successful. At its peak, they had 5 million accounts, 5 million people using this e-gold. It had a total value of about \$2 billion, \$2 billion in the 90s and the early 2000s was still a lot of money. It seemed like it was ramping up. It seemed like everything was going pretty well, and then 9/11 happened, and they got tangled up with all the restrictions under the Patriot Act, money laundering, using money for terrorism, anonymity, and all of this stuff. Basically, they were forced into bankruptcy in 2009 as a consequence of the government going after them for all of that.
- 7.26 In 1997, something called hashcash was invented. Hashcash was a system, a proof-of-work, that was supposed to restrict the spam mail that you got. Imagine if everybody who sent an email got charged a little bit of money for it. So every email you sent cost you, let's say, a fraction of a cent or a cent, something very, very minor, so it wouldn't affect any of our lives. We wouldn't think twice about sending an email. But if you were sending massive spam, suddenly it would cost you, and therefore it would reduce—the idea was to reduce the amount of spam.
- 7.27 Now, sadly, for those of us who get a lot of spam, this never took off. It never was really successful. But this idea of having proof-of-work— you'd have to expend some energy, expend some money, expend something in order to produce the result. This idea, as we'll see, is going to be picked up by Bitcoin. There's a direct relationship here.
- 7.28 From '97, really, throughout the next decade, in the early 2000s, there are all these bulletin boards, chat groups, where people are discussing how to come up with an idea of better money, of digital money, of something that solves all the problems that we just talked about in terms of fiat. How do we get rid of inflation? How do we secure privacy? How do we get speed? How do we get instant everything? How do we do stuff peer-to-peer? Peer-to-peer solves the privacy issue; it solves the speed issue. How do we get out of a centralized authority like a central bank? How do we get away from having to use banks so that we can just do it directly?
- 7.29 There is a lot of thinking, a lot of writing, and there's groups called cypherpunks who are at the forefront of this, a mixture of various libertarians, anarchists, and then technologists who think this is cool and want to see something new happen. And they're writing to each other. You can follow, you can trace the evolution of what has become a cryptocurrency today to these chat groups and to the different developments. And they wrote papers and they analyzed each other's papers, and they would have disagreements and they would support each other and build on top of each other throughout the 2000s, early 2000s.
- 7.30 The origins of cryptocurrency as we know it today come from those chat groups. And they included, by the way, economists, so monetary theorists, people who actually

understood money. And they would chime in and participate and critique and discuss these things.

- 7.31** In 2008, out of one of these chat groups, an unknown developer, or for that matter, a group of developers we don't actually know, named Satoshi Nakamoto, basically implemented a decentralized method. Now, Bitcoin is a great story, partially because of this Satoshi. We don't know who he is. We don't know if it's a "he." We don't know. It might be a group of people. We don't know who they are. They own, for example, still to this day, a lot of Bitcoin. They've never taken any out. If they did, they would be gazillionaires.
- 7.32** Somebody just sold Bitcoin—I was just reading about this yesterday—somebody who bought Bitcoin in 2013 and spent like \$50,000 on it, just sold the entire stash yesterday, a few days ago—\$9.6 billion. A \$50,000 investment: \$9.6 billion. It is, according to this article I read, the highest return ever on an investment that anybody's ever made. You're suddenly a billionaire, 9.6 billion, for sitting on this stuff.
- 7.33** Anyway, there's a real mystery story here about the identity of Satoshi. There was even a trial by somebody who claimed to be Satoshi, and there was a whole trial around it. So this is ongoing in the background as an interesting mystery.
- 7.34** Anyway, in October 2008, Satoshi wrote a paper, a nine-page paper, very academic, describing what he called a Bitcoin peer-to-peer electronic cash system. He posted this to his cryptographic mailing list, to the people in this chat that were communicating with one another. Again, only nine pages long, but it basically outlines the structure and the functioning of what is to become the Bitcoin—the Bitcoin and the Bitcoin blockchain.
- 7.35** There was a lot of debate about this. People are writing about this, they're critiquing it, and he's answering. There's a lot of back and forth going on between Satoshi and a bunch of these people in the chat. They're trying to improve it, trying to figure out what it does and how it would work and what would happen. And so this is all today—people have archives of this, and you can tell who all the people are, many of them are identifiable. Nobody knows who Satoshi is, or nobody's saying who Satoshi is. If they do know who he is, nobody's saying it. And that's pretty amazing. 2008 to today—that's what, 16 years, 17 years? You'd expect it to have leaked by now. Somebody would have said something. So the whole thing is a real—it really is intriguing.
- 7.36** In January 2009, Satoshi releases basically all the code for this new creation, this blockchain, this Bitcoin blockchain. It's open source. Everybody can evaluate it, everybody can read it, everybody can tell what it is. And he also mines the first blockchain block. And what does mining mean?
- 7.37** Mining means solving a puzzle. And solving a puzzle doesn't require any particular thought. It's just a matter of guessing. You have to guess a lot, so you have to have computer power to do a lot of guessing and come up with the solution. Once you solve

the puzzle, a block is created, and you get rewarded for solving the puzzle by getting some Bitcoins. So the first Bitcoins are Satoshi's. As a consequence of his launching this program, the whole code of the program is public. It's open source, available to people.

7.38 During the next two years, 2009 and much of 2010, Satoshi is continuing to update the code, make improvements and communicate with different people in the chat about how this works and what it's doing and getting comments and then improvements and so on. And then at the end of, towards late 2010, he disappears, never to be heard of again. Cool stuff. This could be a great novel.

7.39 He hasn't made any public posts since then. As far as we know, he hasn't touched the code. The management of the Bitcoin system now is in the hands of whoever, in a sense, is interested, is willing. It's open source, people can contribute. Any major, any significant changes have to be by unanimity. There's a whole community of people out there basically responsible for maintaining the Bitcoin system online. All of it is completely transparent. Any change you make, any code you write, everybody can see it. They can vote it down, they can vote it up. There's no—the beauty of the system, part of the beauty of the system, is its transparency.

7.40 So what is Bitcoin? What is Bitcoin? At the end of the day, Bitcoin is a public ledger. It's a place where we keep track about who owns what. If you think about it, we talked about this a little bit with money—money has that ledger feature. You're getting money for stuff you've produced, you spend money. The money is keeping track of what you produced, what you consumed. In a bank account, it's much clearer that there's a ledger there: money goes in, money goes out. Everybody's keeping track of net, what is actually happening to your money.

7.41 Bitcoin functions in a very similar way, and it's very similar to fiat money. It's fiat money without the backing of a government, but it's something that we can use to buy and sell stuff, to transfer, and it keeps track of all that. But the difference is: it's keeping track of all that, not through a bank account. And ultimately, if you think about the bank accounts, who keeps the ledger of all the ledgers? Well, the central bank. There's a big ledger at the central bank keeping the ledgers of all the banks, which then keep the ledgers of all of our accounts.

7.42 In Bitcoin, the ledger is kept in what's called the blockchain. Think of the blockchain—I mean, this is how I think of it because it's the only way I know how to think of it, because I don't know the technical details of this. It's a database. Which Bitcoin belongs to whom, what was it spent on, when was it spent on? Everybody says, "Yes, that's true." It's verified by the network constantly, by people participating. And it's allocating who has the Bitcoins, where are they and what were they spent on.

7.43 It's a database that has the entire record of every single Bitcoin that's ever been produced, how it's been spent, and where it's gone. It's a ledger. You can participate in

this. Your ownership of Bitcoin can be, not anonymous, but you can use a pseudonym. It isn't linked necessarily to your identity, so it preserves its privacy.

- 7.44 It allows participants all around the world to come to a consensus about the state of the ledger every ten minutes. Every ten minutes, in a sense, everybody who's on the network gets polled. "Is this right?" I think it's pretty straightforward. But if somebody's trying to muck with the code, or if somebody's trying to do something fraudulent there, it can be easily caught. It's constantly updating.
- 7.45 There's no centralized server. There's no one place in which the Bitcoin database sits. Fragments of it sit on all the different nodes and all the different computers that people have. Indeed, every computer has a copy of the whole thing so that you can verify that nobody's mucking around with one of them. Because they all have to be consistent. They all have to be the same. They're all networked through the Internet.
- 7.46 Again, it's completely transparent. Every line of code is public. Every entry ultimately is public. You can make updates to this database freely, but they have to be backward compatible. So you can't—you can create new updates that are not backward compatible, but then it branches out and you've created something new. Bitcoin has to be completely backward compatible. Everything that used to work, works in the future.
- 7.47 Think about backward compatible as a new operating system for the Mac that requires a new chip. When Macs moved from, I don't know, Intel to their own Apple chip design, at some point their operating system became native and didn't work for the Intel machines. That's not backward compatible. On the other hand—I actually don't know if Microsoft has always been backward compatible, but Microsoft for years, everything was backward compatible.
- 7.48 But even now, I think there's certain files that you go back—I know I have files on my computer from the 90s. I can't open them. You have to find some special piece of software that knows how to—because they're not backward compatible. The stuff I have today can't open old stuff because it's not compatible with it. And on the blockchain, on the Bitcoin blockchain, everything has to be compatible.
- 7.49 Of course, transactions can move around the world at the speed of light, the speed of electrons. There are no real barriers. There's no centralized authority. Everything is verified by the people participating in the activity of Bitcoin, which is the activity, as we'll see, of mining, or what they call mining.
- 7.50 So what do we have here? We have, in a sense, an intangible digital asset. We have a piece of code. That's what Bitcoin is; it's a piece of code. A piece of code that, for a variety of reasons, which we'll get to, people have given value to. And now that it has value, in a sense, it's an asset. It has some value. It's unlike gold, in a sense, that it's an intangible. And unlike gold, it has no other purpose or no other use other than being this asset, which you can value. We'll get to why you would value it.

- 7.51** It's not an IOU, it's not a bank account. It's everything there is; it's in the Bitcoin. It, hopefully, facilitates transactions. It's transferable digitally through the Bitcoin native ledger system, which is the blockchain. And it can be transferred only by a holder of a key. The key is this long sequence of numbers, very long, and I guess letters and characters. I guess you can create a phrase that mimics that key.
- 7.52** There are people who—and this would have happened to me. If I bought Bitcoin in 2010, 2012, 2013, I would have lost the key. I would have forgotten what the password is. There are people out there looking in dumpsters for their old hard drives because they realize they've got millions of dollars on the hard drive and they've thrown away the hard drive.
- 7.53** Or I know people who've forgotten the key. Once you forget the key, it's done. There's no way to guess; it's true encryption. There's no way to get at it. So people have— this would be tragic. I would have been a gazillionaire, and I couldn't access any of the money, so better that I didn't buy it. So you have to have the key; otherwise, you can't access it. And that's a security, because you'd have to give somebody else your key in order for them to access this. So this is a very powerful password.
- 7.54** It has built-in safeguards so you can't double-spend your tokens. A lot of the problems with digital stuff, because it's just electrons, it's just code, you could duplicate and you could, you could double-spend the same piece. Here you can only spend it once, so it's a secure system in terms of spending.
- 7.55** It functions, even from the very beginning—I think the first transaction ever done with Bitcoin was somebody bought a pizza with it from a friend, from somebody who's also part of this little group that loved Bitcoin and thought it was cool, and they bought a pizza with it. It's used in transactions. We'll get to what transactions and in what context it is used, but it's pretty rare. The idea is that it will be used one day, but it's still not very well used.
- 7.56** So Bitcoin is an asset. It has value because people have given it value. Is it money? Well, if money—if the conclusion we've come to, I think, during this course is that money is a medium of exchange and a common medium of exchange, a medium of exchange that we're using, that is used in our world, then the answer to Bitcoin is no, it's not money.
- 7.57** Now, something could be a medium of exchange that's used sometimes for some things, and those exist. All kinds of mediums of exchange have existed in history like that. But if you're not a common medium of exchange, you just aren't money. You're an asset that can be exchanged, and Bitcoin today is an asset. Now, again, everybody would like it to become money, and a lot of people think it will become money, but it isn't yet. To the extent that they would like it to be money, it is money that they think will be going through, will be able to communicate without going through any government central authority. So it would be free of state control.

- 7.58** When is it—they talk about the government censoring transactions, and we think of censoring in terms of speech, but this is censoring what you can and cannot buy, censoring who can transfer to whom. Censoring here means blocking, not allowing, which happens all the time because the government has all these laws, money laundering and national security and a bunch of other things, that tell us that things that are illegal we can't buy, tell us that you can't use money for that, and they'll put you in jail if you do.
- 7.59** So what is it used for today? What is the primary use in terms of, as a medium of exchange? To the extent that it is, what is it used for today? Well, a big part of it is, it's used to raise money for causes in places where there's a lot of oppression. So if you're like, I don't know, freedom fighter in Belarus or even in Russia right now, the Russian authorities are not going to permit dollars to flow through the banking system into your account. If you're going to spend that account to argue against Putin, that ain't happening. So how do you get money?
- 7.60** Well, people can donate Bitcoin to you, and then you can go to other people, maybe within Russia or maybe in Belarus or whatever, and you can exchange that money for local currency that then you can use in order to do what you do. And you can stay under the radar, particularly if you ultimately use cash. So there are exchanges where you can get the local currency, but you can't get people from overseas to transfer dollars or local currency to you, because then the government will stop it.
- 7.61** So if you look at places like, as I said, Belarus or Cuba. In Cuba, for example, you cannot get credit cards to do online purchases. You can get Bitcoin because you don't have to rely on the Cuban banking system. You can subvert it, and you can use that Bitcoin to buy stuff online with those vendors who sell stuff using Bitcoin. Not a lot, but you can do some.
- 7.62** There are many other countries in the world where the regimes are very oppressive, and where the only way for you to get money in or to get money out is using sometimes cryptocurrency. And Bitcoin is one of the ones you use, not necessarily, I think, right now the dominant one. We'll talk about that. But it is one to use.
- 7.63** For example, if you live in China and you've made a lot of money and you have money, you'd like to get it out of China because you're afraid the Chinese might confiscate it, they might inflate it, they might do a lot of things to it. One way for you to do it is to buy Bitcoin. They won't let you—if you buy dollars, they won't let you ship the dollars out of the country. They have capital controls; a lot of countries have capital controls.
- 7.64** So instead, you could buy Bitcoin, ship the Bitcoin, and the authorities would never know. They'd wonder where your money went, but you could probably explain that in some way. But the fact is that you now have Bitcoin in the United States, and then in the United States, you might be able to open a bank account under a different name or a cousin's name or whatever, and then convert the Bitcoin into dollars and have a

dollar account in the United States. But getting the money from China to the U.S., you can't use the regular pipes, the regular relationships, because the government will stop it. Bitcoin allows you to get around that and get the money to where you want it to go.

- 7.65** So getting money out of places that have capital controls. And capital controls are not only done in authoritarian states. Argentina, before Milei had capital controls, so you couldn't get dollars out. They didn't want dollars to leave the country. They wanted dollars to stay in Argentina. So Argentinians were very good at using Bitcoin and other cryptocurrencies to get the money out. And most of their dollars outside of Argentina; they were not in Argentina.
- 7.66** Canada—you remember the truckers and the trucker demonstration in Canada? You remember that at some point, Trudeau froze their bank accounts, literally froze their bank accounts? Talk about authoritarian. And no money. They were trying to fund this protest, and they had no money. So what they did was they appealed to people to send the Bitcoin. And people did; a lot of people sent the Bitcoin. They got a nice pool of Bitcoin there.
- 7.67** But then they encountered this second problem: how do you convert the Bitcoin into Canadian dollars? To do that, they actually had to go, because Canada is an organized country, they had to go through a Canadian Bitcoin exchange. Private companies, unregulated, supposedly by the government, but the Canadian Bitcoin exchanges were afraid. And they basically said, "We're not distributing this money because we don't want the government coming after us. We don't want the courts coming after us." So ultimately, all that, I think, was returned. Or maybe after the whole crisis went away, it was distributed. But they couldn't get the money. They couldn't access the money.
- 7.68** So even though they were using Bitcoin, they couldn't convert it into Canadian dollars in Canada because it was all centralized. So one of the challenges with Bitcoin is it's completely anonymous, completely free, completely—no government intervention, no fear of government intervention, as long as we're peer-to-peer. But as soon as you start using an exchange, the exchange is regulated. The exchange is a controlled business. It's registered with the government.
- 7.69** Most of us, most of you who probably own Bitcoin, probably own it on an exchange, I don't know, Coinbase or something like that. Well, then all the things that you were promised by Bitcoin, don't hold. That exchange could be seized tomorrow. Your Bitcoin could be seized with it. You can't get it out of the exchange without the exchange participating. So the only way for Bitcoin to be truly what they say it is, truly unreachable by the government, is for you to hold the Bitcoin yourself.
- 7.70** You can get a little USB drive, put your Bitcoin on the USB drive. It's just code. Keep it in a safe at home, then nobody can get at it. But if you keep it at Coinbase, if you keep it in exchange, it's as vulnerable as your bank account is. And not only that, they can

usually figure out who you are. So, in spite of using pseudonyms, the NSA is pretty good at these things. They figure out who you are.

7.71 And they can probably steal it. There's a famous case of—Bitcoin was being used a lot in ransomware. You know what ransomware is? Let's say you run a hospital. I take over your computer systems, lock them up, and say, "You want your computer systems back? Pay me." And the payment was often made in Bitcoin, and then I release the computers, and it goes back to normal. Usually, ransomware—they pay because you can't afford to have your computers down for more than a few hours. Utilities, hospitals are all susceptible to this. That's why over the last few years, they invested gazillions of dollars in trying to beef up their cybersecurity.

7.72 Anyway, on a couple of occasions when this happened, the authorities managed to find the people, find the account, steal the Bitcoin back, which suggests that it's not that anonymous, it's not that safe in that sense, because the NSA could get it back. So, as long as you're using exchanges, these people are not safe. If they take the Bitcoin and immediately put it on little drives and put it in their pocket, the NSA couldn't get it. But as long as they kept it connected to a computer, connected to a network, or connected to an exchange, they figured out ways to get it. So if you do own Bitcoin and you care about this stuff, then be aware.

7.73 Bitcoin's not issued by a business. So this is one of the things about decentralization. It's not issued by a business, even though you might be keeping it at a business, like an exchange, like Coinbase. It's not issued by a business. So new coins are issued by the software, by the source code, by the blockchain, and they're issued as rewards to those who devote their computer power to solving the puzzle. And that computer that solves the puzzle gets the Bitcoin. It's the reward for expending electricity and effort on solving the puzzle.

7.74 There's no effort, really. You set up the computer, you give it a command, and you walk away. And there are thousands of computers all over the world doing this all the time as we speak. One of them will solve the problem, it'll get the Bitcoin, start over again, solve the new puzzle. This is an ongoing, continuous process.

7.75 The program releases quantities of new Bitcoin according to a fixed schedule. So every year there's a certain number of Bitcoins that are going to be released. Every four years, the number of Bitcoins that are going to be released gets cut in half. So that over time, fewer and fewer and fewer. So the opposite of fiat currency, where you're getting more and more and more. Here, you're getting less and less and less. And all of this ends when there are 21 million Bitcoins out in the world. That happens at around 2140, year 2140. So in about 100 and something years, there'll be no more Bitcoins to be released. The total pool of Bitcoins forever is maxed out at 21 million.

7.76 Now, of course, it's smaller than that. Why? Because there's the guy who has his Bitcoins in a hard drive somewhere in the garbage thing. People have lost Bitcoins, but

only 21 million will ever be produced. So there's a finite supply. It's important to understand some of the issues around Bitcoin.

- 7.77** You can't change this. To change this, to change any of the source code, the original code of Bitcoin, would require everybody to agree, everybody who is participating in the network to agree. That's never going to happen. Partially because, let's say, you change the code to now have 50 million Bitcoins instead of 21 million. That would immediately depreciate the value of the Bitcoins. These are all people who own Bitcoins. They're not going to purposefully devalue their own, the value of their own assets.
- 7.78** So people are constantly working on this. There's an algorithm that issues the new releases. These blocks that are created when you solve the puzzle both give you Bitcoin. You also get a fee for having mined the block, a transaction fee. And this block includes all the information of all the transactions that it is aggregating. Now, don't ask me to get into the details of this, it's not worth it. But it's all happening automatically. In other words, as soon as you solve a thing, a block is created. All the new transactions that people have done over the blockchain are integrated into this block; they're confirmed, and you get a reward for creating this block. You get a transaction fee for having created it, and you get new Bitcoin.
- 7.79** For example, when there are 21 out there and there are no more Bitcoins, there'll still be miners, there'll still be people solving the puzzles to get the transaction fee, because solving the puzzle is necessary for the block to be created, which is what facilitates new entry into the ledger, which facilitates actually recording all the transactions.
- 7.80** All right, so where does Bitcoin's value come from? As we speak, Bitcoin is somewhere just under \$120,000 a Bitcoin. You could have bought Bitcoin in its initial phase at about 10 cents. Do the math on that. It's about \$120,000. Where does the value come from?
- 7.81** And part of the question here is: how did it start? How does the first person put a value on Bitcoin when it's barely anything? Well, I think the first Bitcoins were basically bought and sold by enthusiasts who were playing around, who thought this was cool. And to a large extent, I think much of the buying that suddenly happens today is by people who think it's cool and who want to be part of something new and something exciting. But certainly early on, this was all about enthusiasts. This was all about that little group. They were talking about it and thinking about it and trying to figure it out, and they were like, "Yeah, let's buy and sell some stuff. Let's try to buy pizza from Joe. Let's see how this works." Much of that was happening.
- 7.82** And then people caught on, and the number of enthusiasts grew. And then there was this idea of, "Hmm, maybe this could become money. Maybe one day it will be used as a common medium of exchange. Maybe. Fiat money is obviously trash. It's obviously inflated. It's obviously got all these problems. One day, there will be an attempt to

come up with an alternative. Maybe this will be it." I think that is where its value really comes from.

- 7.83** If you do the math, you can do the math and say, "How many Bitcoins would you need to have, or what would the value of Bitcoin need to be"—because the number of Bitcoins is set. We know that—"what would the value of Bitcoin need to be if Bitcoin replaced every dollar in the world economy?" It turns out that it would have to be a lot higher than \$100,000 a Bitcoin. The total market capitalization of Bitcoin right now is just over \$2 trillion. So all the Bitcoins in the world right now are worth about \$2 trillion.
- 7.84** There are not that many left still to produce. I think it's at 19 million. So there's only like 2 million left. So it's not like the number's going to be that much bigger in the future. So \$2 trillion—the market value of gold right now is \$23 trillion. So Bitcoin is still only about a tenth of gold in terms of its market capitalization. And I don't know how much fiat money is out there. A lot. A lot more than any of those numbers.
- 7.85** So you could say, "What is the possibility that Bitcoin really does replace the dollar, becomes the global medium of exchange? And if it does, what would it have to be worth?" Let's say it would have to be with \$10 million, and let's say—I mean, this is how I would do it. And I'm not saying anybody does this. This is how I would do it, how I think about it. \$10 million. What's the probability that that happens? Because it's not a certainty. Nobody thinks it's a certainty. What's the probability that it happened?
- 7.86** We could argue what the probability is going to be. But let's say you say the probability of that happening is 10%, a one-in-ten chance that Bitcoin will replace the dollar. So 10% of 10 million is a million. And then it's going to happen in the future. So that's not—so I had to present value it to today. What's the value today of something that has a 10% probability chance of being worth \$10 million sometime in the future, 20 years in the future, let's say? And then you do present value of it, and that'll give you the value of Bitcoin today.
- 7.87** I don't know what those numbers add up to because I haven't done the exercise because I would put the probability of it becoming a common medium of exchange at a lot lower than 10%. So I'm probably underestimating the value of Bitcoin relative to everybody else in the world out there. But that to me makes sense. And I would add to that—so that's one part of the equation—and then I would add the present value of all its uses today.
- 7.88** What is it worth for somebody to be able to move money out of China? How much would they pay for that? Well, that's Bitcoin. If Bitcoin is making that possible, then it has a positive value. If Bitcoin is making it possible donating money to Belarusian opposition parties, then it has some positive value. Exactly what positive value, I don't know, but that's how you would think about it in order to come up with the number it's expected.

- 7.89** What is its expected purchasing power, given 21 million units if it becomes the common medium of exchange one day? I think that's how people are thinking of it in one way or another, whether they do it explicitly or implicitly. I think that's basically what's going on. Right now, a lot of people, I think there's a base of people who are committed to Bitcoin. They think this is going to become. So their probability that they placed on it becoming the common medium of exchange one day is very high.
- 7.90** I know people like this, that think that it's close to certainty. They are convinced that the fiat money world is going to implode, there's going to be hyperinflation, banks are going to collapse, and we will all revert to Bitcoin because that'll be the best alternative possible. They're never going to sell their Bitcoin. They're going to buy it no matter what the price is. They're going to keep buying as long as they can afford. They're going to accumulate massive amounts of Bitcoin because they are sure that that is the outcome.
- 7.91** I don't know how many people like that there are in the world, but there's a whole ecosystem of them. They're out there, and they drive the price. So every time the price drops, they come in and swoop in and buy it up. As long as they have the money, as long as they have the resources, they will keep doing that. We'll talk about what might change if that happens.
- 7.92** And then there are people who are buying it because they're speculating, "Hey, it's gone up so much and I was left out. I should buy now because it could go up a lot more." So pure speculation. And I think most people right now, I'd say, I don't know, 80% of the people who hold Bitcoin right now, are just speculators. What are they hoping for? They're hoping for—the technical term for this is the greater fool theory. Whether it's truly fools or not is a different question. But the point is, all they're hoping for—they haven't run the math, they haven't done a present value calculation, they're not real believers either. They're basically under the assumption of, "In a year, somebody will buy it from me at a higher price than what I have today." That's it. That's what drives Bitcoin. Price and why it is going up.
- 7.93** A couple of more things on this. Just a side point: one of the challenges with Bitcoin is that actually processing transactions is actually fairly slow and expensive because it all depends on the creation of a new block. So what they built on—and again, don't ask me for the technical details because I don't know them—what they built on is, on top of the blockchain, they built a layer of code called the Lightning layer. In that layer, we can transact with Bitcoin instantaneously at a very, very low transaction cost. And then all of those transactions that are happening on this other layer get aggregated as a ledger entry. There's a ledger that keeps track of all this. And that gets implemented in the blockchain in a block.
- 7.94** It's like what happens—we're transacting with our backing system, I'm buying from you, money is moving around. But then at some point, they go to the bigger ledger

between the banks and they settle. Well, the Bitcoin blockchain is where all the settling happens for the transactions that most people are doing on the Lightning level.

- 7.95** So a lot of times Bitcoin has been—in the past, I've seen debates where this has happened, where people have accused Bitcoin, "Well, you're not that fast, and you're really expensive to transact." Bitcoin has solved that by creating—and this is a new innovation of the last few years—of creating this Lightning layer on top. And you could create all kinds of layers. You'd just have to get a significant buy in from the people. You can't do whatever you want with Bitcoin. There has to be a significant [number of] people who vote. There has to be a significant buy-in.
- 7.96** All right, so let's think of what Bitcoin has achieved to the extent that it's achieved things. So in Satoshi's original document, there are really three things, or in this particular initial plan, there are three things that they're trying to solve for.
- 7.97** They're trying to solve for inflation. So that's one—we'll get to inflation in a minute. They're trying to solve for privacy, that's two, privacy and security. And then they're trying to solve for the high cost of bank-mediated payments and make micro payments, for example, small payments online possible, which was difficult in the previous fiat money system that we have. So let's take these one by one.
- 7.98** Inflation—they've solved the problem in the sense that there's only a finite number of Bitcoins. So they're not going to be depreciated from being inflated. There's no discretion. But there is another problem, which a finite quantity of Bitcoin creates and that is there is a supply of Bitcoin that is set. It doesn't change once we get to 21 million, but even as we get there, it's changing very, very slowly. It's not affected by demand, it's unresponsive to any kind of demand shifts, and as a consequence, it's incredibly volatile.
- 7.99** So the value of Bitcoin is not inflating in the sense that the quantity is. But you don't know actually what it's worth day to day, week to week, month to month, year to year, decade to decade. It's not clear. It's incredibly volatile, and there's no reason to believe it's going to become less volatile because the supply is fixed. So if I suddenly want a lot, not more supply is going to come. And therefore the only way I'm going to get a lot of is by driving up the price. If I want to dump a bunch, if I want to sell a bunch, the only way to sell is for the price to adjust. So on a day-to-day basis, if you look at the chart of Bitcoin, it's very volatile. It's not volatile in the sense that it goes down 50% or up 50%, but even if it goes down 1% a day or up 1%, that's a lot. That's volatile for currency.
- 7.100** It means if I'm buying something online, let's say I'm buying something for \$100 online and I send Bitcoin worth \$100, by the time the other party turns around and realizes and wants to do something with it, that Bitcoin might be worth \$98, might be worth \$102, we don't know. But he sold me something for \$100, let's say his expense is on that. Let's say he has to turn around and pay his vendors a certain amount of money. By the time he turns around to pay his vendors, that Bitcoin might be worth less than

what he has to pay his vendors. He might lose money on the transaction without even knowing it because of the volatility.

- 7.101** Indeed, I think the main reason Bitcoin is not used for transactions, it's not used to buy and sell stuff, is because you don't know what you're buying, and you don't know what you're selling. You don't know the amount. I think it's also not used because people are keeping it because they believe it's an appreciating asset, and they don't want to miss out on the next big rise. If I go and buy a Ferrari with it and then it goes up 50% the next day, I've lost out on the 50%. This is the way people think about these things.
- 7.102** Its volatility is very high, and it's not clear that that solves the problem of inflation because your purchasing power is constantly changing, and that's not a good thing. That creates uncertainty regarding purchasing power, which is very, very damaging for a unit of account for a medium of exchange. It can't function as a medium of exchange as long as it's volatile.
- 7.103** We still don't know—and a lot of people are trying and thinking about this—we still don't know how to create a crypto, a cryptocurrency that will adjust the supply automatically without human intervention to demand. Until we have something like that, this is going to be a problem with crypto. They can change the supply, but then it's human intervention, and we have to worry about the incentives of the humans changing it. To automate an actual system that works like Bitcoin, but doesn't have this inflexibility and doesn't have a finite amount, but then the amount changes over time—again, we'll compare it in the next class.
- 7.104** We'll compare it to gold. But remember, gold adjusts. We mine more. If the price goes up, we can convert jewelry into money if the price is high enough to justify that. That doesn't exist with Bitcoin. That self-correcting mechanism doesn't exist. So what happens with gold is you get no inflation. Sometimes, sometimes prices go up, sometimes prices go down, but on average you basically get zero here. Here, there's none of that.
- 7.105** We said the second achievement was privacy. Again, to really achieve privacy and to really do everything that was promised in terms of privacy, you're going to have to do everything peer-to-peer, keep it out of the hands of the government, peer-to-peer. As soon as you start working through centralized exchanges, privacy can break down.
- 7.106** So it achieves what it set out to be, but because we tend to use centralized exchanges, it doesn't really work that way. One of the interesting features of this is that over 90% of people who hold Bitcoin hold it in centralized exchanges, which is another sign that we don't value privacy that much. We're okay with the government snooping around.
- 7.107** And as we said, you can find, you can figure out who owns what. You can get through the pseudonyms and Bitcoin—the blockchain, one of the strengths of the blockchain, this database, is that it records everything forever. It doesn't go away. It doesn't wipe out past transactions. So if somebody can decipher it, you can tell exactly where each

Bitcoin was every single step of the way. There's a huge cost of deciphering it, but it is possible.

- 7.108** And then finally, the high cost of bank transactions in our current system, again, the cost of doing it on the blockchain, it's quite expensive, but they've solved that problem with another layer. So the price of Bitcoin indeed is very low. And it is true that you can transact cheaply, quickly, and easily on the Bitcoin network if you do it right.
- 7.109** All right, to summarize—Bitcoin today is a speculative asset. It's not money yet. Maybe it will become one day, but it's not yet. It's a speculative asset. And to a large extent, the speculation is about whether it will become money or not one day. Although I think, for many people, the speculation is just: is it going to go up or is it going to go down? I don't think it's more than that.
- 7.110** Bitcoin holders don't use Bitcoin. They store Bitcoin. And while it serves certain functions, very few people are using it for those functions— transfer money out of countries, donate to dissident groups—those are tiny amounts of money in the big picture, in the big scale of things.
- 7.111** There's some worrying signs—we'll get to this later as well, but there's some worrying signs. There's a company called Strategy right now that owns, I think the last time I saw, they own \$70 billion of Bitcoin. And all they do, the whole company, all they do is they store, they have, they hold \$70 billion of Bitcoin. And yet the stock of this company, Strategy, the value of this company, is \$140 billion. So the market is valuing every Bitcoin at twice what it's actually worth in the marketplace, which doesn't make any sense, because if Bitcoin's gonna go up, then the value of the portfolio will go up, and the share then should go up. But there's no point in valuing it now at twice what that is.
- 7.112** If it's one company, that's okay. But now there is a number of companies, they figured this out. You've got, they're called treasury companies or blank check companies, that are going out there and all they're doing is buying a bunch of Bitcoin and then watching their market capitalization, market value go up above the value of the Bitcoin. And now, I just saw there's a Peter Thiel-backed company that is doing the same thing with Ethereum, which is another cryptocurrency, and they expect to get the same effect. It's great if you're minting money that way. I go out into the marketplace, I buy a dollar of Bitcoin, and then I turn around and sell it for \$2 over here. I mean, wow, I want that. So the worrying sign is that there's a lot of speculation going on.
- 7.113** The people who will never sell the Bitcoin, the guys who always hold it and buy whenever there's an opportunity, how do they live? Many of them—no, I'm serious. They borrow money against their Bitcoin. So they use Bitcoin as collateral, and then they borrow and they live off of borrowed money. That's basically how they utilize their Bitcoin. Then maybe they sell a little bit to pay the interest, but that's about it. And they keep borrowing.

- 7.114** Now that's fine as long as Bitcoin's going up. When Bitcoin goes down and you lose your collateral or the bank calls in your loan, now you have to sell. You could get spirals of panic sellers in that environment because a lot of these people are leveraged. A lot of the Bitcoin purchases, in a sense, have been done through debt, or they're living on the debt.
- 7.115** So there are a lot of risky signs around Bitcoin. And I'm not suggesting it's going down because every time I have, it's doubled. So I'm not making any investment advice or suggestions. And by the time this is viewed on the Peterson Academy platform, Bitcoin might be triple the number. I don't know where it's going to be. But as somebody who follows finance, whenever you see froth, whenever you see people doing crazy stuff, not just buying stuff, but leveraging up, companies being valued out of sync, out of what makes sense, you've got to start questioning, you've got to start asking questions.
- 7.116** All right, next class, we're going to talk about, we're going to summarize what we've just done by comparing fiat, gold, and Bitcoin. Then we're going to talk a little bit about other crypto, and we're definitely going to be talking about what the big thing right now is, which I think is going to have, in some sense, a much bigger impact on the world in the short run than Bitcoin is, which is stablecoins.
- 7.117** We'll talk about what stablecoins are—you're going to hear a lot about stablecoins in the next few years, months, years. It's a huge, growing industry. Congress just passed legislation to regulate it. So a lot of companies are stepping in to take advantage of that. So we'll look at that, and then we'll look at central bank digital currencies, which is again, the next big thing out there on the horizon in terms of money.
- 7.118** Do you think it's possible to create a digital currency that solves the problem of inflation and volatility?
- 7.119** Do I think it's possible? I don't know. I think it probably is, but I have no idea how. And a lot of really, really, really smart people are trying to do it. As I'll say when we summarize all our discussion of crypto, look, I believe in competition. I'm not against any particular technology. I have no particular interest in one technology. I just want to see what happens if we actually allow for competition. But we live in a world in which the competition is skewed by government interference.
- 7.120** So, for example, as you'll see, stablecoins are not going to be regulated. So we're going to get one type of stablecoins. We're not going to get other types of stablecoins because those are being deemed inappropriate by regulation. So the government is going to play a huge role in what arises as winners and losers in the crypto space, which is a big and a real problem. If you had—what I will argue for in the end—if you had a real free banking system: let's see what money works, what money evolves. Go back to a market where money evolves out of the market. Then, it could very well be that we ultimately don't choose gold. We choose some new cryptocurrency.

- 7.121** Also I'd say—why do we assume Bitcoin's won? It's been 16 years. Sixteen years is nothing in the course of economic history. We don't know what new technology is around the corner. We don't know what somebody's going to innovate next. It could be that Bitcoin 2.0, something better than Bitcoin arises tomorrow, drives the price of Bitcoin to zero, and that takes off. I'm not saying that'll happen. It probably won't, given the passion with which people believe in Bitcoin, but it could technologically.
- 7.122** So I want to see evolution, I want to see competition. I think, I very much think it's super exciting what's going on, because there are multiple uses for these technologies, probably not as money, but probably for all kinds of other things. The worst-case scenario is that our regulations crush the innovation and crush the possibilities that are available for our future.
- 7.123** There are a lot of cryptocurrencies, hundreds, maybe thousands. What is the distinction between Bitcoin? Is it strictly the limit on the amount and the decentralization? Is there anything more than that?
- 7.124** Well, a lot of them are decentralized. A lot of the cryptos are decentralized. I think, as far as I can tell, and I'm no expert, particularly not on the whole universe of cryptocurrencies, the real distinction is what's called proof-of-work, which we talked about a little bit, versus proof of stake, which is what almost all the other cryptocurrencies have. Proof of stake is something that Ethereum is different. Now, what is the difference as far as I can tell?
- 7.125** What Bitcoin requires you to do to get the block, to get the additional thing, and to participate. If you're going to participate in governance, in determining the nature of the network, in the voting, in the confirmation of the transactions and the blocks, you have to be working. What does work mean? You have to be expending energy to try to solve the puzzles or to participate in the network to create the blocks. So you have to—your participation is contingent on the expenditure of electricity, on using computer power to back it all up.
- 7.126** Ethereum, which I know less about, is based more on something called proof of stake, that is that you can prove that you actually have a stake, that you actually own something, that you have an ownership right on this thing, on this crypto piece. It doesn't require you to actually expend energy the way that Bitcoin does. And it doesn't require unanimity. Ethereum doesn't require unanimity. The voting rights are distributed differently. I don't know exactly how, but they're different than Bitcoin.
- 7.127** One of the things, particularly with Ethereum, Ethereum has a founder who still is there. He runs—there is a company, and it's very much— he doesn't have maybe the votes, but he has the charisma, if you will, the influence to mold the direction in which it goes. Satoshi's not around. There's no founder. There's no one person in Bitcoin. It really is a network effect and really is determined by some consensus, some meeting of minds or agreement that people have to come to in order for this to evolve.

- 7.128** So it's things like that. I'm sure there are a lot of other differences that have to do with technicalities. I'll also say that a lot of things, there are thousands, that we're calling crypto. They're not really crypto in the sense that Bitcoin is. So Bitcoin and Ethereum and maybe a few others are a base layer of crypto. They're the foundation on which everything else runs. They're the blockchains.
- 7.129** But a lot of the other, what we call cryptos, a lot of the coins, are not the same as Bitcoin. They sit on a different layer. They are traded on the blockchain. They might be traded on the Ethereum blockchain, even though they're not Ethereum. And many of them right now are stablecoins. So they are actually coins backed by something redeemable—we'll talk about stablecoins.
- 7.130** There are meme coins. Meme coins are coins that everybody knows are not worth anything. They're basically just a joke. I mean, they're not. People buy them as a keepsake, like that Trump meme coin. It doesn't sit on a blockchain. It's not tradable in any significant way. Or maybe it does trade on a blockchain, but it's not money. It's not pretending to be money. It's just cool to have a Trump coin, I guess, or all kinds of other meme coins that exist out there, where it's basically just cool to have it.
- 7.131** Dogecoin—remember Dogecoin? It was literally a joke somebody put out there, and today it's worth a lot of money. People actually trade it. It sits on a blockchain, but it is not a unit of the blockchain. It just trades on the blockchain. People keep track of it. And it's actually quite popular and worth a lot of money. Yet it was started as a complete joke.
- 7.132** So I have a feeling that it will come a day, I have no idea when, but there will come a day when there'll be a cleansing. A lot of this stuff will disappear, a lot of it will go bankrupt. Just like the dot-com bubble. You remember the dot-com bubble in the 90s. Anybody with a half-baked idea started a dot-com company, got a website, and went public. I was teaching in those days, and my C students would become CFOs of companies right out of graduation in Silicon Valley, and it was insane.
- 7.133** You didn't know when this was going to end, and you didn't know how it was going to end, but you knew it was going to end. And in March 2000, it ended. The Nasdaq went down 75%. Think of that—a major index going down 75%. The Nasdaq didn't get back to its previous level, to its previous high for—what was it—until 2013, 2014, I think? So it took over 10 years to get back to its previous high.
- 7.134** So these things cleanse. I think the same thing will happen here. The same thing, to some extent, happens to every new technology. Right now, there are hundreds, maybe thousands, of AI companies being founded. How many of them will survive? A handful will. But that's normal. That has to happen.
- 7.135** With crypto, there's even more hype, there's even more excitement, and it might take longer to cleanse, but there will be a cleansing. I mean, there will be survivors because it is probably a technology of value. I'm not sure—Ethereum, because it's not as rigid as

Bitcoin, has more to offer. It has things like smart contracts, which Bitcoin doesn't do quite as efficiently. So who wins? Maybe a number of. Maybe Ethereum and Bitcoin will win. I don't know. But at some point, a lot of these junk coins will disappear and go away, who knows when. In the meantime, make money.

Lecture 8: The Future of Money

At a glance — The future of money.

- 8.1** Last lecture, we talked about Bitcoin. Exciting topic and one that's super relevant in the news all the time. I want to now do a thought experiment, in a sense. If the case was that, if fiat money imploded and we had hyperinflation, there was real demand for new money, well, let's think through about what is likely to arise, given the options we have today. We don't know what options will be in existence then; but given the options today, really, in a sense, let's compare a fiat money, which is what we have today, with gold, which we've talked about quite a bit in terms of what preceded fiat money and Bitcoin, which is what a lot of people would like to believe is going to be the replacement. Let's think about some of the differences and similarities, particularly between Bitcoin and gold, as options for fiat money.
- 8.2** So, both Bitcoin and gold are stateless. They don't require a government to issue them and to control them and to determine the quantity. They can evolve out of a marketplace without any kind of central planning and central control. Either one of them could provide an international base money, if you will, for transactions, that's not a creature of any particular national or international central bank-type institution.
- 8.3** Both provide base money that is limited in quantity. I mean, Bitcoin, a known quantity or known declining amount of added additions. Gold, as we described, can fluctuate, but it's never really grown, even at the biggest gold rush, it's never really grown more than 2.5% in any particular year, and usually that is offset by lower growth in future years. On average, it grows at about 1.5% a year over time.
- 8.4** Some of the differences between gold and Bitcoin. Gold is tangible. You can touch it, you can feel it. Bitcoin is not, purely digital. Gold can be used for other things— jewelry, primarily. Some electronics, but primarily jewelry. Bitcoin has no other uses, although, arguably, the ledger, the blockchain, might be able to be used for other things like smart contracts and digital contracts and other features like that.
- 8.5** Maybe one of the biggest differences between the two in terms of their function as money is that gold ultimately settles in gold. You ultimately move gold from one place to another. You might use a ledger, but if we're in two different countries, gold might travel from one place to another. At the end of the day, there is a physicality to it, and something has ownership of a physical asset is shifted.
- 8.6** So while a lot of the transactions in gold can happen electronically, you could easily have a token, a digital token, that represents gold, just like today, we have digital tokens, call them, what we do online, that represent fiat money. We don't literally buy everything with cash. We don't need to buy everything with gold, and we don't even need to buy everything with paper money reflective of gold. We can buy stuff online,

digitally, with digital gold, literally digital gold, digital tokens that are redeemable in gold.

- 8.7** And those already exist, and those are pretty easy to conceive of. Even though those transactions happen digitally and they happen instantaneously, ultimately, the settlement has to happen in physical material. Whereas in Bitcoin, there's no physical material. So all the transactions settle in a sense just through the Bitcoin itself. Bitcoin is a database. It is a ledger, so the ledger settles all the ownership issues on the blockchain.
- 8.8** Gold, because it's physical, needs a custodian. Somebody has to have physical possession of the gold. A lot of the gold today, owned by private individuals, sits in vaults in Switzerland or in Singapore or in London or in New York. People don't own the gold. They don't keep the gold in their home. I mean, some people do, but the vast majority of the gold in the world is held in a few big places where it's stored, primarily by financial institutions.
- 8.9** And given the political risk of owning a lot of gold, I think people who are wealthy, have a lot of gold, probably diversify them—a little bit in London, a little bit in Switzerland, a little bit in Singapore, a little bit in New York—so that if some government decides to confiscate the gold, and that sounds bizarre, government confiscating gold? How could that happen? Remember 1933? 1933, FDR confiscated all the gold in the United States, all gold that you owned as an individual. So we have a precedent. This is not crazy.
- 8.10** Again, one of the advantages of Bitcoin—much harder to confiscate Bitcoin because Bitcoin has no custodian. Now, if you keep it in an exchange, it does, and it could be confiscated. But if you keep it on a drive at home in your pocket, probably can't get access to it. So a Bitcoin is self-custodied, if you choose to do so. You could argue gold is self-custodied if you choose to do so as well. But since it's physical, since it's bulky, much easier to take from you than would be the Bitcoin.
- 8.11** Gold provides purchasing power stability. We've seen that in history. We've got plenty of evidence that that is indeed the case. Inflation is pretty much zero under a gold standard. You get purchasing power stability. Bitcoin does not. It gives you the opposite. It gives you huge purchasing power volatility. And there's really no reason to believe that goes away, even if it becomes the dominant medium of exchange. It's still going to have volatility because its supply is constrained.
- 8.12** So I don't think it would be quite as volatile as it is today, when it's more of a speculative asset. So speculation drives a lot of that volatility, but it still would be volatile, because of its inflexibility. It has no way to adjust. It is a set supply, and that's it. So demand will make the price move quite a bit, whereas with gold, demand would make the price move, but then there would be a correcting factor, which is more gold, less gold, an adjustment to the supply.

- 8.13** So you can see some of the pluses and minuses to each one of these, and there are some. But there are also legal obstacles that would have to be overcome if we wanted to switch either to gold or a Bitcoin standard. So there are real obstacles today, and there's no reason those legal obstacles, to assume they go away. I mean, you could think of a few of them.
- 8.14** One is capital gains taxes. Every time you transact with Bitcoin, and if you make money, that's taxed, the capital gains. Same with gold. Whereas if you have pound sterling and you switch them to dollars and you make money on the transaction because when you bought the pound sterling, they were worth less, you don't pay capital gains. Currencies are not taxed. For that matter, your dollar might be worth more tomorrow than it is today, but that difference is not going to be taxed.
- 8.15** Let's say there's deflation and your purchasing power goes up. That's not taxed. So as long as Bitcoin and gold are viewed as property, in other words as assets and not as a currency, there is a capital gains tax. And that is a real deterrent, particularly when you get this volatility. Some countries just ban cryptocurrency. There are a bunch of countries in the world where you can't use them. They're not usable. China bans cryptocurrency. Why? Because they discovered people were using them to get the money out of China. They did not want the money going out of China, so they banned it.
- 8.16** So nine countries in the world have explicit bans, cannot use it. There are 42 countries that have implicit bans. Basically, you can't use it. They don't tell you outright, but it's clear that you would suffer legal consequences if you tried to use them. In a lot of places, gold-based banking is implicitly barred because, to have a bank, certainly in the United States, but I'm pretty sure this is true everywhere in the world, you need a government license. You can't just start a bank. You can't just start a fintech company. You can't just start a financial entity.
- 8.17** Financial entities that function as banks need to be chartered, need to be registered by the government. And if you come and say, "Oh, my business model is to be a gold bank or to be a Bitcoin bank," they could easily say no, and they will say no most of the time. The reality is that, during the Biden administration, regular banks, normal banks, that dealt with crypto companies, for example, facilitated salary payments, facilitated maybe exchange of crypto into dollars for the crypto companies, those banks were heavily persecuted by the Biden administration, something called Choke Point 2.0, but it was real.
- 8.18** Silvergate, which is a bank in California, which specialized, their whole business model, with servicing the crypto industry, was basically forced to close down. They closed down, they unwound their business and sold everything off and gave everything; they did it in an organized way. Signature Bank in New York was caught up in the Silicon Valley mess and was shut down by the government. And there's good reason to believe they were shut down by the government because they had a lot of crypto accounts.

- 8.19** Elizabeth Warren was indeed on record, very explicitly saying, "I don't want this. This is bad. We shouldn't have this." And within the Democratic Party, certainly, she has a lot of influence and had a lot of influence during the Biden years. So banks that dealt with crypto were discriminated against by the regulators during those years, and there's no reason to believe that that will not happen again. The Trump administration is very favorable towards crypto for a variety of reasons, but that could change on a dime, in this administration or in future administrations. So just getting a license, just becoming a bank, is very difficult.
- 8.20** Now, we do have today gold-backed credit cards. So you can actually open an account with a company, and they will send you a credit card. You open an account by sending them money, of course, buying gold, and they will have a gold deposit for you. And the value of the deposit will change as the value of gold changes. And you have a credit card, and you can go and use the credit card to buy stuff. And the merchant will be paid in dollars, but those dollars will be determined based on, how much gold is sold will be determined based on the value of gold at that given point in time. So everything you do would be linked to gold.
- 8.21** There are even bonds now that you can buy that are gold-denominated. So there is a market out there that is gold-based, but it is limited. It is constrained by many of the laws and regulations that exist in the world. I think you'd probably expect the same thing to happen as Bitcoin becomes more acceptable. There will be more of these kind of things, although with Bitcoin, the challenge, of course, is mostly the volatility. I mean, gold is already volatile, but Bitcoin is quite a bit more volatile than gold is.
- 8.22** And then there are a lot of these, in terms of legal challenges, there's a lot of money laundering laws. We've talked about this a little bit. There's a \$10,000 worth of crypto has to be recorded, just like \$10,000 worth of anything. If you're transacting, it has to be recorded for money laundering. As much as I would love money laundering laws to go away, it's unlikely that they're going to go away.
- 8.23** Restraints on how you use your dollars, how you use your gold, how you use your Bitcoin, how you use any financial asset are still going to exist there, and they're going to be barriers to the introduction of a new type of money. There are all kinds of restrictions today on both gold and Bitcoin or crypto exchanges as a consequence of that.
- 8.24** But really, the biggest barrier, I think, to the introduction of, or to return to a gold standard on the side, if you will, or the introduction of a Bitcoin standard or anything like that in terms of being the medium of exchange, being money, the biggest barrier is the fact that we are pretty comfortable with the money we have. There's a massive networking effect. I know that if I have dollars, I can use them anywhere, certainly anywhere in the United States, but really anywhere in the world.

- 8.25** I can take dollars anywhere in the world, easily convert them into a foreign currency, or maybe even use them in those countries. Some people will accept dollars. Now, you can imagine a world in which that changes, but it would have to be something very dramatic, not the inflation of the 1970s. That didn't destroy the dollar. Not the inflation we just had. That didn't destroy the dollar. The reality is that once people start using a particular currency, it's very hard to shift.
- 8.26** One of the reasons I accept dollars is because I know I can use dollars. And the reason they accept the dollars is because they know they can use the dollars, and on and on and on it goes. And that's a network effect. And you have to reestablish that for a new currency. If Bitcoin wants to become money, you have to be willing to accept Bitcoin because you believe that you can then use Bitcoin. So when it stops becoming an asset that just goes up, when we start using it as money, we have to now have to have some kind of shared belief that it is usable.
- 8.27** And what would that belief be based on? Gold has a little bit more credibility here because it's been used in the past. We know people have always valued gold, pretty much in any culture in the world. Even today, you could take gold coins anywhere in the world and you'd do OK. You could exchange them. So gold has a little bit more credibility here. But even then, now, if I have paper money that is redeemable in gold, how do we establish the network? How do we get scale? Fiat money has a huge advantage.
- 8.28** Now, under high inflation and massive inconvenience, as we said, that could indeed change. So I'll give you some examples. In Latin America, in a number of Latin American countries that had super high inflation in the 1980s and '90s, they switched like that to U.S. dollars. They just killed their currency, replaced it with dollars, dollarized their economy, and people were fine. But it took a massive inflationary period and real distrust of the system for them to shift to something different.
- 8.29** Now, they have to trust the American central bank, but the American central bank is much more trustworthy than most local central banks. We've seen switches in currency, but it's rare, and it's difficult. A lot of countries don't want it. I remember when I was living in Israel, there was high inflation in hundreds of percentage points, and a Treasury secretary, finance minister, suggested dollarizing the economy. He was out of his job the next day. He didn't last 12 hours after he made that statement. Patriotism. You mean we don't have an actual currency? We don't trust our own? It doesn't last very long, so most people are out.
- 8.30** Or take, for example, a good case, which is happening right now before us, which is Venezuela. Venezuela has high inflation. It has a crumbling state infrastructure. It has a corrupt and abusive government. And what you're seeing is that most people have chosen. They could use gold, they could use Bitcoin, but they've chosen, the simplest, easiest, cheapest thing to do is to use dollars. So much of the Venezuelan economy is dollarized, not from the top down, from the bottom up.

- 8.31** People just use dollars. They're easier to get a hold of, and they know that anywhere they go in the world, they've got them, anywhere they go in the world, they can use them, and anybody in Venezuela will accept dollars. That's just a given. Now, it is true that the uses of Bitcoin as a medium of exchange are higher in Venezuela than they are in other places around the world. So some people are using Bitcoin because, again, they can access it. It's secure. The government can't track them.
- 8.32** And there are regions in Venezuela where they actually have gold mines, and there they use gold. So you've got a country that's falling apart from a governance perspective, and different mediums of exchange are rising to replace it. But the dominant one is dollars. Because as bad as we think our currency is, it's a breath of fresh air for most people around the world, given their alternatives. And it's the cheapest, easiest one to use because, again, the network effect, because everybody else uses it, so it's cheaper for you to use it as well.
- 8.33** So the reality is, as long as dollar inflation is reasonable, and I think anything above zero is unreasonable, but from the perspective of the world, it seems like single-digit inflation. As long as it's under 10%, nobody's going to dump their dollars and switch to some other form of money. It's just too much of a hassle. It's too expensive. It's too difficult. It's too risky. And there's no obvious good alternative. Gold is, but again, there's capital gains and there's all these other things that obstruct the ability to use gold.
- 8.34** Most people, as we've seen, by example, don't really care that much about privacy. And if they do, they get a suitcase full of cash. Not more than \$10,000, though. And so, there really is no reason to believe that fiat money is going to be replaced magically by any of these—again, short of a major catastrophe. Now, I'm not saying a major catastrophe is not going to happen. We're running debts like, insane levels of debts in this country. Somehow we're going to have to pay them back. And there's at least a possibility that paying them back will involve hyperinflation or some inflation.
- 8.35** So who knows what the future holds. But I'd say if inflation is below 10%, nothing's going to happen. People will complain and bitch and so on, but they're not going to actually abandon the system for the sake of that. It's either a completely philosophical change happens where we say, "No, we do not want the government anymore to be involved in our money. We do not want the government involved that way in our lives," and then we change ideologically, and ideologically we impose a new system. Or the system has to collapse and then who knows what will arise in its ashes, in its place. It's not clear at all.
- 8.36** It might be something good. Usually when systems collapse, nothing good happens. When Rome collapsed, it wasn't like, "Yes, now we'll establish an amazing society in which we are all going to be rich and prosperous." No, for a thousand years, life sucked. It's a good example with a bit of Bitcoin in El Salvador. So in 2021, the El Salvadoran government actually passed a law that made Bitcoin legal tender in El Salvador. You

could pay taxes with it. You could go to stores, and the stores would accept it, and you can transact with it.

- 8.37** And in order to juice it up, in order to get—remember, El Salvador is a very poor country. In order to get people excited about using Bitcoin, the government actually introduced its own Bitcoin wallet and preloaded it with \$30 worth of Bitcoin and gave it to anybody who wanted it. Free \$30. The idea was, this would get them using Bitcoin. But what actually happened was they converted the \$30 worth of Bitcoin into dollars. Dollars is the currency in El Salvador. They don't have a local currency. They use dollars. They converted 30 Bitcoin into dollars and abandoned the wallet.
- 8.38** There's almost no transactions that go on in El Salvador in Bitcoin. People don't trade it. People don't accumulate it. They're too, dollars is good enough for them, and they're too busy living their lives, and they know dollars, they don't quite understand Bitcoin, even though both, you could argue, are intangible. Neither one of them is backed by anything. But dollars are already there. Dollars already exist. Dollars are already accepted by everybody. Why worry about it? Why even, wallets, digital?
- 8.39** So you actually did not get a significant uptick in use of Bitcoin. And early this year, in early 2025, as part of a deal where El Salvador borrowed a big chunk of money from the International Monetary Fund, the IMF, as part of that deal, they actually eliminated Bitcoin as legal tender in El Salvador. The IMF insisted they do that. They didn't want the uncertainty that it provided. IMF is a conservative organization. It's not the kind of an organization that is going to embrace Bitcoin very easily.
- 8.40** So I think the bottom line is that if people lose faith in fiat currency because of, let's say, high inflation, I still believe that it's the most likely that they pick something that has known quantities, that has worked in the past, and that we fully understand how it actually works, and is kind of a guarantee of zero inflation, which is being gold. I mean, this assumes that we don't just replace one fiat currency with another, which is probably the more likely scenario.
- 8.41** I think I told you that Israel, when it had hyperinflation, basically took five zeros off the currency. Three in one go, and then two in another go. They changed the name of the currency. They didn't change their central bank. Central bank was just there, still there. It issued the currency. They just took zeros off, pretended nothing happened, and then ultimately solved the inflation problem by printing less money. It's pretty easy to solve the inflation problem, once you realize what causes it. They printed less money.
- 8.42** You had lira, shekel, and the new shekel. So today's Israeli currency is new shekel because it's two zeros less than the old shekel. So the solution, usually, to hyperinflation is not a new currency and not a new money unless there's some ideological shift. It's just another fiat currency. And I'll trust this central bank; I don't trust that central bank. So we just shift who I trust.

- 8.43** But gold would probably work if we had the ideological shift. I think it's more understandable than Bitcoin and has a history and is more knowable. But at the end of the day, we don't know what will happen if we have hyperinflation, if we have such a situation. Again, I think the most likely scenario is a different fiat currency, a different fiat money. And if there is an intellectual ideological revolution against fiat currency, then I'm still going to go with, the probability is that it's going to be gold and not some crypto.
- 8.44** But again, who knows? Who knows what new crypto or what other cryptos will come about? I do think Bitcoin has some inherent flaws in the system that make it unlikely to become money in the future. I would put the probability of Bitcoin becoming money a lot lower than 10%. That's just me. One of the things that crypto has done, one of the effects of crypto and of the hype around crypto and the excitement about crypto and the interest in crypto, is that it has spurred a whole bunch of other cryptocurrencies and crypto tokens and crypto altcoins and all kinds of stuff that is related but different.
- 8.45** It's certainly conceivable that one of these other ones rises up and replaces Bitcoin as the leader of alternative currencies. Indeed, half the market capitalization today in the crypto world is Bitcoin, but the other half is other coins, so it's not overwhelmingly Bitcoin. And there's a whole new possibility here of synthetic intangible things, electronic assets that who knows what could be used for and how they could be used and what is in our future.
- 8.46** So it's very hard to predict, given how competitive this industry is and how much innovation is possible in the future. I'm certainly not going to put myself in a position to choose winners and losers. I think we're way early. It's still early in the revolution of crypto and the evolution of crypto. We'll know a lot more in 10 years, we'll know even better in 50 years, exactly what happens with this.
- 8.47** It's just generally, it's unlikely that the first company that goes into an industry is ultimately the winner. Usually, other companies learn from that company and improve and offer a better product and ultimately defeat it. Not always the case, but often the case. To compete with Bitcoin, cryptos are going to have to do things like more security, greater user anonymity. And there are quite a few crypto out there that give you more anonymity than Bitcoin, that cannot be traced, that cannot be found.
- 8.48** There are crypto out there that don't have exchanges, where it's all self, you maintain it. I mean, they have exchanges, but they don't have where you store it. You don't store it anywhere; you only store it yourself. There are also crypto, particularly Ethereum and Solana, which offer other functions, as I mentioned, smart contracts and other things, on top of the function as a cryptocurrency. So it knows, might evolve into something really useful.
- 8.49** I mean, one of the problems with crypto, the whole industry, really, with the exception of stablecoins is we don't yet have the killer app. We don't have the thing that will

make everybody say, "Oh, I have to have crypto because I have to do X." There's nothing that crypto does that you can't do without it. There's nothing that crypto does that is so special, so unique that everybody wants to have it, yet. And maybe there will be something. And at that point, it'll become the thing.

- 8.50** What made computers the killer app, what made the Internet the killer app, was email. That's what set it off. What made VHS tapes killer app, that everybody had to have it, was porn. Reality. That's what leveraged that. And indeed, you could argue that high bandwidth for our homes, impetus for that was porn. The first people to be using video, the first people, I think, using now AI to do is that industry because there's a huge consumer base, like it or not. It's a reality. And they consume it, and they are the first movers. They drive use.
- 8.51** So there's no killer app, yet. There might be, and there probably will be at some point. And I think that's when they become much bigger. But the reality is, I think, that at least for now, I don't know what for now means, how long that means for, fiat is what we have. The network effect is too big. It's too significant. We're used to it. We've accepted the inflation. We have lots of financial tools to deal with inflation. You can buy inflation-adjusted bonds that adjust to the inflation, take it into account, and give you a return above and beyond inflation. Everybody now has a 401k that's a diversified stock portfolio, and stocks do better than inflation, most of the time.
- 8.52** So, yeah, we've adapted. And I think to change our behavior and to abandon the dollar for something new would require something pretty major. But there is one application of crypto within the world of fiat money, without changing the world of fiat money, that has taken off, and I think it's about to take off even bigger, although there are going to be some big problems with it as well, I think, because of the way it's structured and the way it's regulated. And that is stablecoins.
- 8.53** So we just had a bill passed called the GENIUS bill. And don't ask me what GENIUS stands for. It's something, something, something, something, something. Each letter stands for something, but it obviously was put together to spell GENIUS because the actual name doesn't make any sense necessarily. So what is a stablecoin? Now, stablecoins are basically a token, a digital token, that is redeemable into a fixed number of dollars. Each token is worth a dollar. And you go back to the people who issued the token, and they will give you a dollar in exchange for the token.
- 8.54** And the price never changes. It is a dollar. That's what it's worth. It's fixed, so it's a fixed redeemability. Now, you could, it doesn't have to be a dollar. It could be, you could make it redeemable to gold. You could make it redeemable to Bitcoin. You could make it redeemable into anything you wanted. Well, right now you can. But once the GENIUS Act goes into effect, it's not clear. It looks like you will only be able to redeem it into dollars. You'll only be able to fix it into dollars.

- 8.55** So, sadly, if my interpretation of the bill is right, I'm not sure it is, but if it is, then this is going to constrain the market. It's going to regulate, control, standardize, and standardization is something markets like, and eliminate experimentation, eliminate new things that could be done with stablecoins. You'll get one kind of stablecoin. And the stablecoin is basically going to be this; this is what the bill seems to encourage.
- 8.56** And that is, let's say, I'm an issuer of stablecoins. You send me your money, you send me dollars, and I send you tokens in equivalents. You send me \$100, I send you \$100 worth of tokens. And then you can use those tokens to buy yourself stuff. And let's say you go on Amazon, you buy some stuff with the tokens, and then Amazon comes to me and redeems the tokens for dollars, and I send them to Amazon. Think of it like, it's almost like a credit card, right? Almost like a credit card.
- 8.57** The difference is that there's not going to be any charge. Credit cards take how much? 2.5% off of Amazon's, if you use a credit card on Amazon, 2.5% is gone. Amazon loses 2.5% every time you use a credit card versus cash. Can't use cash on Amazon. The token is like cash. There's no 2.5%. We'll get back to Amazon in a minute. How do I make money? I'm giving this out. I've got employees, I've got software, I'm monitoring this, I have to pay Amazon. How am I going to make money? Well, I make money by backing these dollars up with something. So I make money by taking these dollars and investing them.
- 8.58** Now, in order to keep it one on one, that is, that the token is worth exactly a dollar, I have to invest in something that's worth a dollar for every dollar I put out there and won't change, so something very safe. If I buy stocks, some years I'll make a lot of money, some years I won't be able to redeem my tokens. I won't be able to pay it back because stock will go down.
- 8.59** So the regulation now says that you basically have to back your dollars with treasury bills. Government bonds, government bills, really short-term. Can't be long-term because if they're long-term, there's risk. Interest rates can change, value of the bonds can change. Short-term, very little risk. So I might make right now, on a short-term government bill, I can make 4%. So you give me your money, I buy treasury bills, I give you tokens, use the tokens for convenience, and I make 4%. Not a bad deal. 4% as a margin. Can't make a huge amount of money. This is not Apple, 50% margins, but 4% is decent. It's pretty good. I can build a business around that.
- 8.60** You could imagine tokens being incredibly useful for internet transactions. No need for a credit card number, no need for the credit card fees. So you could imagine Amazon giving you a discount if you use a token, rather than a credit card. But here's a better idea. This is an idea the bill, as written, suggests that they don't like this idea. But this is, I think, the best use of the tokens. Imagine if Amazon is the one issuing the token. So you basically send 100 bucks to Amazon, and they give you tokens, and you use those tokens on Amazon.

- 8.61** And they say, if you use the token on Amazon, we'll give you a 5% discount on everything you buy. Why 5%? Well, 2.5%, they don't have to pay the credit card company, and then they're going to split the interest that they make on the treasury bills with you. That'll make you much more of a good customer with Amazon. That'll bond you with Amazon. Now, you can use tokens elsewhere if they accept them, because anybody can redeem them for the dollars, but you don't get the discount over there. But at Amazon, you get the 5% discount.
- 8.62** The same deal they have right now is, if you get an Amazon credit card, I have an Amazon credit card, I get 5% cash back. Why? Because they're not paying the fees and they get the float, just like a credit card company gets a float. And some people carry a balance, and they're subsidizing me who doesn't carry a balance. So they can afford to give me 5% discount. The tokens are even a simpler way for them to give me a discount. And indeed, both Amazon and Walmart have both suggested that they would like to issue stablecoins.
- 8.63** The bill is a little ambiguous about how they're going to treat non-bank stablecoins, leaving that to the regulators to decide. Congress these days, just as an aside, I have to say this, Congress, when they write laws today, in the past, they used to write a law, and it used to be pretty clear exactly, if you read the law, what exactly is allowed, what's not allowed, what's going to happen, what's going to be done, what's going to be this. Everything was clearly written in the law.
- 8.64** Over the last 40 years or so, Congress has shifted to a motto where they write a law saying, "Generally, this is the effect we want to have. We're leaving it to the regulators to decide the specifics of how it's going to happen." So laws are these empty shells filled in by regulators. Regulators generally don't necessarily know what they want. So they invite industry to come and sit down with them and figure out what the best way to regulate the industry would be.
- 8.65** Remember, the regulators get a job when they leave the regulatory agency with the industry. Anyway, they probably came from the industry to begin with. But they sit down and they figure out the regulations, often inspired by what the industry is suggesting. The industry means the big players in the industry who are going to write regulations to keep out the little players. It gets very ugly and very unpleasant very quickly.
- 8.66** But the point is that we don't really know what's going to happen. This bill is going to be phased in over the next 18 months. And until the regulations are written, we won't know exactly how it applies. But there is a suggestion. And if you think about, who are the regulators going to invite in to help them write the regulations? They're going to ask the bankers to come in. They're going to ask JPMorgan to come in, JPMorgan Chase. They're going to ask Citibank. They're going to ask the bankers to come in.

- 8.67** And do the bankers want competition from Amazon or Walmart? Probably not. And they're probably not going to invite Amazon or Walmart because they're not financial institutions. What do they know about financial regulations? So we'll see how it all plays out, but that's my prediction, that it becomes much more harder for Walmart and Amazon to do this, even though they're the ones who make the most sense.
- 8.68** Indeed, Amazon or Walmart should have been allowed years ago to have their own banks. Our financial system would be a lot more effective if they just had their own banks. They're not allowed to have that. Walmart tried years and years ago to get a bank charter. No way. But it would have increased convenience dramatically if our big retailers could actually own banks. So it's going to be interesting. JPMorgan Chase, even though Jamie Dimon, the CEO, has been skeptical of crypto generally, has already announced that they intend to try to do a stablecoin.
- 8.69** Visa and MasterCard, who are the big losers here, they're going to have their own stablecoin. So they have already announced that they want in, and they're going to try to do. My guess is American Express will probably do the same thing. They'll also try to figure out ways to differentiate the credit cards, to give you more perks, to get you, to entice you to use credit cards because they're now going to face competition, which they've never really faced before, from these kind of coins.
- 8.70** And then you've got a bunch of crypto companies that have stable coins right now and that now have to modify their business models to fit the regulation because they're not functioning as a regulation. For example, there's a tether gold. You can get now a token that is backed by gold and its value fluctuates with the value of gold. Well, it seems like, again, we don't know because the regulations haven't been written, that that would be illegal, and they could only do it based on treasuries. But maybe not. Maybe gold is liquid enough that they're willing to accept that.
- 8.71** So the key is that if all of us come to the company that issued the stablecoin all at once with our tokens, they will be able to pay all of us out in a reasonable time. You can always sell treasury bonds. Treasury securities are easy to sell. There's a huge liquid market. We'll get to that in a minute. There's huge liquid markets. They could redeem us. Gold is a pretty liquid market. So maybe the argument is that, yes, you could be redeemed because gold is liquid. But given the bias against gold, I find that I'm skeptical. But again, we won't know until we see the actual regulations.
- 8.72** Now, here's one of the big challenges. Right now, treasury bills are 4%. What were treasury bills in 2013, 2014? Half a percent? 1%? I mean, these are the zero interest rate years. If you remember, the Fed drove interest rates to zero. How do you issue stablecoins and make money if all you're making on your treasury bills is half a percent or 1%? This is the problem, by the way, banks face. How do they make money? If they buy treasury bills as assets, they can't pay you any interest on your deposit. And I'm sure they would like to charge you negative interest. They'd like to have you pay for the privilege of keeping your money.

- 8.73** And in some European countries, that actually happened. You had to pay to have your money in a checking account, because they couldn't make money in any other way, because interest rates were so low. How do these stablecoins function when interest rates go down a lot? Well, one option for them will be to replace their treasury bills with bonds that are longer term. Longer term, interest rates go higher. But then they're harder to sell, less liquid, and they fluctuate with interest rates.
- 8.74** What if interest rates go up a lot? Now the value of the bonds go down, and now they don't have a one-on-one link anymore. The bonds are worth less. It's exactly what happened to the banking industry. We talked about this when we talked about Silicon Valley Bank. The value of their bonds went down. They couldn't redeem all their checking accounts when there was a run on the bank.
- 8.75** So it seems to me that stablecoins are creating exactly the same situation. How do you redeem? And how do you back them up? And yeah, right now it's easy, 4%, but now is not going to exist forever. We don't know what the situation is going to be, and what's the volatility going to be like as interest rates go up and down? If Trump gets his way, and Powell lowers interest rates by 3%—he won't, but if he did— now we're at 1%. Stablecoins all go out of business. They can't make any money. Fire everybody, shut it down.
- 8.76** And you can't do that. You can't have an industry that, oh, yeah, flourishes when interest rates are high and then just disappears when interest rates are low. That won't work. So a lot of uncertainty around these things. A lot of potential, but a lot of uncertainty. And this idea of stablecoins going bust is, we've seen it happen a couple of years ago in 2022, May of 2022, so three years ago now.
- 8.77** Terraform Labs, which issued a stablecoin called TerraUSD, went bust, and people lost hundreds of millions of dollars on this. And now they had this very sophisticated way of trying to keep the parity of \$1 token versus redeemability. They had some algorithm that managed the whole process. It was very complicated. It's not even worth getting into. The point is, everybody thought this was perfect, and people bought into it, and people had the stablecoins, and it broke. The redeemability broke. They couldn't redeem their money. And the whole thing was wiped out.
- 8.78** It wasn't necessarily fraud. It was just a bad algorithm. It was just placing wishes above fact, pretending that something was real when it wasn't. And again, hundreds of millions of dollars were lost. And if you remember, a bunch of digital companies went bankrupt in that period, 2022. Remember when Bitcoin prices went down, everything went down, and FTX was massive. A crypto exchange, Celsius, Voyager Digital, a number of these things all went bankrupt, primarily because of a lack of liquidity or because Bitcoin had gone down and nobody expected. Bitcoin never goes down. So nobody expected Bitcoin to go down.

- 8.79** And they had basically assumed Bitcoin would be high, and they'd issue tokens based on Bitcoin. And now Bitcoin was down, and they couldn't redeem the tokens. So still a lot here. This is still a very early, very infant industry. There's a lot that has to be worked out. And from everything I know of studying financial regulations over many, many years, and financial regulations in this country have been extensive since the founding of the country, but particularly since the Great Depression, the probability that regulators will get this right in terms of helping to establish the industry properly and safely and everything goes well, is about as close to zero as you could get.
- 8.80** They're just not that good at it. They just have—and nobody is. Central planning is no good. And the chance that this ultimately leads to some financial crisis down the road? Very high. You'll remember this day, this class, when there's a stablecoin financial crisis. "He told us so." All right. So there's a lot that's going to happen in the marketplace. There's a lot interesting happening in the space. Again, I think government dabbling in it, the industry loves it. Industry likes regulations because it keeps out new players and it keeps the current players in place and it gives them rules to play by.
- 8.81** But it usually crushes innovation and creates crises. But right now, everybody's really excited. It's one of the reasons Bitcoin is so high. People are excited about crypto, but there's still a lot to be done before we know what this particular legislation is going to do. There are two more bills related to crypto in the House of Representatives and the Senate right now that relate to Bitcoin and other cryptos, and what's in them is going to determine, again, the shape of the industry and then how they get applied through the regulatory process.
- 8.82** And generally this whole space is in flux. And it's very hard to tell right now who the winners and losers, and what makes sense or what doesn't make sense, is going to be, until we get clarity around the regulatory environment and then we get a shakeout, a competitive shakeout, of the players involved. It's a risky space to be playing in. For those who play in these spaces, it's risky. Some people are going to make a lot of money.
- 8.83** There are other applications out there that are not crypto, but that are being used to facilitate faster and better transaction capabilities. And a lot of those are being done by governments. They control, they're fiat countries. In a fiat country, it's easiest for them to do. I'll just give you an example. One example of this is from Brazil. Right now, Donald Trump is really going after these guys, so it's in the news. It's called Pix, P-I-X. And it's an instant payment system that has been introduced by the Brazilian central bank. It makes it possible to do peer-to-peer money transfers.
- 8.84** Again, the Central Bank controls the ledger. They figure out who owns what, and they move the money around appropriately. So it doesn't have to go through your bank account. It's directly with the central bank. They have control over the ledgers of the banks. It was developed by the central bank of Brazil, so it's not a private company. It's

instant transfers, works 24/7, pretty much any amount. You can pay companies, you can pay other people, you can pay the government, you can pay your taxes with this.

- 8.85** It's just an account. You put in how much you want to pay, you scan a barcode, so it goes to the right person, and you just pay it. It's a little bit like what we use here, Venmo and things like that, but this is much more widespread. And Venmo, you can only use up until a certain amount of money. You can't send large amounts of money. Here, it's pretty much all transactions can be done on this system. Very low cost, very easy to use. And indeed, a lot of Brazilians who never had a bank account in their life are now using this Pix, and they're now using this to transact.
- 8.86** It connects to financial institutions, it connects to central banks, fintech companies, and it unifies all of that in one system. It has pretty robust encryption, so it's secure, although they're going to break the encryption and there's going to be fraud. There already has been, and there will be more of it. And the reality is that 70% of Brazilians are using this, many of whom have never had a bank account. And it's basically now the way in which you pay for everything in Brazil.
- 8.87** Credit cards are out, and this is what is pissing off our President; like, Visa and MasterCard, two solid American companies, are being priced out. Pix is beating them, and he doesn't like that. So in the name of Visa and MasterCard, he's going after the Brazilians for this. It disadvantages U.S. payment providers, he's claimed. It harms U.S. commerce. And it's an attempt by BRICS, BRICS is Brazil, Russia, India, China, South Africa, to replace the dollar as a reserve currency. I mean, it's all a little hyped. But anyway, he's excited about this.
- 8.88** But there's no privacy. Central bank knows everything, can access everything that you spend money on. Nobody cares. It's fiat currency, still the same money. But people love it because it's convenient and easy and quick. And this is why fiat currency is so powerful right now. It's because it's so easy, quick, easy to use. It doesn't cost us anything to transact. With a Venmo, PayPal, and we only make those better, why bother with all these really complicated, hard stuff that don't even, hardly understand how they work? This is straightforward and simple. And this is why, once you have really an ideological shift away from central banking and fiat currencies, it's hard to believe that it goes away, it goes anywhere.
- 8.89** The next big thing, which central banks are already experimenting with, the Federal Reserve has dabbled with, but the GENIUS Act actually prohibits this from happening in the United States, at least for now, is central bank digital currencies. A central bank digital currency is basically a digital version of a dollar that the central bank issues. Basically, it will go hand in hand with elimination of cash. They deal with, you get rid of cash and you only use this digital, it's on your phone or it's anywhere, and you'll use this as money.

- 8.90** But of course, because it's a piece of code, that piece of code now can be tracked completely by the government. They can tell every transaction that you make; there's zero privacy. It represents a direct liability of central banks. It's not going through your bank checking account. The central bank is issuing these directly to the population. Maybe it's not clear exactly in exchange for what. Maybe it goes through the banking system in exchange for reserves somehow. But basically, you will be interacting. Each one of us will have a little ledger at the central bank with our transactions in them.
- 8.91** It's going to be secure. It's going to be encrypted digital currency, resistant to fraud. The other aspect of it, which is interesting, is it'll be programmable. You might have \$100 on your phone with a program that says you have to spend it within the next week. Let's say the government is trying to stimulate the economy, and they believe that stimulating the economy is getting you to spend money. Then they might say, "We're not just going to send you a check like in COVID, and you can then put it in savings, God forbid. We're going to send you a check, this digital currency, and you have to spend it today." That'll stimulate the economy.
- 8.92** Or think about a digital currency. It's sitting on your phone, \$100, and it has, coded into it, can't spend it on alcohol, can't spend it on porn, can't spend it on cigarettes. I don't know, you can invent the things. Sugar, can't spend it on sugar unless it's in Coca-Cola, and then it's OK. That's a joke relevant to July 2025, maybe not relevant later on. They will be able to tell what you can and cannot spend your money on because it'll be coded. It won't connect if it's not.
- 8.93** Of course, the main player here in terms of piloting a centralized central bank digital currency is China. They love this. They have a pilot program. It's called digital yuan. It's crypto. It's controlled by the government. Crypto in a sense that it's cryptography. It replaces, the idea it will replace, it's still a pilot, but it will replace physical cash. Imagine combining this with what China has today, which is a social score. You get a score. The score tells you if you can get a mortgage or not get a mortgage. It tells you how good of a person you are from a Chinese perspective, from a Chinese government perspective.
- 8.94** Now, imagine your cash is connected to that, and they say, "You've been bad this week. You can't use your money. No candy for you this week." It's also considering using this for international payments. It's very easy and simple to use. Digital transactions are very simple and easy. It's not clear if other countries will like the idea that it's programmable and the Chinese government has control over it. We'll see if other countries buy into that. But China has a lot of muscle, so they might be able to force certain countries to do it.
- 8.95** Of course, it's all about control and surveillance for the Chinese. There's a big fear that the Fed will do the same thing, although it's unlikely the Fed would put in the surveillance stuff that the Chinese would. But Congress, as far as I can tell, in this GENIUS act, there's a prohibition on central bank digital currency, but that could change like that, if Congress was convinced otherwise.

- 8.96** So, where are we? Our money is, if you had money, it's not going to change anytime soon, we're going to see all kinds of innovations, some good, some not so good. Central bank digital currency might not be a good innovation, but we will see a lot of innovation, primarily related to how fast we can transact, how simply we can transact. We'll probably see stablecoins. Whether the stablecoins will be stable or not is going to be a good question. But stablecoins have the potential to increase the efficiency of transactions, if they're structured right and regulated right and controlled right.
- 8.97** Ideally, they would be regulated by a market process, but that has been taken away from the market at this point. It's unlikely, without a catastrophe, that we switch away from a fiat currency. We are linked to the dollar. Indeed, the dollar is only going to get, I think, more popular as other countries abandon their own currencies because they do a pretty bad job doing it. Milei has promised to dollarize the Argentinian economy, to go off the peso, to shut down its central bank, and to dollarize.
- 8.98** And indeed, what he's actually promised is, in addition to the dollar, and this will be an interesting experiment, is to actually create competition in Argentina. To create a situation in Argentina where you'll be able to trade, or you'll be able to buy and sell goods in dollars, in gold, in Bitcoin, in a variety of different medium. We'll see, again, how we structure this and how it all comes together. But I'd say the only promise in the world right now for competitive currencies, for actually creating competition and seeing which one wins, is Argentina.
- 8.99** And I'd say that is really what we'd want, if we're really interested in money and a good money. What we want is competition. What we want is—and I think ultimately you'd have one winner. I don't think competition would lead to multiple players because there's a huge advantage, huge network effect, to having one money. But let the market determine what that money would be, as it did historically. Gold arose and silver arose as monies because they made the most sense, because they were most productive. They were the most efficient. They were most efficacious, and they were the ones people trusted the most.
- 8.100** We would have never, and indeed, there's never been a case in history, where the market has evolved a fiat currency. Fiat currency can only come about by being imposed by a government. So let's bring back competition. And crypto should be part of that. And let's see which one wins out. Maybe we'll get a little taste of that in Argentina, but it would be nice to get it on a larger scale.
- 8.101** More importantly, what I want you to get out, I'm hoping you got out of this class, is really the importance of money, the importance of getting it right, the danger of getting it wrong, but also the importance of money for our lives. Now, we take it for granted. We use it to buy stuff. But having good money is crucial to our ability to save, to produce, to create. Good money really is a reflection of our productive ability. It is a store of the values we have created. The work that you do is captured, frozen in time, if you will, by money, by good money. It's an immense value to our lives.

- 8.102** I hope that you get out of this course that money is not the root of all evil, and that even love of money is not at the root of all evil. If love means understanding something's nature and really appreciating it for what it really is, not just desiring it for the sake of desire, that's not love, that's desire—but to love something, to know something, is to respect it and to know where it comes from. And money comes from value creation.
- 8.103** In a fiat world, we think money is just printed. It's just created out of thin air. But it doesn't represent thin air. It represents the labor and work and entrepreneurship and thought and creativity and imagination of countless productive individuals. And in your case, your money represents your own productive ability, your own mind in action, if you will. It's a beautiful thing. Thank you.
- 8.104** So you've talked a lot about different eventualities of money. I wonder what role Gresham's law has on this. Is fiat money always going to be the go-to money because of that?
- 8.105** No.
- 8.106** Bad money drives out good?
- 8.107** Yeah, but under what circumstances? So Gresham's law is not bad money drives out good. Bad money drives out good, if the bad money and the good money have the same face value and accepted as having equal value. So let's have two coins, both gold, one, and they're both stamped with, let's say, \$100. So two coins, each with a stamp of \$100. One is pure gold, one is diluted. But the legal laws out there say I have to treat them both as \$100.
- 8.108** Which one am I going to keep in my pocket and which one I'm going to spend? I'm going to keep the one that's pure gold. I'm going to spend the bad money. But if the two coins were, this one's \$100 and it's pure gold, and this one's only worth \$80 because it's not pure gold, and I can spend it as if it's \$80 and this one's spent as \$100, then it's not clear that this will drive out that. Then they both will function because they each have the value appropriate to what they are.
- 8.109** And in circumstances where I can tell which one's pure and which one's not, the bad money will be melted down and created into good money because nobody is going to accept the bad money. So good money will drive out bad money. So Gresham's law applies only when the exchange rate between the good money and the bad money is fixed, and I can't change it. And in this case, for example, it's equal. And yet the one—
- 8.110** I had some examples. I think there was a period where you had two types of silver coins in the United States. You had the Spanish silver coin, which was basically pure and undiluted. And you had other silver coins transacting who were diluted. And yet, they all were at the same nominal value and all were treated the same. Well, what happened is the Spanish silver coins disappeared. They all went to Spain or they went

somewhere else. Nobody would use them in the United States because you could get less for them, because everything was priced as if it was bad money, so bad money dominated.

- 8.111** So it's only under that condition does bad money win out. Fiat money wins out today not because it's bad money, not because of Gresham's law. It wins out today because it's legal tender, and you can't use other money. You can't set up a bank where I issue gold certificates. I can't set up because there's capital gains. They won't give me a bank charter. I won't get approved. And people have to use this other thing that's not quite clear what it is. And my fiat money is pretty satisfactory right now. So I'd rather use the gold.
- 8.112** I mean, there's a sense in which maybe Gresham's law applies, in the sense that I'm using the gold to save and I'm using my fiat money to speculate. There's a sense in which that is true. But you also have to question, is gold money today? Gold is not money today. Gold used to be money. It could be money, but it's not money. Gold is a speculative asset like Bitcoin. It's a different type of speculative asset, but still speculative asset. It's why it's so volatile.
- 8.113** And does it match inflation? It doesn't. Not exactly. It matches something. When things get bad, people buy gold. I think it's a hedge against the end of the world. It's a hedge against things. If civilization disappears, gold will have value. So you want to have a little bit of gold, just in case civilization disappears. But it's not treated as money today. So fiat money is there because legally, it's the only money that is really allowed. So Gresham's law, like many of these things, are contextual. So it has a particular context in which it applies. Other questions?
- 8.114** Do you think it's possible that with our current fiat money system and, say, Bitcoin would combine with how we've been moving towards digital currency recently, taking the more positive attributes or the better attributes from each of them, say, like the cap on Bitcoin?
- 8.115** No, I don't see it because they're really two completely different systems. And why would the government ever want to cap? The whole point of fiat currency is they don't cap, is they can do whatever they want with it. Now, I do think that fiat currency will become more digital. That is, you will see, as we saw with Pix in Brazil, you'll have more payment systems. You'll have it, you might get central bank digital currencies, whether we like it or not. We might ultimately get that. If not, we'll get something else. These stablecoins are all basically being run on a blockchain. So keeping track of who owns what is all on a blockchain. So that's, in a sense, some kind of combo.
- 8.116** So you're going to get some influence of crypto on to how we use fiat currency. But I don't think the government is going to give away its ability to manipulate the currency. That is not going to walk away. I mean, I wish; it would be great. I mean, a simple way for them to do that, we talked about this when we talked about central banking, is for

the government to say, OK, central bank, here's the rule you have to follow when you issue currencies, like a Taylor rule, or you have to follow gold, or you have to, something. There are all kinds of proposed rules.

- 8.117** You can't deviate from that rule, no matter if there's a crisis or not a crisis or whatever. But what we've learned, if we've learned anything from history, is every time there's a crisis, they'll deviate. No matter what rule you give them, they'll deviate, as long as you give them the discretionary power to do so. That's why you have to take it out of the hands of government, because as long as it's in the hands of government, they will find ways to cheat. There's too much at stake, too much power at stake.